



Waveserver 5

Waveserver 5 Installation

OS 2.2

323-3001-200 - Standard Issue 1
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CHAPTER 1

New in this release

This technical publication contains information on how to install the Waveserver 5 chassis and related equipment for OS 2.2.

For more information, refer to the *Waveserver 5 User Guide*, 323-3001-100.

Issue 1

The following new hardware are introduced in Waveserver 5 for OS 2.2:

- Waveserver WL5e 2x800G C-Band Multi-Baud Submarine 12xQSFP28/QSFP-DD Module (186-3120-900)
- Waveserver WL5e 2x800G C-Band Multi-Baud Premium 12xQSFP28/QSFP-DD Module (186-3121-900)
- 400GBASE-LR4, SMF, 10Km, QSFP-DD (160-9603-900) pluggable module

CHAPTER 2

Observing product and personnel safety guidelines

This section contains safety guidelines that you must follow for personal safety and to operate the equipment correctly.

Ciena documentation contains precautionary messages and safety procedures that refer to specific tasks or conditions. You must read and follow all precautionary messages before you start to work on the equipment.

This section contains the following topics:

- [Audience](#)
- [Precautionary messages](#)
- [Safety standards](#)
- [Laser radiation—optical transmission systems](#)
- [Labeling](#)
- [Grounding standards](#)
- [Using optical fibers](#)
- [Working with systems cooled by forced air](#)
- [Working with power](#)
- [Dissipating static electricity](#)
- [Preventing module damage](#)
- [Controlling equipment access](#)
- [Controlling equipment environment](#)

Audience

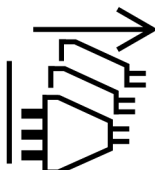
Personnel working directly on the equipment must be:

- trained, authorized, and qualified to carry out the tasks required
- able to follow safety guidelines specific to the product and all local customer-specific safety procedures

Precautionary messages

To prevent personal injury, equipment damage, and service interruptions, follow all precautionary messages in Ciena documentation and all local safety standards required by your company.

The following precautionary messages appear in Ciena documentation.

	CAUTION Risk of electric shock Disconnect all power sources before servicing to avoid shock hazard.
	 ATTENTION Risque de choc électrique Débranchez toutes les source d'alimentation avant l'entretien afin d'éviter tout risque de choc.

	DANGER Risk of personal injury A precautionary message with this symbol indicates a risk of personal injury.
--	--

	DANGER Risk of electrical shock A precautionary message with this symbol indicates a risk of personal injury caused by an electrical hazard.
---	--

	DANGER Risk of electrical shock A precautionary message with this symbol indicates a risk of electrical shock and that you must wait the specified amount of time for the power to dissipate from the module before touching it.
---	--

	DANGER Risque de décharge électrique Un avertissement montrant ce pictogramme indique un risque de décharge électrique. Vous devez observer le délai prescrit avant d'entrer en contact avec le module, pour laisser suffisamment de temps à la charge de se dissiper.
---	--



DANGER

Risk of laser radiation exposure

A precautionary message with this symbol indicates a risk of personal injury caused by exposure to a laser beam.



WARNING

Risk of burn

A precautionary message with this symbol indicates a risk of personal injury caused by a hot surface.



CAUTION

Risk of interruption to service

A precautionary message with this symbol indicates a risk of service interruption or equipment damage.



CAUTION

Risk of damage to modules

A precautionary message with this symbol indicates a risk of static damage to modules. Use antistatic protection.



CAUTION

Hazardous moving parts

Keep fingers and other body parts away.



CAUTION

Parties mobiles dangereuses

Tenir les doigts et les autres parties du corps éloignés.



CAUTION

Risk of electric shock

Disconnect all power sources before servicing to avoid shock hazard.



CAUTION

Risque de choc électrique

Débranchez toutes les sources d'alimentation avant l'entretien afin d'éviter tout risque de choc.

Safety standards

The following sections outline the safety specifications.

Safety compliance

Ciena network elements conform to all relevant safety standards. Waveserver 5 meets or exceeds the following standards:

- IEC 62368-1 Edition 2
- IEC 60825-1 Edition 3.0 (2014)
- IEC 60825-2 Edition 3.2 (2010)
- EN 62368-1:2014
- EN 60825-1:2014
- EN 60825-2:2004/A1:2007/A2:2010
- CAN/CSA-C22.2 No. 62368-1-14
- UL 62368-1 Edition 2

See the “Operating specifications and standards compliance” section of the *Waveserver 5 User Guide*, 323-3001-100 for details about Waveserver 5 safety compliance and environmental and electromagnetic specifications.

Restricted access location

Waveserver 5 is designed for use in restricted access locations only. A restricted access location is a location that meets the following conditions:

- Access can be gained only by service personnel or by users who have been instructed about the reasons for the restrictions applied to the location and about any precautions that shall be taken
- Access is through the use of a tool or lock and key, or other means of security, and is controlled by the authority responsible for the location

Safety label

Waveserver 5 is a Class 1 laser product/optical fiber communications system, (per IEC 60825-2, Annex A).

Waveserver 5 optical interfaces may pose a laser safety hazard of 1M (IEC 60825-1) and applicable modules are marked as indicated in the following figure.

Figure 1 Laser hazard level 1M label

**Laser radiation—optical transmission systems**

Ciena optical products use laser or light-emitting diode (LED) sources that emit light energy into optical fibers. This energy is within the red (visible) and infrared (not visible) areas of the electromagnetic spectrum.

Laser radiation hazards

When operating the product normally, with all optical connectors in position and correctly terminated, the optical radiation is completely enclosed. The system is a Class 1 (IEC)/Class I (FDA) product, regardless of the power transmitted within the optical fiber.

If you have unterminated optical cables (breaks in the fiber-optic cable or disconnected connectors) the output from modules containing optical transmitters can be greater than Class 1 (IEC)/Class I (FDA).

When used and operated as per the procedures described herein, the maximum radiation level encountered does not exceed the accessible emission limit (AEL) for Class 1 (IEC) or Class I (FDA). The worst case hazard level that may be encountered (as defined by IEC 60825-2) is Hazard Level 1M, and defined by ANSI Z136.2 as a Service Group 1. Because the system is intended to be installed in a restricted access location, no additional engineering design features are required to prevent exposure to the laser radiation.

The following statement (English and French versions shown) refers to all modules and components bearing an IEC 60825-2 Hazard Level 1M label. For further details on optical power and wavelength for each instance, refer to [“Module wavelength, maximum optical output power, and laser hazard level” on page 10.](#)



Labeling

The Waveserver 5 system and modules have been labeled in accordance with IEC 60825-1, IEC 60825-2, and FDA Laser Notice 56 (Dated May 8, 2019).

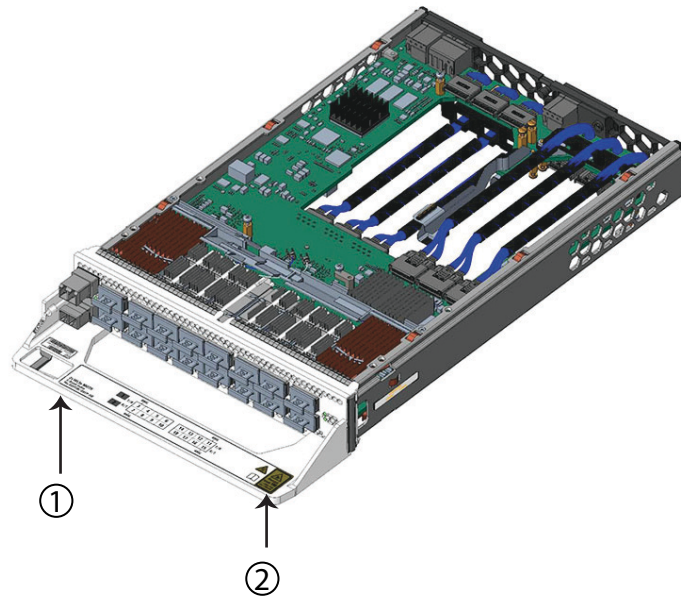
Module labels

The elements of a typical module explanatory label are as follows:

- laser warning symbol
- warning text explaining the risk associated with the hazard from exposed optical fiber ends and unterminated connectors. This warning text indicates that a hazard can exist on either the fixed or free end of a connector depending on the direction of transmission.
- a laser hazard level statement in accordance with IEC 60825-2:2004 + A1:2006 + A2:2010

For laser hazard labels found on the Waveserver 5 modules, refer to the following illustration (actual faceplate rendering may be different):

Figure 2 Location of laser hazard level 1M label (on the module handle)



Legend

1 Module handle

2 Hazard level label

The following labels apply to Waveserver 5 equipment as listed in [Table 1 on page 10](#).

Figure 3 Laser hazard level 1M



Figure 4 Laser hazard level 1 (label may not appear on the product)



The Waveserver 5 modules use lasers that may exceed the AEL for Class 1, and therefore require wavelength and optical power information at the aperture. Additionally, the information in the following list is provided in this section for every optical module and optical pluggable transceiver—refer to specific information in [Table 1](#):

- wavelength, shown as a nominal value or dense wavelength division multiplexing (DWDM) range of values, such as in the following examples:
 - 1310 nm nominal indicates a single wavelength in the range of 1260 nm to 1380 nm
 - 1550 nm nominal indicates a single wavelength in the range 1525 nm to 1625 nm
 - 1510 nm to 1625 nm indicates a range of multiple discrete DWDM wavelengths
 - 1471 nm - 1611 nm indicates a range of multiple discrete CWDM wavelengths
- the maximum accessible optical output power at the faceplate/aperture, after a fiber break (that is, in consideration of IEC 60825-1 classification and the impact of embedded laser safety features as an Optical Fiber Communications System per IEC 60825-2).

For the name and the publication date of the standards used to assess the equipment, refer to the “Safety specifications” section in the *Waveserver 5 User Guide*, 323-3001-100.

The following table lists the wavelengths, maximum output power, and hazard level labels for the modules and pluggable modules supported on Waveserver 5.

Table 1 Module wavelength, maximum optical output power, and laser hazard level

Module	Wavelength	Maximum output power	Hazard Level
186-3100-900	1528-1567 nm	< 100 mW	1M/ Figure 3 on page 9
186-3100-901	1528-1567 nm	< 100 mW	1M/ Figure 3 on page 9
186-3101-900	1528-1567 nm	< 100 mW	1M/ Figure 3 on page 9
186-3101-901	1528-1567 nm	< 100 mW	1M/ Figure 3 on page 9
186-3102-900	1528-1567 nm	< 100 mW	1M/ Figure 3 on page 9
186-3102-901	1528-1567 nm	< 100 mW	1M/ Figure 3 on page 9
186-3110-901	1570-1611 nm	< 100 mW	1M/ Figure 3 on page 9
186-3111-901	1570-1611 nm	< 100 mW	1M/ Figure 3 on page 9

Module	Wavelength	Maximum output power	Hazard Level
186-3112-901	1570-1611 nm	< 100 mW	1M/ Figure 3 on page 9
186-3120-900	1528-1567 nm	< 100 mW	1M/ Figure 3 on page 9
186-3121-900	1528-1567 nm	< 100 mW	1M/ Figure 3 on page 9
186-3302-900	1528-1567 nm	< 100 mW	1M/ Figure 3 on page 9
Pluggable modules			
180-3530-900	1528-1567 nm	< 100 mW	1/ Figure 4 on page 9
160-9400-900	800 nm	<2.31 mW (3.5 dBm)	1/ Figure 4 on page 9
160-9401-900	1310 nm	< 25 mW (14 dBm)	1/ Figure 4 on page 9
160-9402-900	1310 nm	< 10mW (10 dBm)	1/ Figure 4 on page 9
160-9403-900	1310 nm	< 25 mW (14 dBm)	1/ Figure 4 on page 9
160-9404-900	1310 nm	< 4.46 mW (6.5 dBm)	1/ Figure 4 on page 9
160-9405-900	1310 nm	< 4.46 mW (6.5 dBm)	1/ Figure 4 on page 9
160-9406-900	1310 nm	< 1.58 mW (2.0 dBm)	1/ Figure 4 on page 9
160-9431-900	1310 nm	<15.6 mW (11.9 dBm)	1/ Figure 4 on page 9
160-9460-003	850 nm	< 0.5 mW (-3 dBm)	1/ Figure 4 on page 9
160-9460-005	850 nm	< 0.5 mW (-3 dBm)	1/ Figure 4 on page 9
160-9460-007	850 nm	< 0.5 mW (-3 dBm)	1/ Figure 4 on page 9
160-9460-010	850 nm	< 0.5 mW (-3 dBm)	1/ Figure 4 on page 9
160-9460-015	850 nm	< 0.5 mW (-3 dBm)	1/ Figure 4 on page 9
160-9460-020	850 nm	< 0.5 mW (-3 dBm)	1/ Figure 4 on page 9
160-9460-050	850 nm	< 0.5 mW (-3 dBm)	1/ Figure 4 on page 9
160-9460-100	850 nm	< 0.5 mW (-3 dBm)	1/ Figure 4 on page 9
160-9600-900	1271 nm, 1291 nm, 1311 nm, 1331 nm	< 10 mW (10 dBm)	1/ Figure 4 on page 9
160-9603-900	1271 nm, 1291 nm, 1311 nm, 1331 nm	<12 mW (11.1dBm)	1/ Figure 4 on page 9
160-9660-003	850 nm	< 0.75 mW (-1.25 dBm)	1/ Figure 4 on page 9
160-9660-005	850 nm	< 0.75 mW (-1.25 dBm)	1/ Figure 4 on page 9
160-9660-010	850 nm	< 0.75 mW (-1.25 dBm)	1/ Figure 4 on page 9

Module	Wavelength	Maximum output power	Hazard Level
160-9660-020	850 nm	< 0.75 mW (-1.25 dBm)	1/ Figure 4 on page 9
160-9660-100	850 nm	< 0.75 mW (-1.25 dBm)	1/ Figure 4 on page 9

Grounding standards

For the list of grounding standards, refer to the “Power and grounding specifications” in *Waveserver 5 User Guide*, 323-3001-100.

For details related to grounding, see [“Grounding requirements” on page 15](#).

Using optical fibers

Optical fibers are either single mode or multi-mode. The information in the following paragraphs applies to all optical fibers.

Handling optical fibers

When you work with optical fibers, you must take the following precautions:

- Wear eye protection such as safety goggles or safety glasses with side guards when you install optical fibers.



DANGER

Risk of laser radiation exposure

Do not look directly into the optical beam. Invisible light can severely damage your eyes. Keep all optical connectors capped.

- Do not look into the opening of an optical fiber, or the opening of an optical fiber connector, if the optical fiber is active or the unit has the power turned on.
- Avoid direct exposure to optical fiber ends or optical connector ends where you can access the laser signal directly.
- Clean your hands after you handle optical fibers. Small pieces of glass are not always visible and can damage your eyes.



DANGER

Risk of eye injury

If you have a piece of a glass in your eye, get medical assistance immediately.

- Do not handle pieces of optical fiber with your fingers. Use tweezers or adhesive tape to lift and discard any loose optical fiber ends.

- Wear chemical-resistance gloves (latex, vinyl/PVC or nitrile type) when you clean optical connectors. The chemical-resistance gloves prevent direct contact with the isopropyl alcohol and prevent contamination of the ferrules with skin oils.
- Place all optical fiber clippings in a plastic container provided for that purpose.
- Handle optical fibers with caution. Place the optical fibers in a safe location during installation.
- Protect all optical fiber connectors with clean dust caps at all times.
- Follow the manufacturer's instructions when you use an optical test set. Incorrect calibration or control settings can create hazardous levels of radiation.

Splicing optical fibers

When you must look at a spliced optical fiber with a small magnifier, take the following precautions:

- Power off all laser sources to the optical fiber or disconnect the remote optical fiber end from the laser sources before you start splicing. Make sure that all laser sources remain disconnected or have the power turned off. The laser sources can be in a central office, on subscriber premises, or in a remote location.
- Disconnect all optical test sets from the optical fiber before you start splicing.
- Use only the optical instruments approved by your company.

Repairing optical fibers

When an accidental break occurs in the optical fiber, do the following:

- Report the location of the damaged optical fiber to both the central office and the field repair personnel.
- Power down all laser sources to the optical fiber or disconnect the remote optical fiber end from the laser sources. The laser sources can be in a central office, on subscriber premises, or in a remote location.

Working with systems cooled by forced air

When you work with systems cooled by forced air, you must take the following precautions:

- Wear safety goggles or safety glasses with side guards, for protection against airborne debris, including glass fiber particles. Always wear eye protection when working near the shelf air exhaust, or when inspecting or replacing a cooling fan module or shelf air filter.



DANGER

Risk of eye injury

Wear eye protection such as safety goggles or safety glasses with side guards when you work with fan modules, air filters, or in proximity to the shelf air exhaust.

- If you remove a cooling fan module from Waveserver 5 to inspect or replace it, you must replace it within 30 seconds to ensure that the modules equipped in the shelf do not overheat and fail while the fan module is being replaced. Make sure that you have a replacement cooling fan module ready before removing the cooling fan module.



CAUTION

Risk of personal injury

The fan blades on cooling fan modules rotate at high speed while operating. If you are removing an active fan module from a shelf, partially remove it then wait a few seconds for the fan blades to stop rotating before handling it. Do not hold or carry a fan module in a manner that could cause detrimental contact to the fan blades or any backflow louvers.

Waveserver 5 is cooled by forced air pulled through the system by cooling fan modules. The modules in the system depend on a constant flow of air in order to operate efficiently and reliably. The following guidelines apply:

- Do not allow Waveserver 5 equipped with active modules to operate without any working cooling fan modules for more than 30 seconds as some modules may overheat and malfunction or fail.
- Do not block the air exhaust or operate the system in a fully enclosed cabinet or room that will re-circulate exhaust air back into the system.
- Do not operate Waveserver 5 with empty slots, empty subslots, or empty pluggable transceiver ports/cages as it affects shelf cooling. You must equip unused slots and subslots with appropriate filler modules and protect all unequipped transceiver ports/cages with appropriate dust covers.

Working with power

This section includes information on the following topics:

- [Grounding requirements](#)
- [Bonding topologies](#)
- [Consequences of incorrect grounding](#)
- [Working with DC power](#)
- [DC power modules](#)

Grounding requirements

Read and follow this precaution.



DANGER
Risk of electrical shock and equipment damage
 Grounding is mandatory to satisfy local electrical codes/regulations for the safe use of the equipment.

Ground the bay/cabinet to the common building network (CBN), isolated bonding network (IBN) or ETSI Mesh Bonding Network ground/protective earth.

Correct grounding of the equipment bay/cabinet is mandatory in accordance with electrical safety standards and mitigates the electrostatic discharge (ESD) and operational concerns described in [“Consequences of incorrect grounding” on page 17](#).

Also, ground every shelf and peripheral tray as described in the related procedure.

Marked locations on the bay and/or shelf bearing the Protective Earthing Conductor are appropriate locations for attaching grounding hardware (for example, approved cables or lugs)—see the following illustration.

Figure 5 Protective Earthing Conductor (IEC 60417-5019)



Bonding topologies

Bonding is intended to create equipotential nodes throughout the office installation. Marked locations bearing the symbol illustrated in [Figure 5](#) identify sites that can be used to bond equipment but are not intended for safety grounding. The Protective Earthing Conductor can be connected using either of these topologies to satisfy the mandatory (safety) grounding dependencies for the Waveserver 5.

Waveserver 5 supports connection to and installation in the following equipment bonding topologies (additional detail on these topologies and their relationships is covered in Telcordia GR-295-CORE):

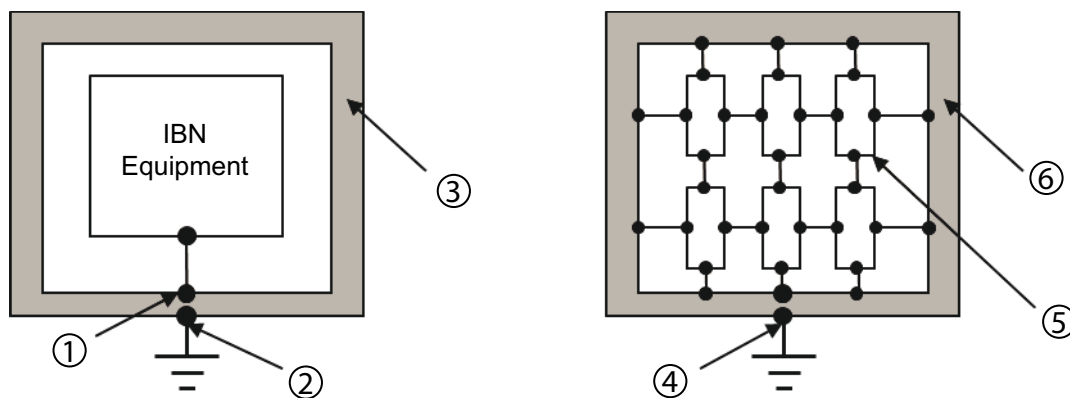
- **CBN**—Common Bonding Network – This is the principal means for effecting bonding and earthing inside a telecommunication building. It is the set of metallic components that are intentionally or incidentally interconnected to form the principal BN in a building. These components

include: structural steel or reinforcing rods, metallic plumbing, AC power conduit, ACEG conductors, cable racks, and bonding conductors. The CBN always has a mesh topology and is connected to the grounding electrode system.

- IBN—Isolated Bonding Network – A bonding network that has a single point of connection (SPC) from the network element (for example, shelf or bay) to either the common bonding network or another isolated bonding network. All IBNs will have a connection to earth through the SPC.
- Mesh-BN—A bonding network in which all associated equipment frames, racks, and cabinets, and usually, the DC power return conductor, are intentionally bonded together and at multiple points to the CBN. Consequently, the mesh-BN augments the CBN.
- There are multiple additional variations of these topologies, for example, Mesh-IBN, Star-IBN.

The following illustration shows the key differences and relationships among the bonding topologies.

Figure 6 Bonding topologies—CBN, IBN, Mesh



Legend

- | | | | |
|---|-------------------------|---|-------------------------|
| 1 | SPC | 4 | Office Principle Ground |
| 2 | Office Principle Ground | 5 | Mesh-BN Equipment |
| 3 | CBN | 6 | CBN |

- There are two DC Return implementations supported within the Bonding Network topologies (in [Figure 6](#)):
 - Isolated DC Return (DC-I) – An implementation of the battery return(s) where the network element has a separate DC return(s), isolated from other BNs until a single connection point, often at the power source,

where connection to the CBN is implemented. This is the default configuration for all Waveserver shelves, with the battery return(s) isolated from the local framework ground. The only common connection between the battery return(s) and CBN is at the SPC which is consistent with the IBN topology.

- Common DC Return (DC-C) – An implementation where the battery return(s) is/are bonded to the surrounding BN, either in a Mesh or Isolated topology. This is often implemented as a bond to the local framework or bonding hardware. Waveserver is compatible with the DC-C configuration by connecting the shelf Power Input Module battery returns to the local framework ground with a strap or bond (procured as Installation-Related Material) that is appropriately sized for the shelf's total DC return current. With this topology, the common return feeder bonding the framework to ground must be appropriately sized for the total DC return current.

ATTENTION

The DC power module return lead, L+, must be earthed (connected to ground) at some point within the building consistent with the applicable power infrastructure (DC-C or DC-I) to satisfy safety requirements.

Consequences of incorrect grounding

The Protective Earthing Conductor is a key dependency of the mandatory safety requirements in most countries. Waveserver 5 is independently certified as safe with the dependency that the equipment (bay, shelf) is grounded at one or more of the approved bonding sites noted in the preceding section as a Protective Earthing Conductor. If not connected or if incorrectly connected, there is a risk of shock or damage due to high energy fault currents (should these arise). Local electrical codes/regulations typically require strict adherence to all safety instructions during installation—the equipment must be grounded for safe use.

Additionally, grounding the shelf will ensure that all external surfaces (including the ESD jack locations) are of sufficiently low impedance to non-intrusively reduce static charge build-up that may occur when operating, repairing, or installing new equipment in the shelf. When correctly grounded, the ESD jack locations work effectively in coordination with good ESD mitigation practices (see [“Dissipating static electricity” on page 19](#)) to mitigate the risk of damage to the equipment. For the list of related standards, see the “Power, grounding, and noise references” section in the *Waveserver 5 User Guide*, 323-3001-100.

Working with DC power



DANGER

Risk of electrical shock

The battery feeds can be at a high enough potential to constitute a shock hazard. Read and understand the power procedures you are performing. Take necessary precautions and use the appropriate insulated tools when working with power.



WARNING

Risk of personal injury, energy hazard

The battery feeds are capable of supplying very high current which, during an unintentional short, can cause burns. Read and understand the power procedures you are performing. Take necessary precautions and use the appropriate insulated tools when working with power.

When you install power feeds to the product input terminals or when you perform routine power maintenance, make sure that you do the following:

- Read and understand the power procedures you are performing.
- Take the necessary precautions and use the appropriate insulated tools to perform any tasks.

A continuously lit green “Power OK” LED on the faceplate of a field-replaceable power input module indicates that the power input circuit is functioning correctly. If the green “Power OK” LED is lit, there is voltage detected on the associated feeder in the correct polarity and the power is passing through the power input breaker to the backplane. The power input modules have one green “Power OK” LED for each fan module.

If the shelf is directly powered without protection by a fuse panel or breaker panel, then breakered power input modules (rated for the lesser of the shelf or feeder rating) must be used. Additionally, a power disconnect device would be required to protect installers from shock and energy hazards during servicing.

DC power modules

The following table specifies the wire gauge per maximum wire length from the Waveserver 5 –48Vdc power and fan module (186-3041-900) to the power distribution area. Ciena recommends using AWG 6 (minimum), AWG 4, or AWG 2 (maximum) cable. The cable insulation must be rated for a minimum of 90°C. De-rating due to the bundling of power cables occurs at ambient temperatures of 45°C and higher.

Table 2 Cable sizes and distances from -48 Vdc power and fan module to the power distribution area

Wire gauge	Maximum cable length from -48Vdc power and fan module (186-3041-900) to power distribution area
Imperial	
AWG 6	5.9 m (19 ft)
AWG 4	9.4 m (31 ft)
AWG 2	14.9 m (49 ft) (33.6 mm ²)
Metric	
16 mm ²	7.1 m (23 ft)
25 mm ²	11.1 m (36 ft)
35 mm ²	15.5 m (51 ft)

Ciena recommends using the 2-hole lugs listed in the following table to connect the DC power module to the DC power plant.

Note: You must use 90-degree power lugs to prevent cable interference with equipment installed directly below Waveserver 5.

Table 3 Recommended lugs for DC Waveserver 5 installations

Ciena part number	Supported cable size	Supplier	Supplier part number
N0026841	2 AWG	Hubbel/Burndy	YAV2CL2NT10FX90
N0133907	4 AWG	Hubbel/Burndy	YAV4CL2NT10FX90
N0025200	6 AWG	Hubbel/Burndy	YAV6CL2TC10FX90

Dissipating static electricity

The level of static electricity increases on your body when you move around or come into contact with other charged surfaces. Excessive levels of static electricity can damage equipment. You must either wear a properly functioning heel grounder (that attaches to your leg and foot) and/or an anti-static wrist strap, or another personal grounding device when you work on any of the following:

- Network element shelves (including the metal frame and cover)
- Cables connected to modules
- Modules



DANGER

Risk of equipment damage

Heel grounders or similar worn footwear attachments work when the floor is designed to dissipate static electricity. If the properties of the floor are unknown or in doubt, use a wrist-strap and make sure it is connected to a piece of electrostatic discharge (ESD) grounding equipment before proceeding with any maintenance or installation activity.

Any one of the previously mentioned grounding devices dissipate electrostatic charges to the ground quickly and safely. Use grounding devices correctly to eliminate the ESD threat you pose to the equipment.

When you wear an anti-static wrist-strap or a heel grounder, you must make sure the grounding straps are in contact with a moist part of your skin. Connect the grounding cord to the grounding plug on a grounded fixture of the product you are working on, such as the shelf ESD jack or bay mounted ESD jack. Grounded fixtures are accessible on most Ciena products.

The following guidelines provide optimal electrostatic discharge protection:

- Install bays on conductive floor coverings.
- Provide conductive shoes, anti-static wrist-straps, and heel grounders to all personnel working on the equipment.
- Maintain local environmental conditions so that relative humidity around equipment to be serviced is in excess of 20% (preferably higher than 40%). This lowers the threat of developing damaging electrostatic levels.
- You can implement an ESD training and control program that educates personnel on the hazards of ESD and simple mitigation procedures that can easily be applied.

Preventing module damage

In this section, the generic term “module” is used to refer to all types of modules, including pluggable modules.

All modules are subject to damage by rough handling or by electrostatic discharges. Review the following sections for more information on how to prevent damage to modules.

If Ciena ships modules separately from Waveserver 5, then these modules can be shipped in shielded containers marked with the following symbol:

**CAUTION**

OBSERVER LES PRÉCAUTIONS. MANIPULER LE MATÉRIEL SENSIBLE À L'ÉLECTROSTATIQUE EN SUIVANT LES MESURES DE SÉCURITÉ

FOLLOW PRECAUTIONS FOR HANDLING ELECTROSTATIC SENSITIVE DEVICES

Handling, installing, storing, or replacing modules

When you handle, install, or replace modules, you must take the following precautions:

- Wear an anti-static wrist-strap, a heel grounder, or another personal grounding device before you remove a module from its package or from the shelf.
- Follow installation and removal procedures for each module. Make sure to understand and perform each precautionary message in these procedures (for example, opening/closing the latches of the module simultaneously).

**CAUTION****Risk of equipment damage**

Do not touch any solder points, pins, connectors, or components including any integrated electrical cables or optical fibers when handling a module. Contact with surfaces other than handles, faceplates, latches/ejectors, thumbscrews, module guides, or module outer shells may result in contamination that will increase the risk of early failure.

Touching pins, connectors, integrated cables or fibers may degrade performance or cause permanent damage. When handling a module, hold it only by the faceplate, pull handle, module guide, or module outer shell as directed.

During unpacking or packing of a module, handle by the module guide or outer edges only if absolutely necessary. Manipulate latches, ejectors, and thumbscrews only as described in the applicable procedure.

- Inspect all pin connectors on each module for damage before use.
- Inspect each module for damage before insertion.
- Store un-installed modules separately in a shielded box.
- Do not stack modules on or against each other.
- Do not force modules into their packaging material.
- Do not store several modules in the same container, store each module in its respective container.

- Allow each module to reach room temperature before you insert the module into the shelf.
- Protect all optical connectors of the transmit (Tx) and receive (Rx) optical modules with clean dust caps at all times.
- Protect module pluggable transceiver ports/cages by leaving all dust covers in place until immediately before a transceiver is installed into a module. If you are not immediately replacing a pluggable transceiver, ensure that the empty port/cage in the module is protected by the appropriate dust cover.
- Store pluggable modules in their protective static-dissipative containers (when not in use) to prevent damage to the exposed connector terminals.

Leave spare modules in the original shielded containers until you need the modules. To prevent damage to modules in storage, follow procedures that prevent:

- Accumulation of dirt or dust on the pin connectors
- Damage to the printed-circuit board or its components
- Warpage (of printed-circuit boards stored in areas where the humidity can exceed 95% and the temperature can exceed 70°C)

Transporting modules

When you transport modules, pack each module in its original shielded container and padding, or in an electrostatically shielded bag. If you lose the original material, place the module in a shielded bag and use another container with sufficient padding.

Controlling equipment access

The Waveserver 5 system is intended to be installed in a restricted access location. In North America, location restriction must be in accordance with articles 110-16, 110-17, and 110-18 of the National Electrical Code, ANSI/NFPA No. 70. Outside North America, the location restriction must be in accordance with the international safety standard IEC 62368-1 Edition 2. In the context of these requirements, a restricted access location is further defined as a location:

- Where access is restricted to only trained personnel
- To which unsupervised members of the general public are not admitted

Controlling equipment environment

For the list of environmental, electromagnetic, and airflow specifications, refer to the “Environmental and electromagnetic specifications” section of the *Waveserver 5 User Guide*, 323-3001-100.

Air flow specifications

The approximate Waveserver 5 air flow with the fan modules operating at minimum and maximum speeds are as follows:

- Minimum fan speed (normal office temperatures): 200 CFM
- Maximum fan speed: 320 CFM

CHAPTER 3

Site planning and preparation

Before installing Waveserver 5, review the information in this section, “Chassis descriptions and technical specifications”, “Module descriptions, equipping rules, and specifications”, and the “Operating specifications and standards compliance” sections of the *Waveserver 5 User Guide*, 323-3001-100.

This section contains the following information and procedures you need to install Waveserver 5.

- [Equipment frames and cabinet selection](#)
- [Recommended access clearances](#)
- [Required materials for installation, powering, and removal](#)
- [Mounting setback options](#)
- [Torque values](#)
- [Bracket kits](#)
- [Fiber and cable management](#)
- [List of procedures](#)

Equipment frames and cabinet selection

Waveserver 5 can be installed in telecom and data center racks and in cabinets up to 800 millimeters deep.

Telecom equipment frames are typically 19-in or 23-in EIA or ETSI frames that are 24-in deep or less, with two uprights (posts). Telecom frames have uprights that are fixed in a base. The uprights are available in a range of spacings and upright setbacks, including uprights centered in the base, uprights rear-justified, and two rear-justified racks installed back-to-back.

Data center frames are typically 19-in or 23-in EIA frames that are 24-in deep or greater, with four uprights. The four uprights that can be fixed or adjustable. Fixed 4-post data center racks are available in a range of spacings and upright setbacks.

Refer to [“Mounting setback options” on page 27](#) for a list of mounting setback options for different frame types.

Installation on a 2-post or 4-post EIA/ETSI equipment frame or on a 4-post equipment frame requires two people.

Installation on a 2-post or 4-post EIA/ETSI equipment frame requires either two people or use of the Temporary Shelf Support Kit version 186-1315-900. Installation on a 4-post equipment frame requires either two people or use of the Temporary Shelf Support Kit. For more information, refer to the *Temporary Shelf Support Kit version, 186-1315-900*.

Recommended access clearances

Ensure that the installation site meets the minimum clearances for accessing equipment from the front and rear, along with proper ventilation and air flow.

Waveserver 5 must be installed at a height of 2 m (6.5 ft) or less.

The following table provides the required access clearances for a Waveserver 5 installation.

Table 4 Access clearances

Access clearances	
Location	Recommended clearance
Front	Sufficient space to work comfortably. Ensure that air intake vents are not obstructed.
Rear	Sufficient space to work comfortably. Ensure that air exhaust vents are not obstructed.
Vertical	1/32-in between each Waveserver 5 installed in an equipment frame.

Power requirements

Verify that the installation site meets the power distribution and protection requirements for the Waveserver 5 system. For more information, see the “Power and grounding specifications” section of the *Waveserver 5 User Guide*, 323-3001-100.

Cabling requirements

When preparing the site for network connections, take into consideration the required cabling for each interface. For more information, see the “Ordering” section in *Waveserver 5 User Guide*, 323-3001-100.

Mounting setback options

The following table details the mounting setback options for Waveserver 5 on supported frame types.

Table 5 Supported mounting setback options

Frame Type	Width	Upright Spacing	Setback Options
Data Center	• 19-in.	• 28-in. - 32-in.	Flush mount (Note 1 , Note 2)
Data Center-Style for Deep Equipment Frames	• 23-in.	• 36-in. to 42-in.	Flush mount (Note 1 , Note 2)
Data Center	• 19-in.	• 24-in. to 36-in.	Flush mount (Note 1 , Note 2)
Data Center	• 23-in.	• 24-in. to 36-in.	Flush mount (Note 1 , Note 2)
4-Post Telecom	• 19-in. • 23-in.	• 3-in. to 24-in.	• 100-mm • 5-in. • Flush-mount (Note 2)
	• ETSI	• 3-in. to 24-in.	• 100-mm (Note 2)
2-Post Telecom	• 19-in. • 23-in.	• 3-in. to 24-in.	• Front-mount: 100-mm • Front-mount: 5-in. • Front-mount: flush-mount • Mid-mount: 3-in. post depth • Mid-mount: 5-in. post depth (Note 2 , Note 3)
Note 1: The flush-mount position is recommended for data center racks since there is typically a limited amount of space in front of the uprights. Fiber extends a minimum of 3.1-in. beyond the front of the uprights. Keep the space in front of the frame clear to prevent accidental fiber damage. Note 2: If you are installing Waveserver 5 in a cabinet, ensure that you use a setback that leaves adequate space for the door to close after the fiber managers are installed. Note 3: Both front-mount and mid-mount installation options are supported. However, for stability, front-mount installations require a minimum of 5-in. upright frame depth.			

Required materials for installation, powering, and removal

The following list itemizes the required Ciena-supplied materials for the installation, powering, and removal of Waveserver 5.

One of the following Waveserver 5 hardware Pick-To-Order (PTO) kits:

- Waveserver 5 Chassis w/AC Power Modules, Ctrl, Fan Modules (K86-3500-900)

- Waveserver 5 Chassis w/DC Power Modules, Ctrl, Fan Modules (K86-3501-900)

For more information about the available PTO kits, see the *Waveserver 5 User Guide*, 323-3001-100.

One or more of the following Waveserver 5 service modules (SMs):

- Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD (186-3100-900)
- Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-901)
- Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD (186-3101-900)
- Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-901)
- Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD (186-3102-900)
- Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-901)
- Waveserver WL5e 2x800G L-Band Submarine 16xQSFP28/QSFP-DD Module (186-3110-901)
- Waveserver WL5e 2x800G L-Band Premium 16xQSFP28/QSFP-DD Module (186-3111-901)
- Waveserver WL5e 2x800G L-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3112-901)
- Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD Module (186-3302-900)
- Waveserver WL5e 2x800G C-Band Submarine 12xQSFP28/QSFP-DD Module (186-3120-900)
- Waveserver WL5e 2x800G C-Band Premium 12xQSFP28/QSFP-DD Module (186-3121-900)

For more information about the available SMs, see the “Ordering” section of the *Waveserver 5 User Guide*, 323-3001-100.

Ensure that you install filler modules (186-3031-900) in all empty slots. See the “Ordering” section of *Waveserver 5 User Guide*, 323-3001-100.

One of the following Waveserver 5 bracket and installation kits, depending on the frame type and size:

- Waveserver 19-inch Bracket Kit (186-1300-900)

- Waveserver 23-inch Bracket Kit (186-1301-900)
- Waveserver ETSI Bracket Kit (186-1302-900)
- Waveserver 19-inch, 23-inch, and ETSI Bracket Kits (186-1303-900)
- Waveserver Data Center Rack Kit, 19-in Wide x 24-36-in Deep (186-1306-900)
- Waveserver Data Center Rack Kit, 23-in Wide x 24-36-in Deep (186-1307-900)

For details about the bracket kit contents, see [“Bracket kits” on page 30](#).

The following list itemizes the required customer-supplied materials for the installation, powering, and removal of Waveserver 5:

- Short-handled #1 Phillips screwdriver
- Long-handled #2 Phillips screwdriver
- 1/4-in Phillips screwdriver (for DC installations)
- Hex driver
- Rack screws
- Two AC power cables (for AC installations)

The AC power modules are equipped with unique Saf-D-Grid connectors. The following cables are supported and must be ordered from Ciena:

- AC Power Cord, Saf-D-Grid to C20, 12AWG, 250V, 2m (NTK955GA)
- AC Power Cord, Saf-D-Grid to Open, 12AWG, 250V, 3m (NTK955GB)
- AC Power Cord, Saf-D-Grid to C20, 12AWG, 250V, 2m, white (NTK955GC)
- AC Power Cord, Saf-D-Grid to C14, 14 AWG, 250V, 2m, black (NTK955GD)
- AC Power Cord, Saf-D-Grid to CEE7/7, 12 AWG, 250V, 2m, black (NTK955GE)
- AC Power Cord, Saf-D-Grid, left routed to C20, 12 AWG, 250V, 2m, black (NTK955GF)
- AC Power Cord, Saf-D-Grid, right routed to C20, 12 AWG, 250V, 2m, black (NTK955GG)
- AC Power Cord, Saf-D-Grid, left routed to C20, 12 AWG, 250V, 2m, white (NTK955GH)
- AC Power Cord, Saf-D-Grid, right routed to C20, 12 AWG, 250V, 2m, white (NTK955GJ)

Refer to the “Ordering” and the “Power and grounding specifications” sections of the *Waveserver 5 User Guide*, 323-3001-100 for details.

- Two DC power cables (for DC installations)
- 6 AWG (16 mm²) wire for the chassis ground
- For chassis ground: single-hole lug for front-mounting ground points, 2-hole lug for all other ground points
- Wire strippers
- Wire cutter
- Crimp tool
- For DC installations, Ciena recommends using the lugs listed in [“Recommended lugs for DC Waveserver 5 installations” on page 19](#) to connect the DC power module to the DC power plant.

Note: You must use 90-degree power lugs to prevent cable interference with equipment installed directly below Waveserver 5.

Torque values

The following table lists the torque values used in the installation procedures covered in this technical publication.

Table 6 Fastener torque values

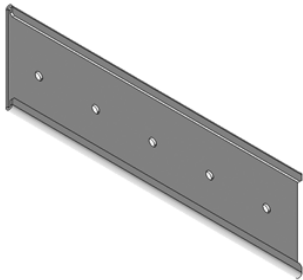
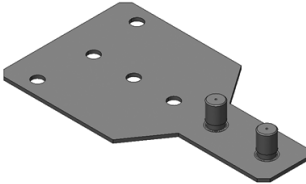
Fastener type	Torque for installation	Torque for inspection
M6 (installing shelf to rack)	50 lb-in. (5.65 N-m)	40 lb-in. (4.52 N-m)
#10-24	42 lb-in. (4.75 N-m)	33 lb-in. (3.73 N-m)
#10-32	20 lb-in. (2.26 N-m)	-

Bracket kits

The following tables list the contents of the Waveserver 5 bracket kits.

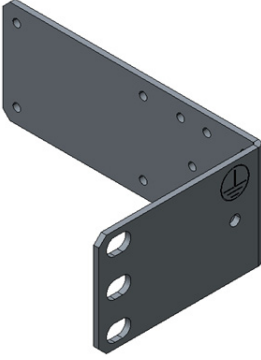
- [“Contents of Waveserver 19-in. Bracket Kit \(186-1300-900\)” on page 31](#)
- [“Contents of Waveserver 23-in. Bracket Kit \(186-1301-900\)” on page 32](#)
- [“Contents of Waveserver ETSI Bracket Kit \(186-1302-900\)” on page 34](#)
- [“Contents of Waveserver Data Center Bracket Kit, 19-in wide x 24-36-in deep \(186-1306-900\)” on page 35](#)
- [“Contents of Waveserver Data Center Bracket Kit, 23-in wide, 24 - 36-in deep \(186-1307-900\)” on page 37](#)

Table 7 Contents of Waveserver 19-in. Bracket Kit (186-1300-900)

Part in Bracket Kit 186-1300-900	Quantity	View
19-in. Front Bracket (410-6851-001)	2	
19-in. Rear Bracket, Sliding (410-7036-001)	2	
Rear Bracket Guide (410-7037-001)	2	
Ground Bracket (410-6903-002)	1	
Phillips Countersink Fastener (N0045005) (includes spares)	10	

Part in Bracket Kit 186-1300-900	Quantity	View
3.5 mm long M3 screw Wafer Head with patch Nickel finish (312-3730-CS1) (includes spares)	12	
Nut, Hex, Keps, External Tooth Washer (N0041806) Note: For use with ground cable, if installed.	2	

Table 8 Contents of Waveserver 23-in. Bracket Kit (186-1301-900)

Part in Bracket Kit 186-1301-900	Quantity	View
23-in Front Bracket (410-6853-001)	2	
23-in. Rear Bracket, Sliding (410-7036-002)	2	

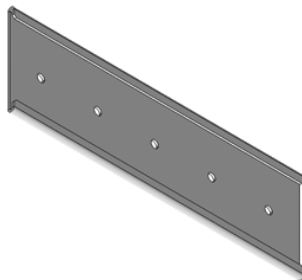
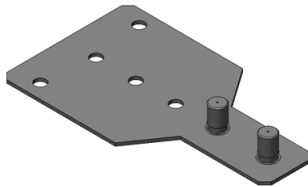
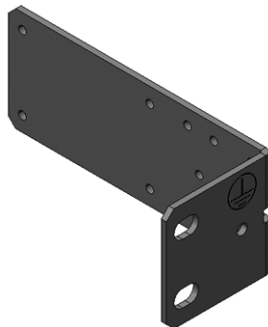
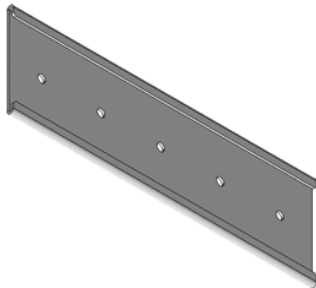
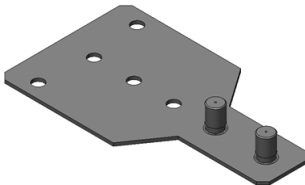
Part in Bracket Kit 186-1301-900	Quantity	View
Rear Bracket Guide (410-7037-001)	2	
Ground Bracket (410-6903-002)	1	
6mm long M3 screw with lock washer (SEMS) zinc clear chromate finish (N0038285) (includes spares)	10	
3.5 mm long M3 screw Wafer Head with patch Nickel finish (312-3730-CS1) (includes spares)	12	
Nut, Hex, Keps, External Tooth Washer (N0041806) Note: For use with ground cable, if installed.	2	

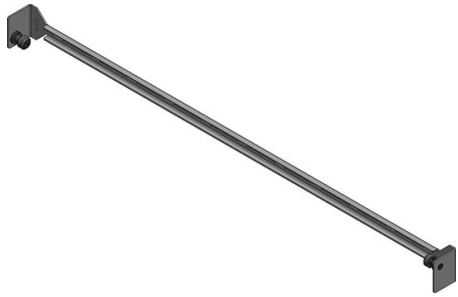
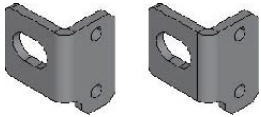
Table 9 Contents of Waveserver ETSI Bracket Kit (186-1302-900)

Part in Bracket Kit 186-1302-900	Quantity	View
ETSI Front Bracket (410-6852-001)	2	
ETSI Rear Bracket, Sliding (410-7036-003)	2	
Rear Bracket Guide (410-7037-001)	2	
Ground Bracket (410-6903-002)	1	

Part in Bracket Kit 186-1302-900	Quantity	View
6mm long M3 screw with lock washer (SEMS) zinc clear chromate finish (N0038285) (includes spares)	10	
3.5 mm long M3 screw Wafer Head with patch Nickel finish (312-3730-CS1) (includes spares)	12	
Nut, Hex, Keps, External Tooth Washer (N0041806) Note: For use with ground cable, if installed.	2	

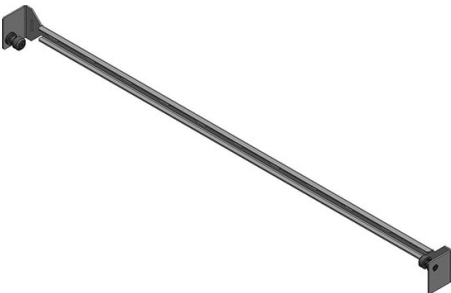
Table 10 Contents of Waveserver Data Center Bracket Kit, 19-in wide x 24-36-in deep (186-1306-900)

Part in Bracket Kit 186-1306-900	Quantity	View
Front Bracket Guide (410-8189-001)	2	
Rear Bracket Guide (410-8190-001)	2	

Part in Bracket Kit 186-1306-900	Quantity	View
Cable Bracket (optional) (410-8191-001)	1	
T-Bar (42T-2970-001)	2	
Mounting Bracket (410-7588-001)	2	
Right and Left Mounting Brackets (410-6735-001 and 410-6735-002) (not used for Waveserver 5 installations)	2	
Phillips Pan head External Lock Washer (312-0630-M06)	10	
Phillips Flat head Machine Screw (312-1030-M05)	10	

Part in Bracket Kit 186-1306-900	Quantity	View
Nut, Hex, Keps, External Tooth Washer (N0041806) Note: For use with ground cable, if installed.	2	

Table 11 Contents of Waveserver Data Center Bracket Kit, 23-in wide, 24 - 36-in deep (186-1307-900)

Part in Bracket Kit 186-1307-900	Quantity	View
Front Bracket Guide (410-8193-001)	2	
Rear Bracket Guide (410-8194-001)	2	
Cable Bracket (410-8191-001)	1	

Part in Bracket Kit 186-1307-900	Quantity	View
T-Bar (42T-2970-001)	2	
Mounting Bracket (410-7774-001)	2	
Right Mounting Bracket (not for Waveserver 5 installations) (410-7773-001)	1	
Left Mounting Bracket (not for Waveserver 5 installations) (410-7773-002)	1	
Phillips Panhead External Lock Washer (312-0630-M06)	14	
Phillips Flathead Machine Screw (312-1030-M05)	10	
Nut, Hex, Keps, External Tooth Washer (N0041806) Note: For use with ground cable, if installed.	2	

Fiber and cable management

Waveserver 5 ships with fiber management clips for routing and securing fibers and cables at the front of the shelf.

The fiber management clips are installed on both sides of the chassis and can also be used to manage cables. Fibers can be bundled with Velcro ties if necessary. Fibers installed in modules on the left of the chassis are routed to the left, and fibers installed in modules on the right of the chassis are routed to the right.

The fiber management clips are segregated to allow for the separation of copper and fiber, if required.

Route cables connected to the rear Access Panel management ports (DCN3, ILAN1, and ILAN2) through the fiber manager to the right. Route cables connected to the front Control Processor (DCN1, DCN2, and RS-232 serial console) through the fiber manager to the left.

Ensure that routed cables do not obstruct Waveserver 5 system level and module LEDs.

Note 1: DCN4 and DCN5 are not supported in the current release.

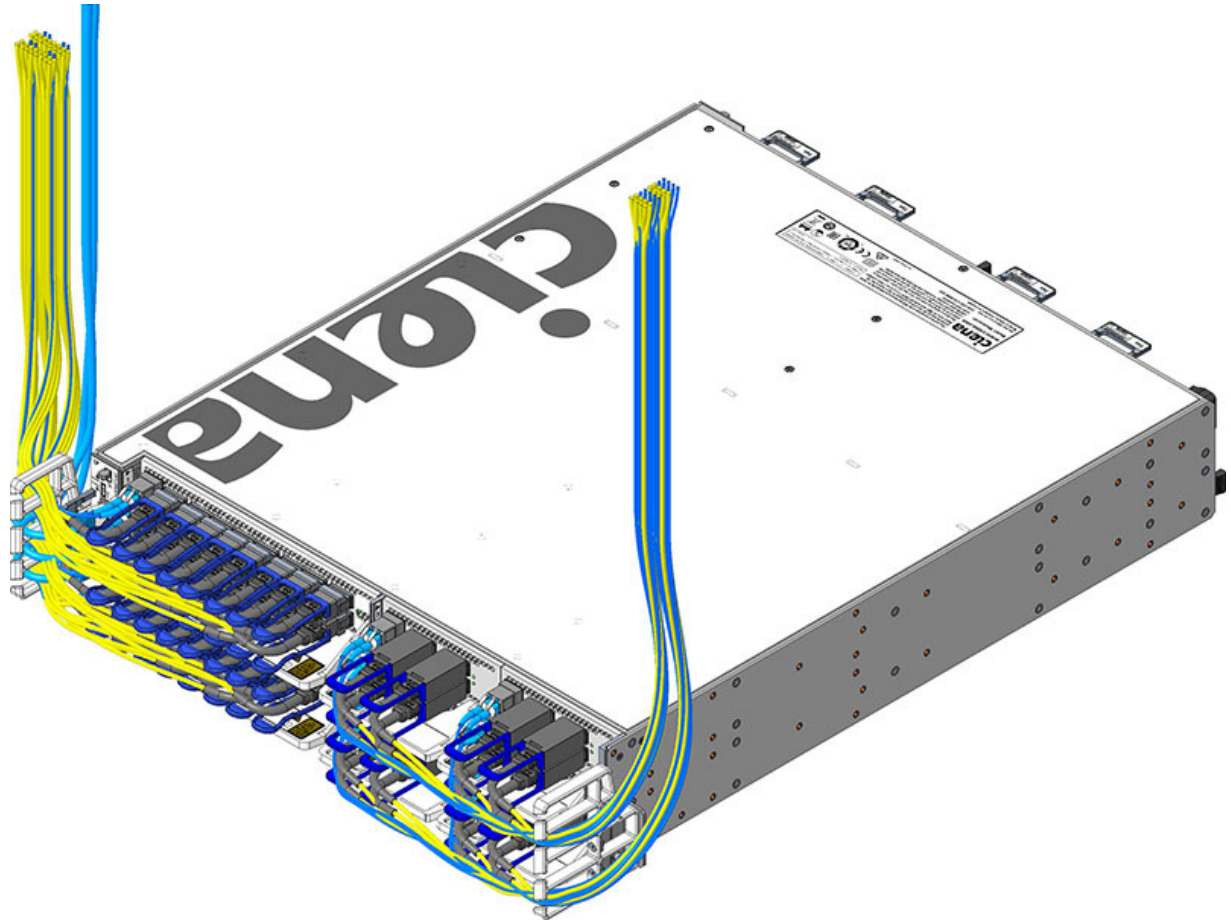
Note 2: DCN2 supports wayside-channel mode only. For more information on SLTE comms, see the “Comms channel for 6500-based SLTE configurations” section of *Waveserver 5 User Guide*, 323-3001-11.

The following figures show front and top views of the chassis with cables routed.

Figure 7 Cables routed at the front of the Waveserver 5 chassis



Figure 8 Cables routed — top view (single-width modules shown for illustration purposes only)



Note: Only double-width modules are supported in the current release.

List of procedures

This section provides the following procedures:

- [Procedure 1, “Grounding guidelines—Applying non-oxidizing compound to compression lugs, cables, and surfaces”](#)
- [Procedure 2, “Connecting the rack ground to the office ground”](#)

Procedure 1 Grounding guidelines—Applying non-oxidizing compound to compression lugs, cables, and surfaces

Use this procedure to apply non-oxidizing compound to compression lugs, cables, and surfaces.

Overview

Make sure that you apply non-oxidizing compound, as required, when you prepare compression lugs, cables, and surfaces for grounding. Grounding steps are included in various procedures in this technical publication.

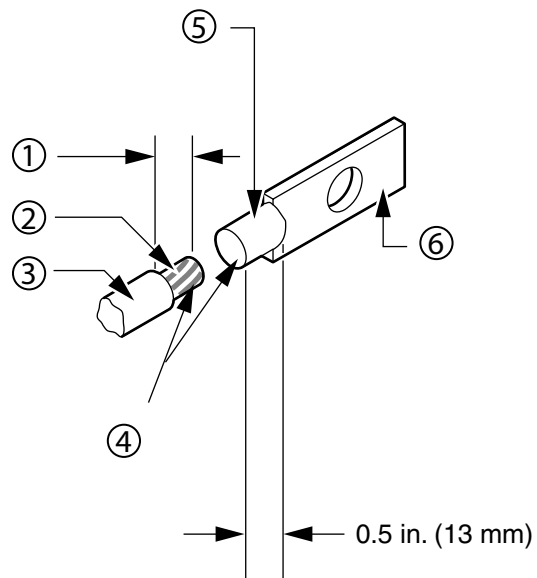
As appropriate, apply non-oxidizing compound to

- any type of compression lug used with power, ground, and battery cables
- the following surfaces
 - system ground point on the bay
 - ground terminal
 - power input and return terminals
 - battery input and return terminals

Review the following illustrations

- for compression lugs and cables
 - [“Preparing a one-hole compression lug and cable” on page 42](#)
 - [“Preparing a two-hole compression lug and cable” on page 43](#)
 - [“Preparing a ring lug and cable” on page 43](#)
- for surfaces
 - [“Preparing a two-hole lug and system ground point” on page 44](#)
 - [“Example of connecting the shelf ground to the bay ground” on page 45](#)
 - [“Preparing a two-hole lug and power input terminal” on page 46](#)
 - [“Two-hole lugs connected to battery input and return terminals” on page 46](#)

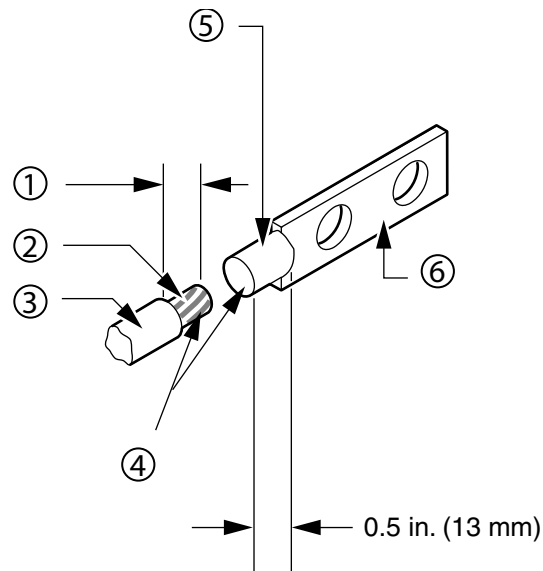
Figure 9 Preparing a one-hole compression lug and cable



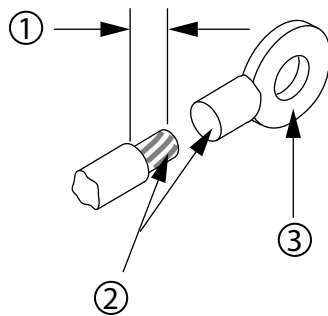
Legend

- 1 0.5 in. (13 mm)
- 2 Copper wire
- 3 Insulated copper wire

- 4 Apply a light coat of non-oxidizing compound on the stripped copper wire and inside the compression lug barrel.
- 5 Lug barrel
- 6 Crimp-type one-hole lug

Figure 10 Preparing a two-hole compression lug and cable**Legend**

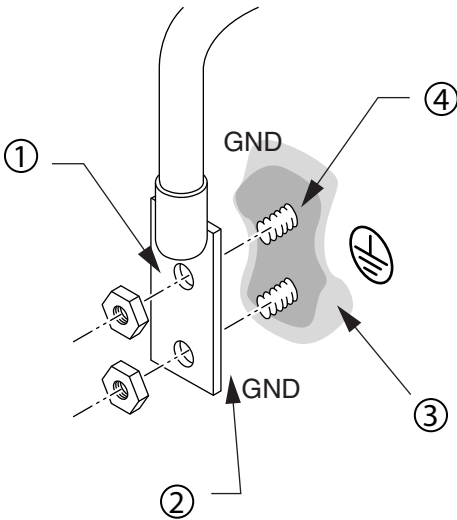
- | | | | |
|---|-----------------------|---|---|
| 1 | 0.5 in. (13 mm) | 4 | Apply a light coat of non-oxidizing compound on the stripped copper wire and inside the compression lug barrel. |
| 2 | Copper wire | 5 | Lug barrel |
| 3 | Insulated copper wire | 6 | Crimp-type one-hole lug |

Figure 11 Preparing a ring lug and cable**Legend**

- | | | | |
|---|---|---|-------------------------|
| 1 | Strip required amount | 3 | Crimp-type one-hole lug |
| 2 | Apply a light coat of non-oxidizing compound on the stripped copper wire and inside the compression lug barrel. | | |

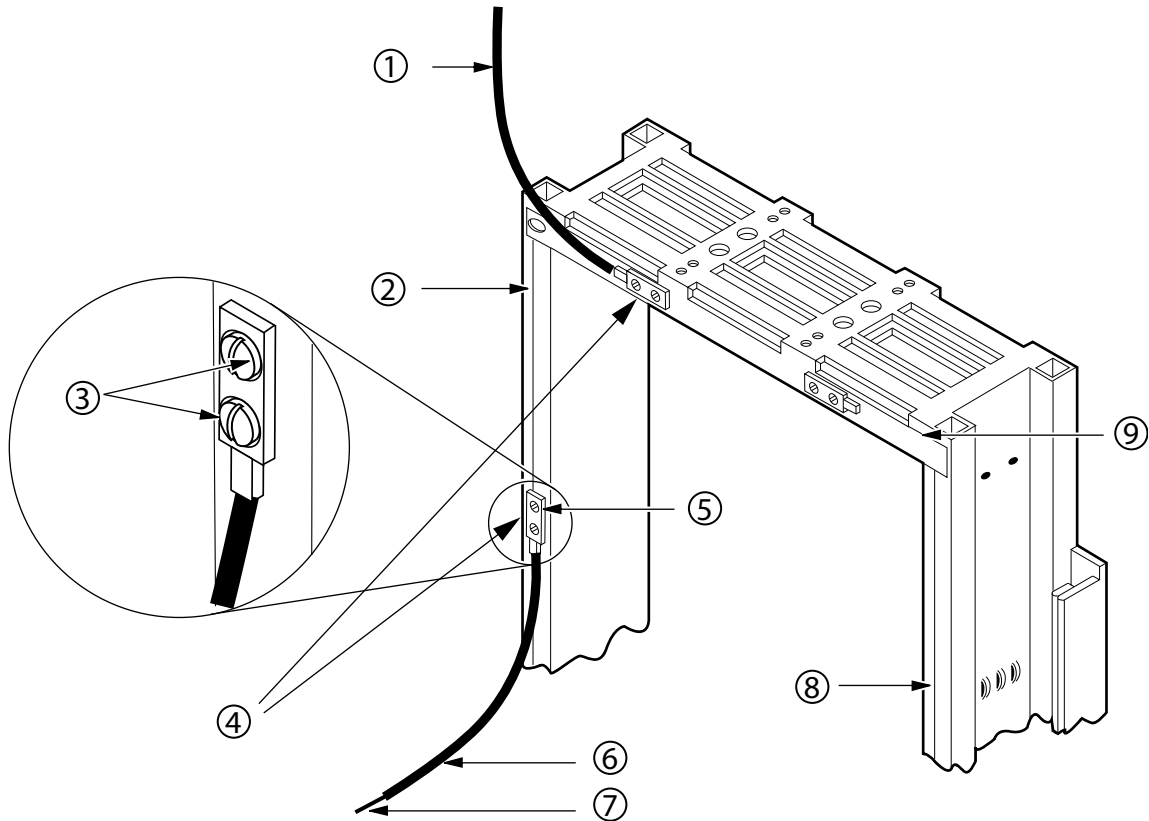
Note: The instructions in the following figure also apply to a one-hole lug.

Figure 12 Preparing a two-hole lug and system ground point



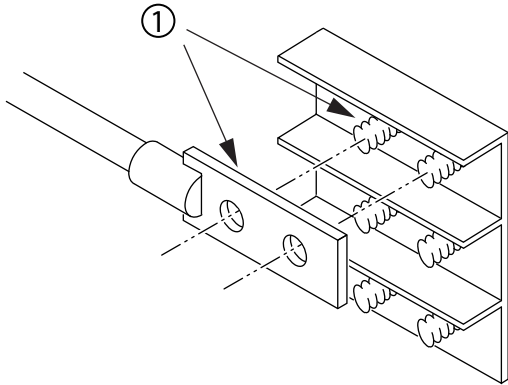
Legend

- | | | | |
|---|--|---|--|
| 1 | Compression lug | 3 | If system ground point does not have a metallic grounding strip, then if necessary remove paint on sheet metal under lug (recommended) |
| 2 | Lightly coat non-oxidizing compound on compression lug and a metallic grounding strip that connects to system ground point | 4 | Lightly coat non-oxidizing compound on compression lug and a metallic grounding strip that connects to system ground point |

Figure 13 Example of connecting the shelf ground to the bay ground**Legend**

- | | | | |
|---|---|---|---|
| 1 | To frame ground collector or single point building ground | 5 | Ground lug |
| 2 | Metallic ground strip | 6 | Shelf ground wire |
| 3 | Pan head screw | 7 | Strip 0.5 in. of insulation from this end |
| 4 | Note: If appropriate, lightly coat antioxidant paste on the compression lug and metallic grounding strip and ground bar that connects to the system ground point. | 8 | Ground bar |

Figure 14 Preparing a two-hole lug and power input terminal

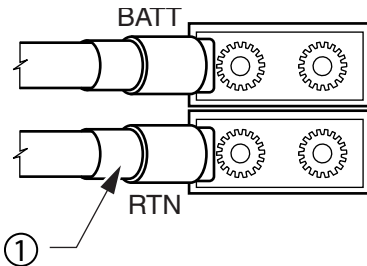


Legend

- 1 If appropriate, lightly coat non-oxidizing compound on compression lug and input terminal

Note: Figure 14 also applies to the power return terminal.

Figure 15 Two-hole lugs connected to battery input and return terminals



Legend

- 1 Heat-shrink tubing

ATTENTION

Make sure that you follow the specific detailed steps included in the related procedures in this technical publication.

The following steps are provided as general guidelines only.

Steps

Preparing a cable and compression lugs

- 1 Strip the required amount of insulation from the end of the cable to expose the copper wire.

- 2 Apply non-oxidizing compound to the
 - exposed copper wire of the cable, and
 - inside the compression lug barrel
- 3 Insert the prepared exposed copper wire into the prepared compression lug.
- 4 Crimp the compression lug to the cable as required.
- 5 Repeat [step 1](#) to [step 4](#) for the other end of the cable, as applicable.
Some cables have one compression lug pre-installed.
- 6 If you are working with a ground cable, see [step 7](#) to [step 12](#).
If you are working with a power cable, see [step 13](#) to [step 17](#).
If you are working with a battery cable, see [step 18](#) to [step 22](#).

Preparing compression lugs, ground terminal, system ground point, and securing a ground cable

- 7 If appropriate, lightly coat non-oxidizing compound on the rear of the compression lug and the ground terminal.
- 8 Secure the compression lug to the ground terminal as required.
- 9 Prepare the system ground point as follows.
 - If the system ground point has a metallic grounding strip, then lightly coat non-oxidizing compound on the metallic grounding strip.
 - If the system ground point does not have a metallic grounding strip, it is recommended to remove the paint on the sheet metal, and then lightly coat non-oxidizing compound on the prepared sheet metal area.
- 10 Lightly coat non-oxidizing compound on the rear of the compression lug on the other end of the ground cable.
- 11 Place the prepared compression lug up against the metallic grounding strip or the prepared sheet metal area, which connects to the system ground point.
- 12 Secure the lugged ground cable in place to the system ground point, as required.

You have completed installing the lugged ground cable with the non-oxidizing compound, as required.

Preparing compression lugs, power input terminal and return terminal, and securing power cables

- 13 If appropriate, lightly coat non-oxidizing compound on the rear of the compression lug and the input terminal on the power component.
- 14 Place the prepared compression lug up against the input terminal of the power component.
- 15 Secure the lugged power cable in place to the input terminal of the power component as required.
- 16 If appropriate, lightly coat non-oxidizing compound on the rear of the compression lug at the other end of the power cable. Secure this end of the power cable to the appropriate power supply, as required.
- 17 Repeat [step 13](#) to [step 16](#) for the return power cable.

You have completed installing the lugged power cables with the non-oxidizing compound, as required.

Preparing compression lugs, battery input terminal and return terminal, and securing battery cables

- 18** If appropriate, lightly coat non-oxidizing compound on the rear of the compression lug and the input terminal on the battery.
- 19** Place the prepared compression lug up against the input terminal of the battery.
- 20** Secure the lugged cable in place to the input terminal of the battery, as required.
- 21** If appropriate, lightly coat non-oxidizing compound on the rear of the compression lug at the other end of the battery cable. Secure this end of the battery cable to the appropriate power supply, as required.
- 22** Repeat [step 18](#) to [step 21](#) for the return battery cable.

You have completed installing the lugged battery cables with the non-oxidizing compound, as required.

Procedure 2 Connecting the rack ground to the office ground

Use this procedure to connect the rack ground to the office ground.

Requirements

Make sure that;

- the equipment rack is installed and secured according to manufacturer instructions and the EDP, IDP, or equivalent site/network engineering plans, as applicable
- you have the following equipment:
 - wire cutters, No. 6 AWG (16 mm²) - #1/0 AWG
 - pliers
 - crimping tool, No. 6 AWG (16 mm²) - #1/0 AWG
 - nut driver/socket, 5/16 in.
 - torque wrench
 - No. 6 AWG (16 mm²), 7-strand copper insulated conductor grounding wire (as required)
 - grounding lug
 - non-oxidizing compound

Review and follow any additional instructions provided with the rack or cabinet being installed and used.

Overview

Throughout this procedure, apply non-oxidizing compound to exposed rack metal and exposed wire over No. 6 AWG (16 mm²) to prevent oxidation. For details, see [Procedure 1, “Grounding guidelines—Applying non-oxidizing compound to compression lugs, cables, and surfaces”](#). If the rack is painted, follow company procedures to remove the paint from the area where you will fasten the lug to ensure a proper ground.

For information on equipment-grounding practices, see [“Grounding standards” on page 12](#).

Precautions



DANGER

Risk of electrical shock and equipment damage

Grounding is mandatory to satisfy local electrical codes/regulations for the safe use of the equipment.

Ground the rack to the common building network (CBN), isolated bonding network (IBN), or ETSI Mesh Bonding Network ground/protective earth. For details, see [“Grounding requirements” on page 15](#).

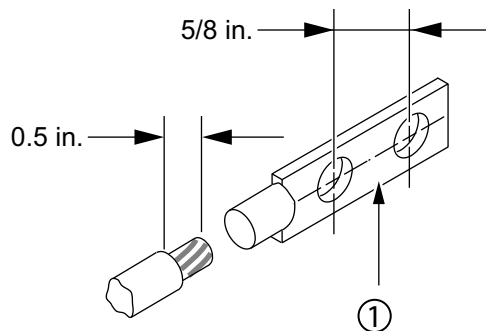
Correct grounding of the equipment rack is mandatory in accordance to electrical safety standards and mitigates the electrostatic discharge (ESD) and operational concerns described in [“Consequences of incorrect grounding” on page 17](#). For additional details, see [“Grounding requirements” on page 15](#).

Also, ground every shelf and peripheral tray as described in the related procedure. Make sure that you perform the grounding steps in this procedure.

Steps

- 1 Check to make sure that a suitable rack ground location exists on the rack.
- 2 Route the ground conductor from the office ground point to the closest ground location on the rack.
- 3 Strip 0.5 in. (13 mm) of insulation from the conductor and crimp the ground lug onto the conductor.

Figure 16 Ground lug

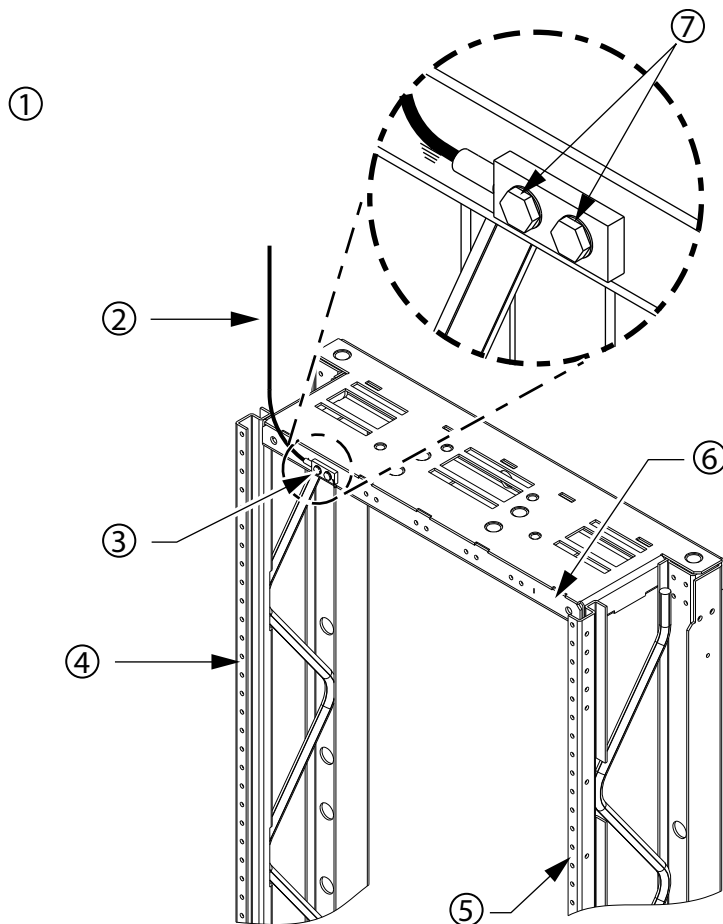


Legend

- 1 Crimp-type two-hole lug

- 4 Attach the ground wire crimped connector to the rack ground location. See the following figure. The hardware stack-up must be as follows: bolt, split washer (or Belleville washer), flat washer, and lug.

Figure 17 Example of connecting the rack ground to the office ground (CBN or IBN)



Legend

- | | | | |
|---|--|---|-----------------------|
| 1 | You must connect the other end of the ground cable to the CBN or IBN ground. | 5 | Metallic ground strip |
| 2 | To frame ground collector or single point building ground | 6 | Ground bar |
| 3 | Ground lug | 7 | Hex head screws |
| 4 | Metallic ground strip | | |
- 5** Bond the equipment as required based on the appropriate bonding topology and industry standards.

Note: Bonding should not be confused with Protective Grounding and is intended to create equipotential nodes throughout the office installation. Marked locations bearing the symbol illustrated in the following figure identify sites that can be used to bond equipment but are not intended for safety

grounding. The Protective Earthing Conductor can be connected using either of these topologies to satisfy the mandatory (safety) grounding dependencies for the shelf.

Figure 18 Protective Earthing Conductor (IEC 60417-5019)



For related details, see [“Bonding topologies” on page 15](#).

- 6** After you have completed connecting the rack ground to the office ground, you are ready to install and ground Waveserver 5.

CHAPTER 4

Installing Waveserver 5 and components

Use the following procedures to install Waveserver 5 and components.

List of procedures

The following table provides a list of installation procedures for Waveserver 5.

Table 12 Installation procedures — Waveserver 5

Procedure	Comment
Procedure 3, “Unpacking Waveserver 5 equipment”	Required
Procedure 4, “Installing and grounding Waveserver 5 in a 2-post 19-in. or 23-in. telecom frame”	Dependent on frame type
Procedure 5, “Installing Waveserver 5 in a 4-post 19-in., 23-in, or ETSI telecom frame”	Dependent on frame type
Procedure 6, “Installing and grounding Waveserver 5 in a data center 19-in. wide, 24-36-in. deep equipment frame”	Dependent on frame type
Procedure 7, “Installing and grounding Waveserver 5 in a data center 23-in. wide, 24 - 36-in. deep equipment frame”	Dependent on frame type
Procedure 8, “Installing the power and fan modules and connecting power cables to Waveserver 5”	Required
Procedure 9, “Testing power to Waveserver 5”	Required
Procedure 10, “Installing the access panel”	Required
Procedure 11, “Installing the control processor”	Required
Procedure 12, “Installing or removing a filler module”	Required if your configuration includes filler modules.
Procedure 13, “Installing or removing Service Modules (SMs)”	Required
Procedure 14, “Installing or removing pluggable transceivers”	Perform as required or when referred to by another procedure.

54 Installing Waveserver 5 and components

Procedure	Comment
Procedure 15, "Routing cables from/to modules on the Waveserver 5"	Required
Procedure 16, "Connecting Active Optical Cables (AOC)"	Perform as required or when referred to by another procedure.

Procedure 3 Unpacking Waveserver 5 equipment

Use this procedure to unpack Waveserver 5 equipment.

Requirements

Make sure you have a

- #1 Phillips screwdriver
- Wire cutter
- Flat stable surface on which to place Waveserver 5 and components

Note: Retain all PIP components and fasteners for future repair and/or return.

Overview

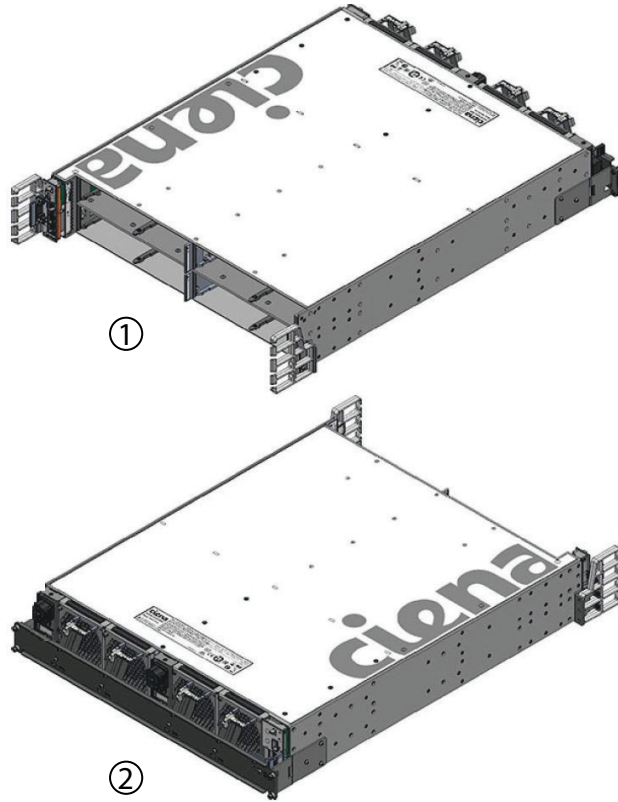
Use this procedure to unpack Waveserver 5 equipment. The Waveserver 5 equipment can be shipped in the packs-in-place (PIP) kit (186-3051-900), as follows:

- Packs-in-place (PIP) kit (186-3051-900) — shipped in a cardboard box with the rear modules secured in the chassis with shipping brackets, the Control Processor secured at the front of the chassis with a shipping bracket, and the remaining modules shipped in a separate box

Note: Service modules are shipped in separate packaging.

See the following figure for an example of the PIP kit.

Figure 19 Waveserver 5 PIP kit 186-3051-900



Legend

- 1 Shipping brackets installed on CP 2 Rear shipping brackets installed on power and fan modules and AP

Precautions



CAUTION

Risk of damage to the chassis, modules, and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and modules or transceivers.



CAUTION

Risk of equipment damage

Holding the Waveserver 5 shelf incorrectly can damage the chassis. Ensure to handle the shelf only by left/right outside edges to prevent damages to the equipment.

Steps

- 1 Visually inspect the shipping container. If it is damaged, inform the shipping company and Ciena immediately.
- 2 Carefully remove the equipment from the shipping container and protective packaging by holding the left and right sides of the shelf, lifting it straight up and place it on a flat, stable surface. See the following figures for correct and incorrect ways of handling the equipment.

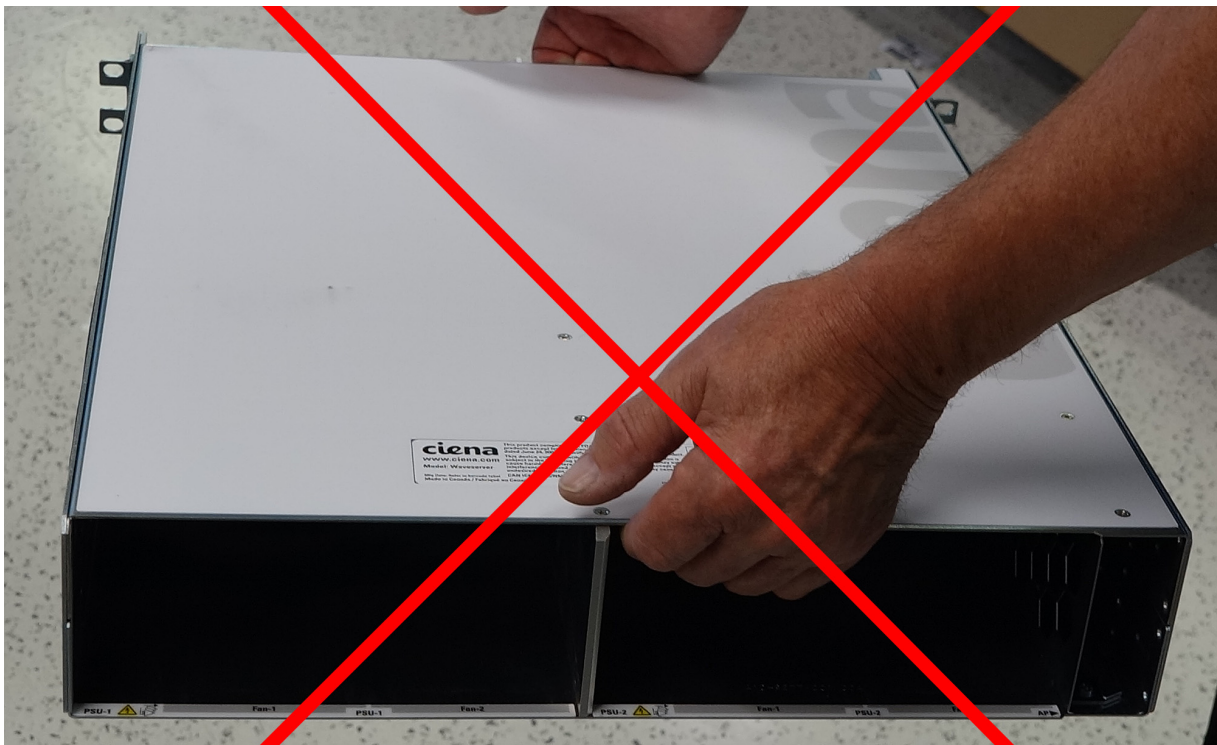
Figure 20 Waveserver 5 chassis correct handling



Figure 21 Waveserver 5 chassis incorrect handling (1)



Figure 22 Waveserver 5 chassis incorrect handling (2)



- 3 Visually inspect the equipment to ensure that all components have been included and are undamaged. The following tables itemize the contents of each Waveserver 5 pick-to-order (PTO) kit.

Table 13 Pick-to-Order hardware kits

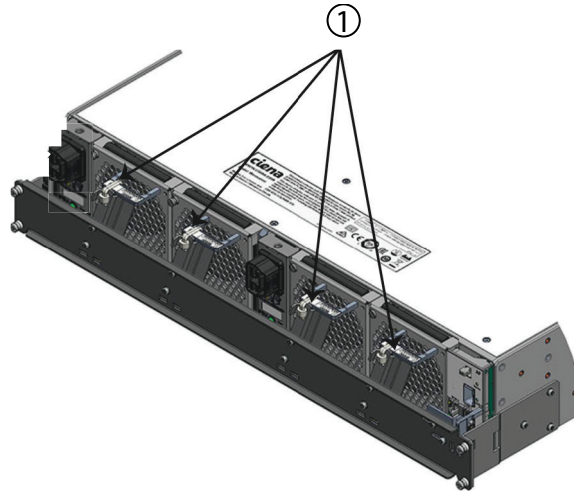
Ordering code (Note)	PTO hardware and software kit description	Code	Description	Qty
K86-3500-900	Waveserver 5 Chassis w/ AC Power Modules, Ctrl, Fan Modules	186-3001-900	Waveserver 5 Chassis	1
		186-3010-900	Waveserver 5 Control Processor Subsystem	1
		186-3020-900	Waveserver 5 Access Panel	1
		186-3040-900	Waveserver 5 100-277Vac/ 240Vdc Power and Fan Module	2
		186-3051-900	Waveserver 5 Shipping Brackets and Packaging	1
K86-3501-900	Waveserver 5 Chassis w/ DC Power Modules, Ctrl, Fan Modules	186-3001-900	Waveserver 5 Chassis	1
		186-3010-900	Waveserver 5 Control Processor Subsystem	1
		186-3020-900	Waveserver 5 Access Panel	1
		186-3041-900	Waveserver 5 -48Vdc Power and Fan Module	2
		186-3051-900	Waveserver 5 Shipping Brackets and Packaging	1
Note: Filler modules (186-3031-900 (Waveserver 5 Double Slot Filler) are not included in the PTO kits and must be ordered separately, if required. See the “Ordering” section of the <i>Waveserver 5 User Guide</i> , 323-3001-100 for more information.				

- 4 Remove the plastic bag of fasteners and/or fiber managers that is attached to the rear shipping bracket with a cable tie.

Removing the rear shipping brackets

- 5 Remove the cable ties that secure the rear shipping brackets to the power and fan module handles.

Figure 23 Power and fan modules secured to rear brackets with cable ties

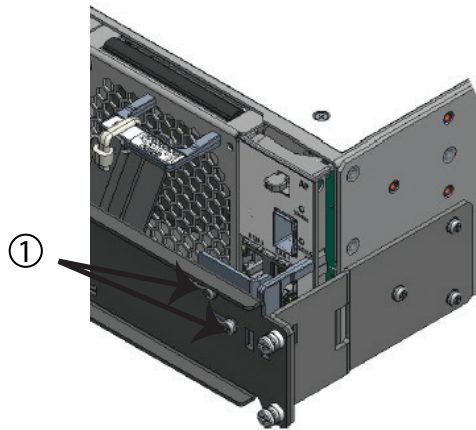


Legend

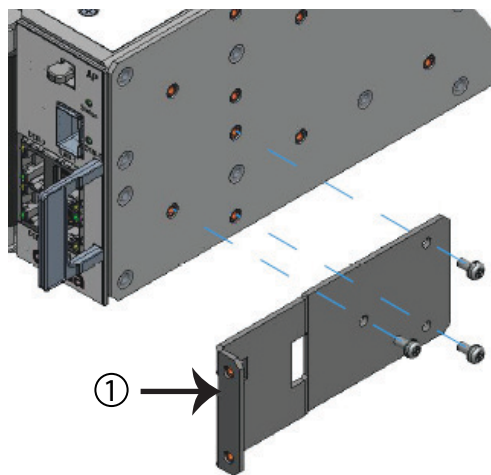
- 1 Cable ties securing brackets to the power and fan module handles
- 6 Remove the bottom four screws that secure the rear brackets to the power and fan module handles.
- 7 Cut the cable ties that secure the fan modules to the bracket.
- 8 Rotate the retention brackets in a clockwise direction and rest them on the top ledge of the rear shipping bracket. This action releases the power and fan modules.

Note: If you are unable to rotate the retention brackets, loosen the top four screws on the rear shipping bracket.
- 9 Unfasten the two screws that secure the shipping bracket to the Access Panel.

Note: Re-tighten the AP retention bracket to the rear rail for future repair and/or return.

Figure 24 Shipping bracket secured to Access Panel**Legend**

- 1 Screws securing access panel to shipping bracket
- 10 Using a Phillips screwdriver, unfasten the two panel fasteners on each side of the rear shipping bracket.
- 11 Remove the rear shipping bracket from the back of the chassis.
Note: Re-install the power and fan module bracket screws on the rear shipping bracket for future repair and/or return.
- 12 On each side of the chassis, unfasten the three screws that secure the rear mount shipping brackets using a Phillips screwdriver.

Figure 25 Removing the rear mount shipping brackets from chassis sides**Legend**

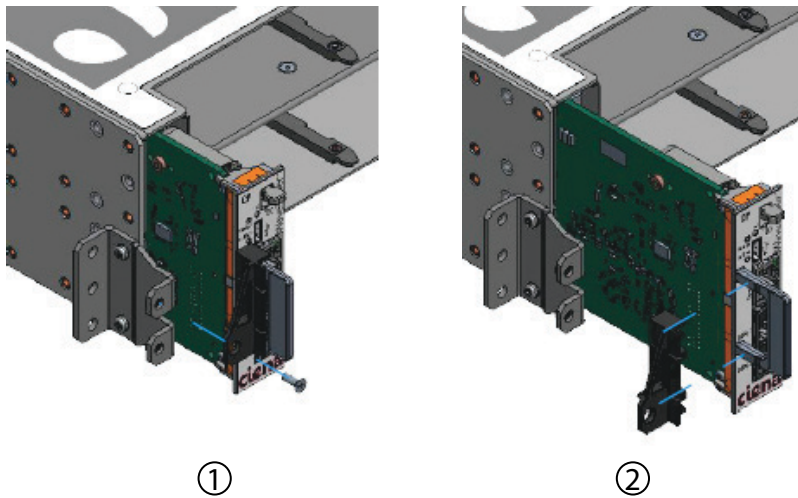
- 1 Rear mount shipping bracket, one each side

Removing the front shipping brackets

Note: Waveserver 5 PIP kit 186-3051-900 is shipped with only the CP secured by shipping brackets at the front of the chassis.

- 13 If a front shipping bracket is present, for each row of modules, unfasten the three screws that secure each bracket to the module handles using a Phillips screwdriver.
- 14 Remove the cable tie that secures the CP to the front shipping bracket. Unfasten the screw that secures the CP clip/bracket assembly to the chassis flange using a Phillips screwdriver. Unclip CP retention the bracket from the CP.

Figure 26 Removing shipping brackets from the CP

**Legend**

- | | | | |
|---|--|---|-------------------------------------|
| 1 | Unfasten screw that secures the CP clip/bracket assembly to the chassis flange | 2 | Unclip CP retention bracket from CP |
|---|--|---|-------------------------------------|
- 15 If required, unfasten the four panel screws (two on each side) on the front shipping bracket using a Phillips screwdriver.
 - 16 If required, remove the front shipping bracket from the chassis.
 - 17 Remove the modules from the chassis and place them on a flat, stable surface.

- 18** Waveserver 5 is ready to be installed in the equipment frame. Depending on the type of frame being used for installation, perform one of the following procedures:
- [Procedure 4, “Installing and grounding Waveserver 5 in a 2-post 19-in. or 23-in. telecom frame”](#)
 - [Procedure 5, “Installing Waveserver 5 in a 4-post 19-in., 23-in, or ETSI telecom frame”](#)
 - [Procedure 6, “Installing and grounding Waveserver 5 in a data center 19-in. wide, 24-36-in. deep equipment frame”](#)
 - [Procedure 7, “Installing and grounding Waveserver 5 in a data center 23-in. wide, 24 - 36-in. deep equipment frame”](#)

Procedure 4 Installing and grounding Waveserver 5 in a 2-post 19-in. or 23-in. telecom frame

Use this procedure to install and ground a Waveserver 5 chassis in a 2-post 19-in. or 23-in. telecom frame.

Requirements

Make sure that

- at least two people are available to perform this procedure.
- the installation site meets all safety and electrical requirements.
- for power disconnection after installation, the male end of each AC power cable must remain accessible at the socket-outlet connection to the power source.
- you have a secure ladder for installations higher than shoulder height
- the power supply and fan modules, CP, and AP have been removed from the Waveserver 5 chassis and placed on a flat, stable surface. See [Procedure 3, “Unpacking Waveserver 5 equipment”](#).
- all tools and bracket kit parts required for installation are available, see [“Required materials for installation, powering, and removal” on page 27](#).
- you have 6 AWG (16 mm²) wire for the chassis ground.
- the mounting setback requirements have been determined, see [“Mounting setback options” on page 27](#). If you are installing Waveserver 5 in a cabinet, ensure that you use a setback that leaves adequate space for the door to close after the fiber managers are installed
- dust covers are installed on all Waveserver 5 optical ports.
- the 19-in. Bracket Kit (186-1300-900) ([“Contents of Waveserver 19-in. Bracket Kit \(186-1300-900\)” on page 31](#)) or the 23-in. Bracket Kit (186-1301-900) ([“Contents of Waveserver 23-in. Bracket Kit \(186-1301-900\)” on page 32](#)) is available.

Note: Both front-mount and mid-mount installation options are supported. However, for stability, front-mount installations require a minimum of 5-in. upright frame depth.

Precautions



DANGER

Risk of laser radiation exposure

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.

**CAUTION**

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and modules or transceivers.

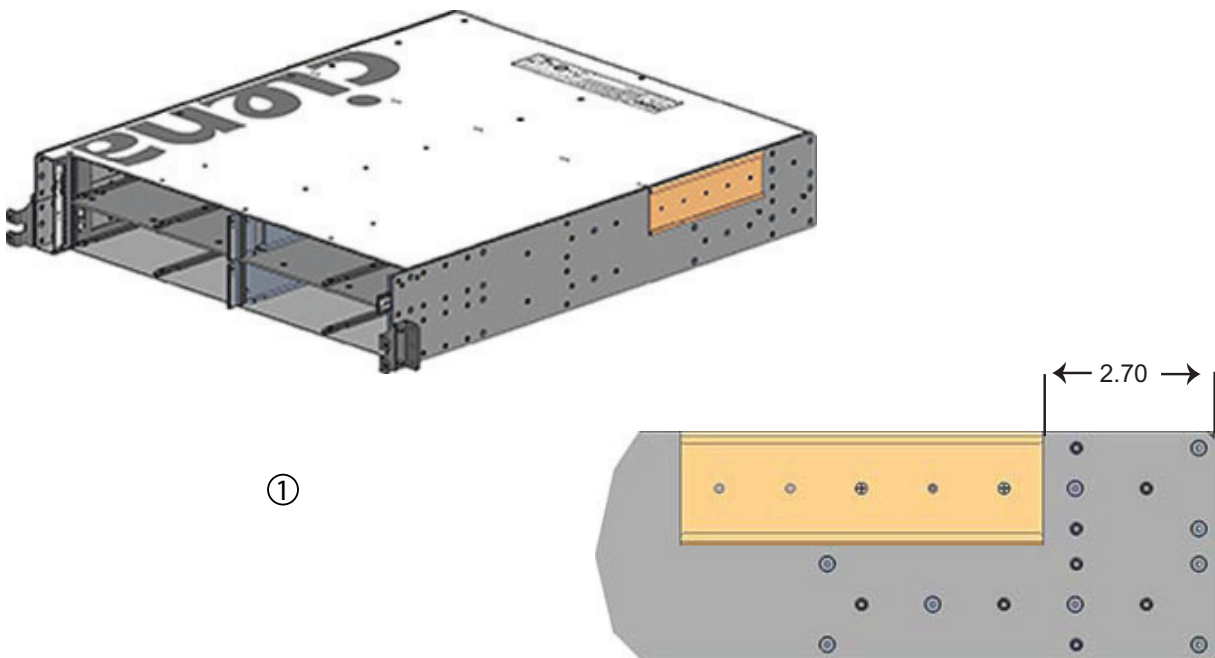
Steps

- 1 Install the rear bracket guide on the Waveserver 5 chassis.
Position the rear bracket guide so that its pilot holes are aligned with the pilot holes on the chassis, and use three (3) wafer head machine screws (312-3730-CS1) to secure it. Do not over-tighten the fasteners.

The sides of the Waveserver 5 chassis have two identical sets of mounting points on the upper and lower sections. You can use either the upper or lower sets of mounting points, however Ciena recommends using the upper row for chassis installation and the lower row for the installation of other accessories.

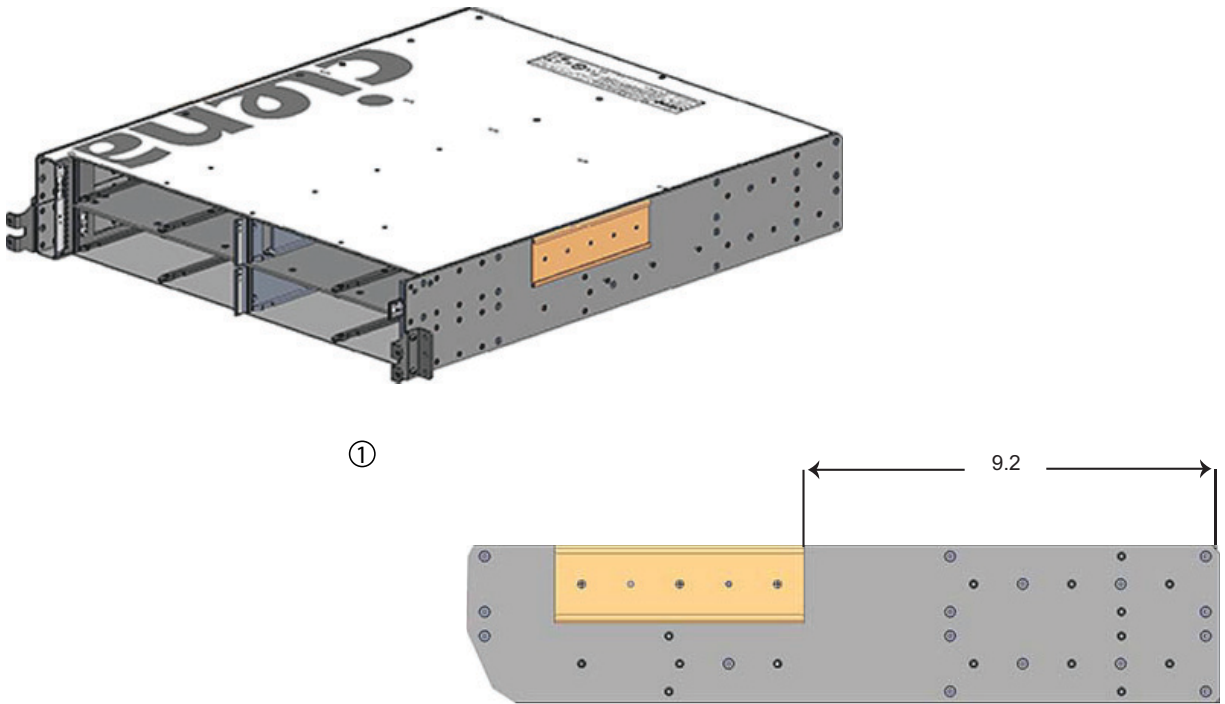
Use one of the two recommended mounting locations, depending on the rack type and setback required. For 2-post racks, Ciena recommends using the flush-mount installation option. See the following figures for examples.

Figure 27 Rear bracket guide installation — mid-mount location

**Legend**

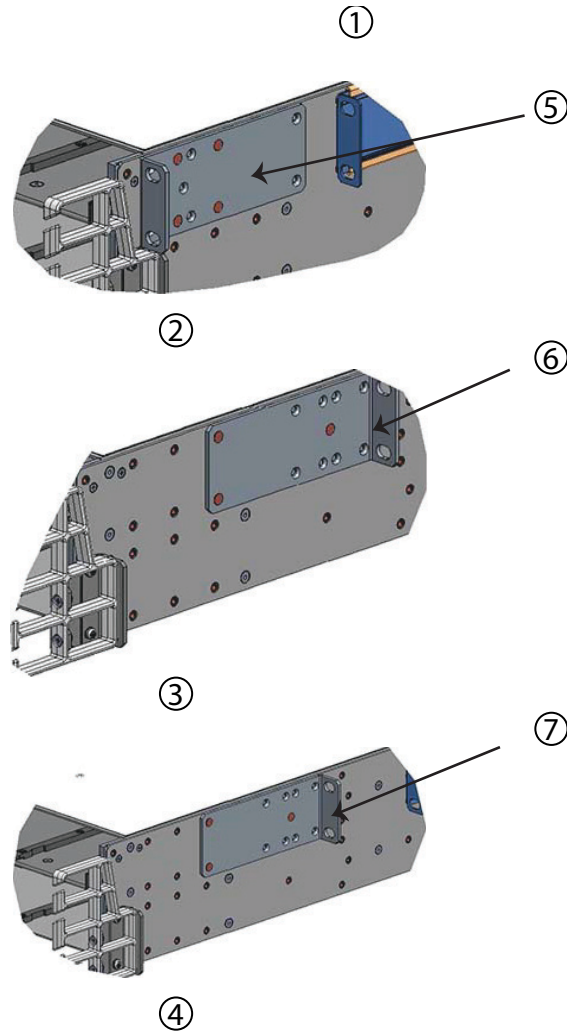
- 1 Recommended location for most racks and setbacks.

Figure 28 Rear bracket guide installation — flush mount location



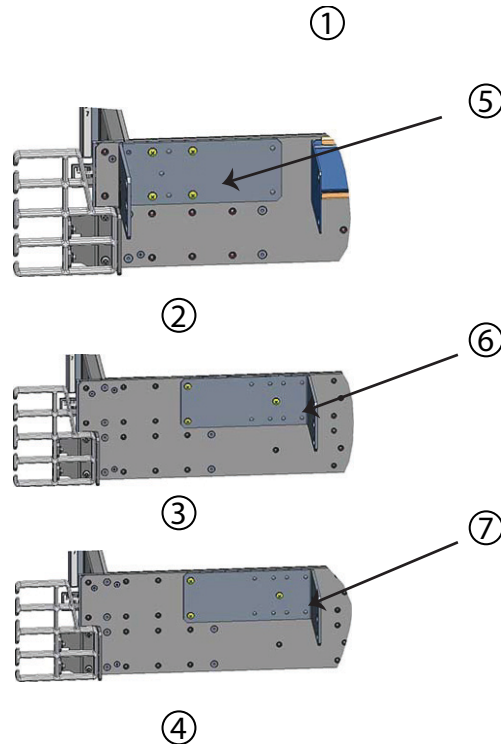
Legend

- 1 Recommended location for 2-post racks with flush-mounting requirement
- 2 Install the front brackets on the Waveserver 5 chassis.
Position the front bracket so that its pilot holes are aligned with the pilot holes on the chassis for the required mounting position, and secure using N0045005 countersink fasteners.
For flush-mount installations, use 4 x N0045005 countersink fasteners per bracket. For mid-mount installations, use 3 x N0045005 countersink fasteners per bracket.
Note: Flush-mount, 2-channel racks with a 3-in flange depth are not supported.
The following figures illustrate the different front bracket mounting setback options for 19-in and 23-in frames. The rear brackets are shown for illustration purposes only, and are installed in a later step.

Figure 29 Front bracket mounting options for 2-post 19-in racks**Legend**

- | | | |
|---|---------------------------------------|----------------------------|
| 1 | Front bracket mounting locations 2- 5 | Front bracket 410-6851-001 |
| 2 | Flush mount for 5-in channel rack | Front bracket 410-6851-001 |
| 3 | Mid-mount for 5-in channel rack | Front bracket 410-6851-001 |
| 4 | Mid-mount for 3-in channel rack | |

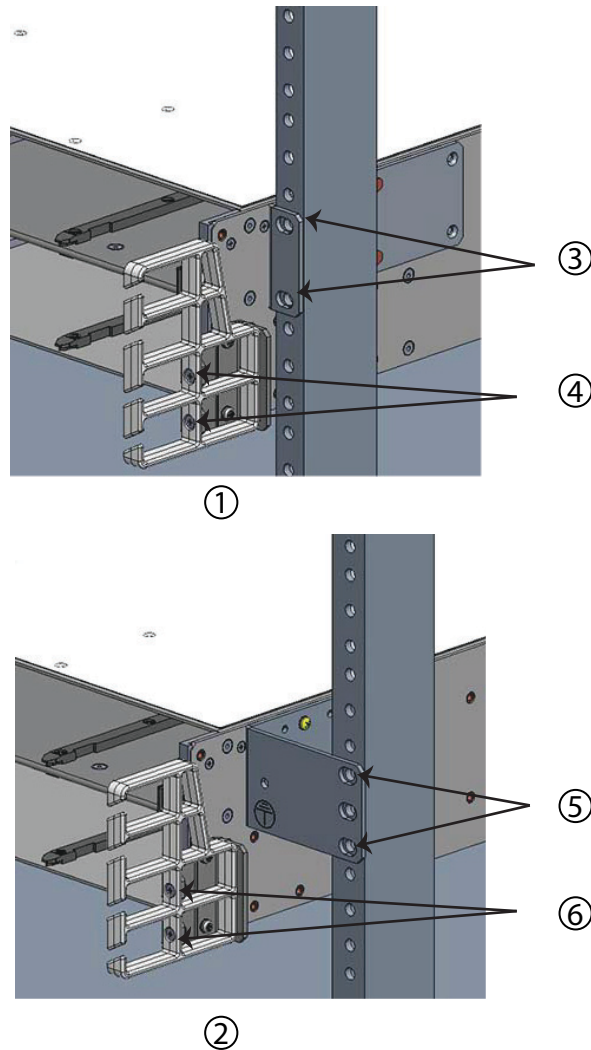
Figure 30 Front bracket mounting options for 2-post 23-in racks



Legend

- | | | |
|---|---------------------------------------|----------------------------|
| 1 | Front bracket mounting locations 2- 5 | Front bracket 410-6853-001 |
| 2 | Flush mount for 5-in channel rack | Front bracket 410-6853-001 |
| 3 | Mid-mount for 5-in channel rack | Front bracket 410-6853-001 |
| 4 | Mid-mount for 3-in channel rack | |

- 3** Secure the front frame-mounting brackets to the equipment frame.
- With one person at the front of the frame and another at the rear, hold the chassis securely in position in the frame. Attach the front brackets to the frame with two (2) rack screws per bracket. Do not over-tighten the screws.

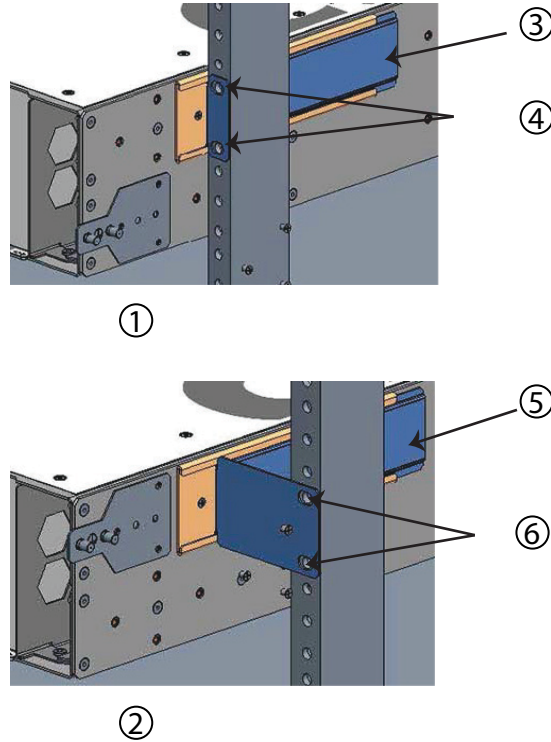
Figure 31 Installing the front frame mounting brackets on the frame**Legend**

- | | | | |
|---|--|---|-----------------------------|
| 1 | Front brackets installed on 19-in rack | 4 | Fiber manager fasteners |
| 2 | Front brackets installed on 23-in rack | 5 | Two rack screws per bracket |
| 3 | Two rack screws per bracket | 6 | Fiber manager fasteners |

- 4** Install the rear brackets and secure them to the equipment frame.
Slide a rear bracket into the rear bracket guide on the chassis, and attach the bracket to the frame with two (2) rack screws. Do not over-tighten the screws.

The following figure shows the rear brackets installed on the equipment frame. The ground bracket is shown for illustration purposes only and is installed in a later step.

Figure 32 Rear bracket installation — 19-in and 23-in racks



Legend

1	19-in rack	4	Fastened with 2 rack screws
2	23-in rack	5	Rear mounting bracket 410-7036-001
3	Rear mounting bracket 410-7036-001	6	Fastened with 2 rack screws

- 5 If required, install the ground bracket (410-6903-002).

Note: A chassis ground is required for all DC installations and AC installations requiring GR-1089-CORE compliance. For AC installations that do not require GR-1089-CORE compliance, the chassis ground is recommended but not required.

If you are installing Waveserver 5 Then

on a 2-post 19-in. telecom equipment frame

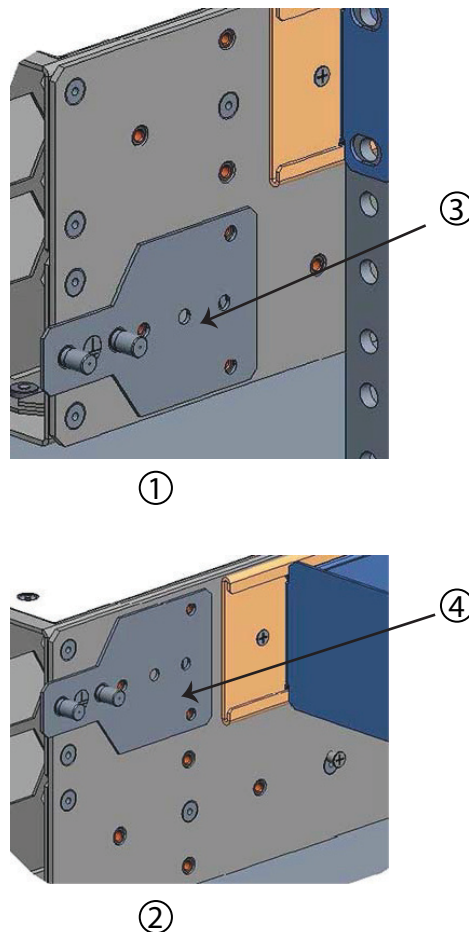
- Align the pilot holes on the ground bracket with the pilot holes on the chassis rear on either the right or left side and fasten the bracket using 3 wafer head machine screws (312-3730-CS1). See the following figure.

- The ground bracket can be mounted in the upper or lower position.

on a 2-post 23-in. telecom equipment frame

- Align the pilot holes on the ground bracket with the pilot holes on the chassis rear on either right or left side and fasten the bracket using 3 wafer head machine screws (312-3730-CS1). See the following figure.

Figure 33 Ground bracket installed

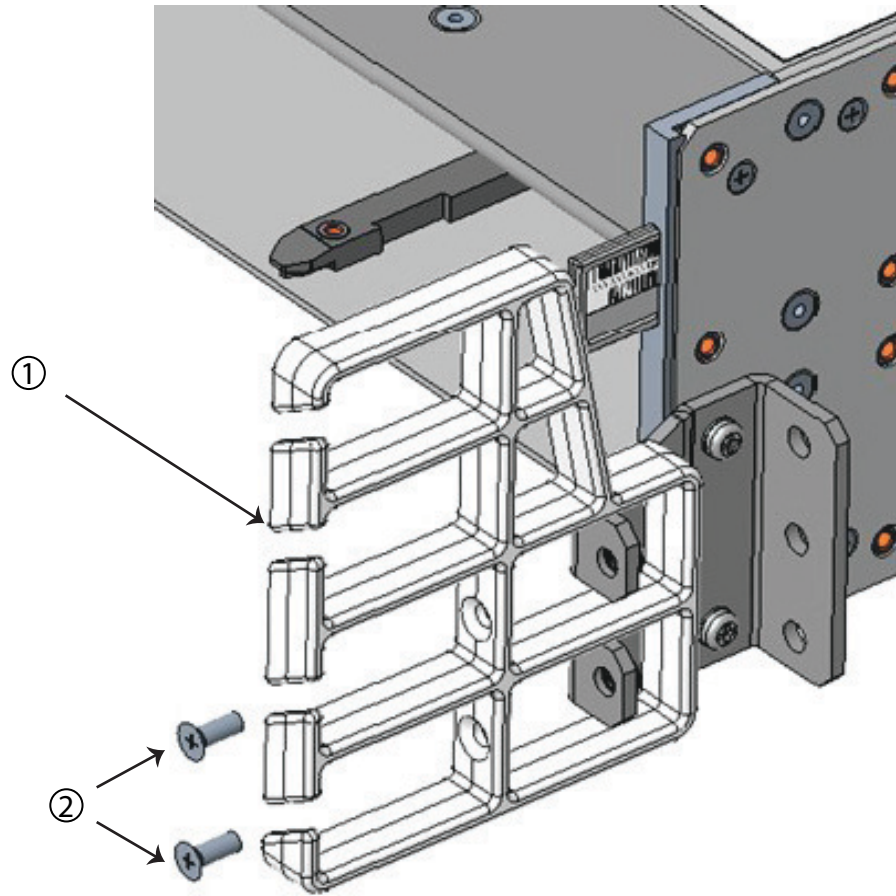


Legend

- | | | | |
|---|--|---|-----------------------------|
| 1 | Ground bracket installed on 19-in rack | 3 | Ground bracket 410-6903-002 |
| 2 | Ground bracket installed on 23-in rack | 4 | Ground bracket 410-6903-002 |

- 6 Install the chassis ground.
 - For a 2-post 19-in. bracket, fasten the ground lug using one or two Keps nuts (N0041806) and torque to 29 lb-in (3.3 N-m).
 - For 2-post 23-in. bracket with the ground bracket installed on the chassis rear, fasten the lug using one or two Keps nuts (N0041806) and torque to 29 lb-in (3.3 N-m).
- 7 Route the ground wire from the ground bracket and fasten it to an equipment frame ground location.
- 8 Install the fiber managers on the chassis, if required.

Fasten the fiber managers to the brackets on left and right sides of the chassis. Use the two flat head screws (312-1030-M08), included in the bag with the fiber managers and shipped with the chassis. Torque to 3 lb-in (.339 N-m).

Figure 34 Fiber manager installation**Legend**

1 Fiber manager

2 Flat head screws

You have completed this procedure. Next, perform [Procedure 8, “Installing the power and fan modules and connecting power cables to Waveserver 5”](#).

Procedure 5 Installing Waveserver 5 in a 4-post 19-in., 23-in, or ETSI telecom frame

Use this procedure to install and ground Waveserver 5 in a 4-post 19-in., 23-in., or ETSI telecom frame.

Requirements

- Two people are available to perform this procedure.
- The installation site meets all safety and electrical requirements.
- The male end of each AC power cable must remain accessible at the socket-outlet connection to the power source for power disconnection after installation.
- All tools are available, see [“Required materials for installation, powering, and removal” on page 27.](#)
- The mounting setback requirements have been determined, see [“Mounting setback options” on page 27.](#)
- You have 6 AWG (16 mm²) wire for the chassis ground.
- The power and fan modules, CP, and AP have been removed from the Waveserver 5 chassis and placed on a flat, stable surface. See [Procedure 3, “Unpacking Waveserver 5 equipment”.](#)
- Dust covers are installed on all optical ports.
- The bracket kit required for installation is available. Refer to one of the following tables:
 - [“Contents of Waveserver 19-in. Bracket Kit \(186-1300-900\)” on page 31](#)
 - [“Contents of Waveserver 23-in. Bracket Kit \(186-1301-900\)” on page 32](#)
 - [“Contents of Waveserver ETSI Bracket Kit \(186-1302-900\)” on page 34](#)

Precautions



DANGER

Risk of laser radiation exposure

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.

**CAUTION**

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and modules or transceivers.

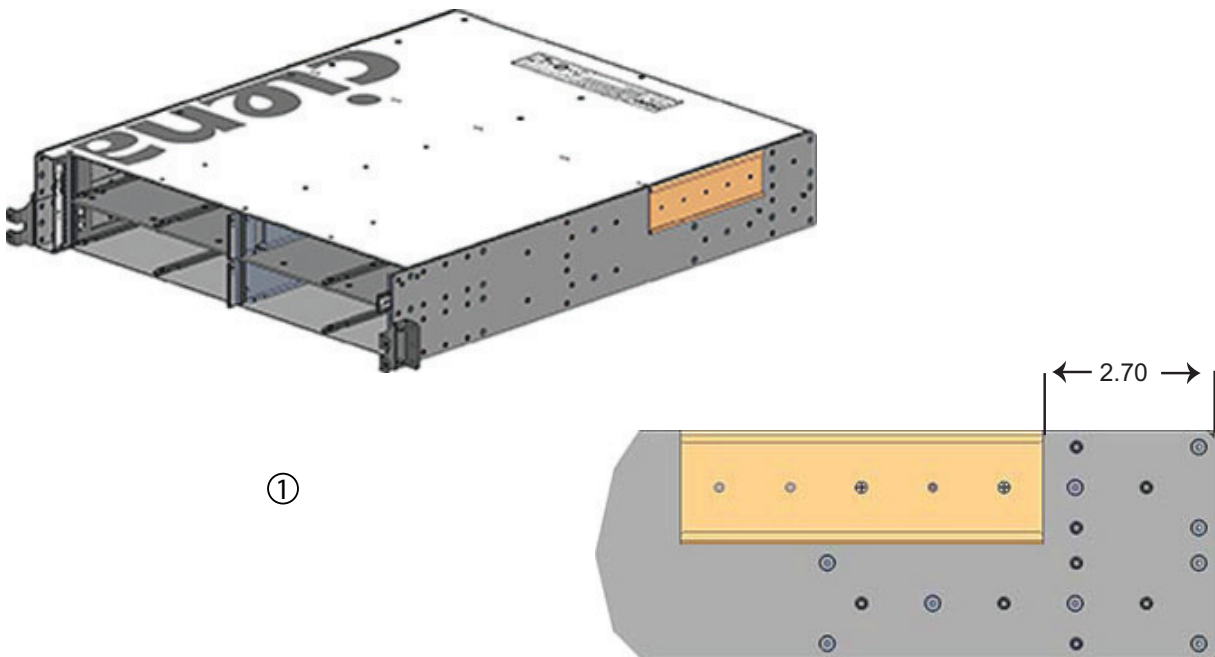
Steps

- 1 Install the rear bracket guide (410-7037-001) on the Waveserver 5 chassis. Position the rear bracket guide so that its pilot holes are aligned with the pilot holes on the chassis, and use three (3) wafer head machine screws (312-3730-CS1) to secure it. Do not over-tighten the fasteners.

The sides of the Waveserver 5 chassis have two identical sets of mounting points on the upper and lower sections. You can use either set, however Ciena recommends using the upper row for chassis installation and the lower row for the installation of other accessories.

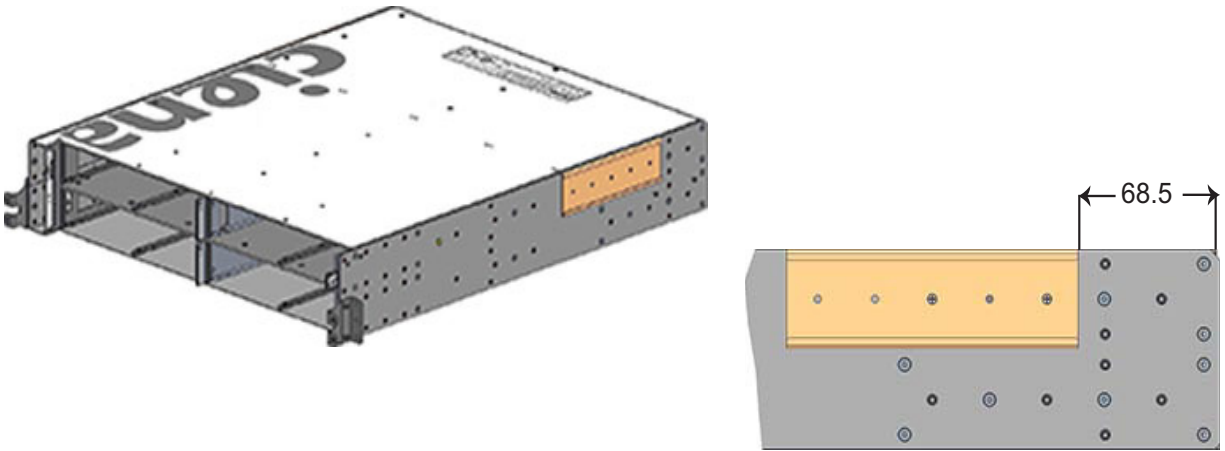
The recommended mounting locations depend on the rack type and setback required. See the following figures for examples.

Figure 35 Rear bracket guide installation — 19-in and 23-in racks, mid-mount position

**Legend**

- 1 Recommended location for most racks and setbacks.

Figure 36 Rear bracket guide installation — ETSI rack

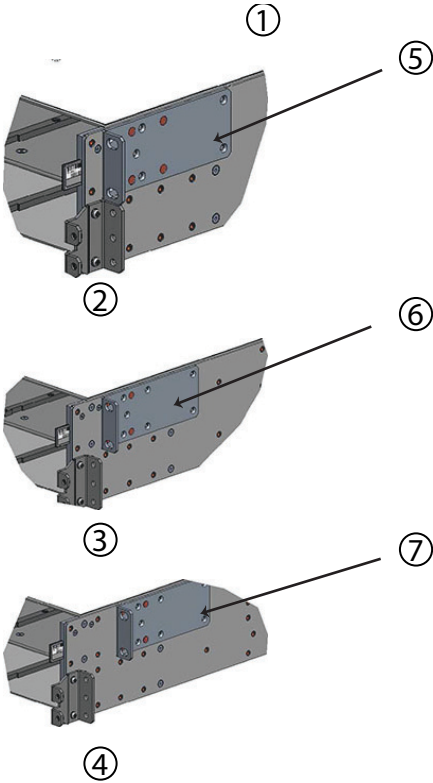


- 2 Install the front brackets on the Waveserver 5 chassis.
Position the front bracket so that its pilot holes are aligned with the pilot holes on the chassis for the required mounting position. Secure the brackets with the following fasteners:
 - For 19-in and 23-in racks, secure with 4 Phillips countersink fasteners (N0045005) per side
 - For ETSI racks, secure with 4 SEMS external tooth washers (N0038285) per side

The following figures illustrate the different front bracket mounting setback options for 19-in, 23-in, and ETSI frames. The rear brackets are shown for illustration purposes only, and are installed in a later step.

Note: ETSI brackets support a 100 mm(4 in.) or 127 mm (5 in.) standard setback.

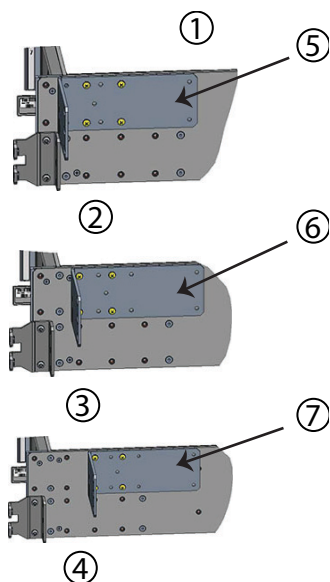
Figure 37 Front bracket mounting setback options — 19-in racks



Legend

- | | | | |
|---|--|---|----------------------------|
| 1 | Front bracket mounting setbacks - 4-post 19-in racks | 5 | Front bracket 410-6851-001 |
| 2 | Flush mount | 6 | Front bracket 410-6851-001 |
| 3 | 100 mm setback | 7 | Front bracket 410-6851-001 |
| 4 | 5-in setback | | |

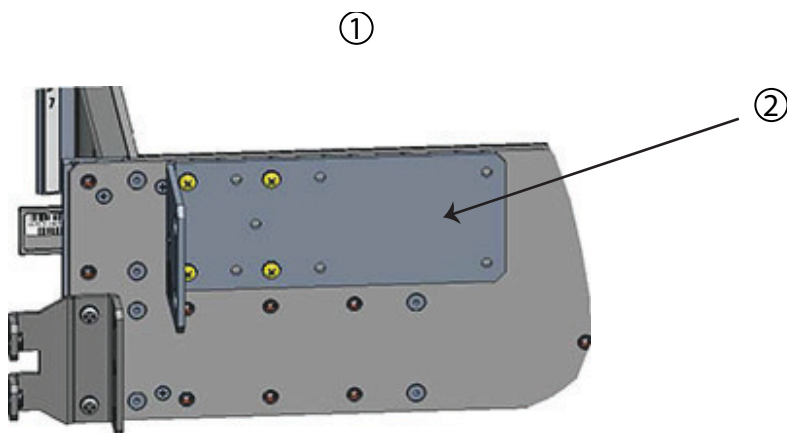
Figure 38 Front bracket mounting setback options — 23-in racks



Legend

- | | | | |
|---|--|---|----------------------------|
| 1 | Front bracket mounting setbacks - 4-post 23-in racks | 5 | Front bracket 410-6853-001 |
| 2 | Flush mount | 6 | Front bracket 410-6853-001 |
| 3 | 100 mm setback | 7 | Front bracket 410-6853-001 |
| 4 | 5-in setback | | |

Figure 39 Front bracket mounting setback — ETSI racks

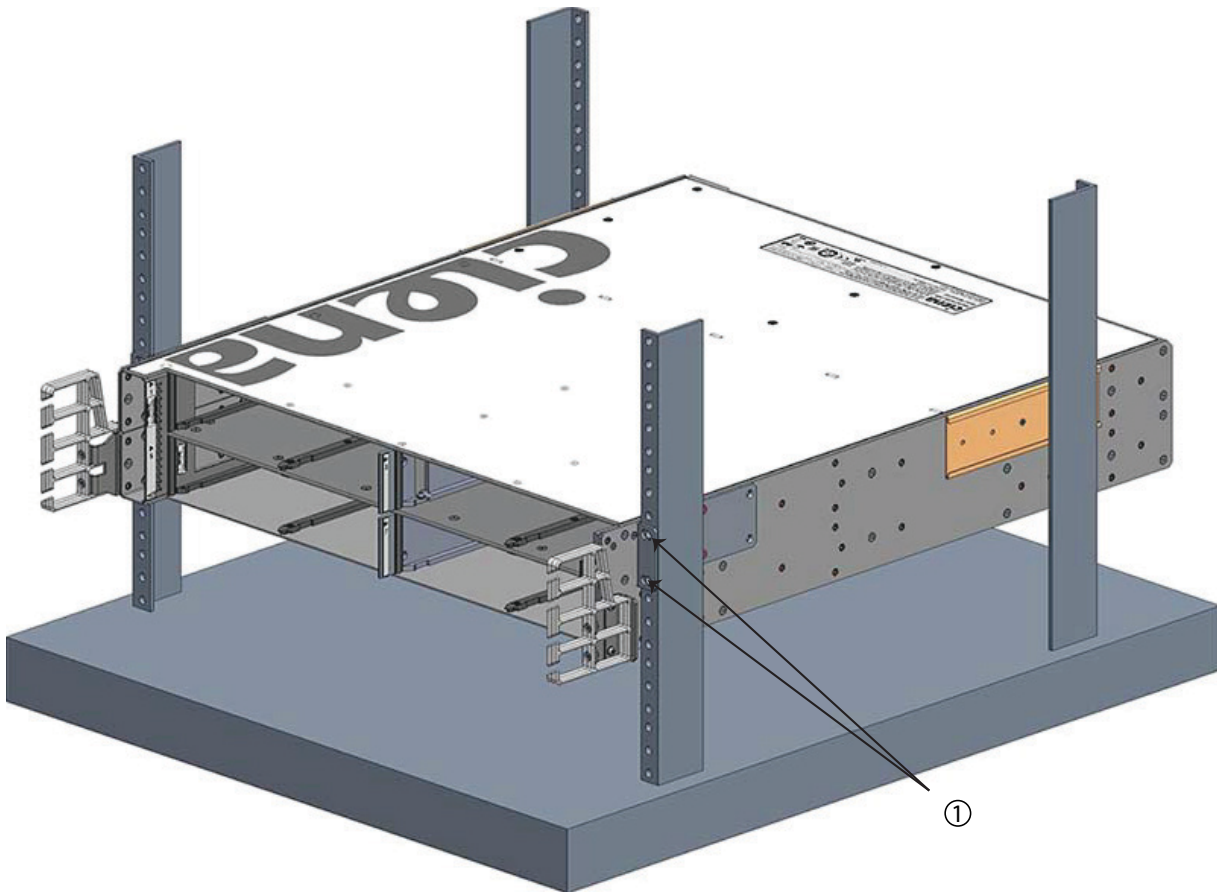


Legend

- | | | | |
|---|---|---|----------------------------|
| 1 | Front mounting bracket setback - ETSI racks | 2 | Front bracket 410-6852-001 |
|---|---|---|----------------------------|

- 3 Secure the front frame mounting brackets to the equipment frame.
With one person at the front of the frame and another at the rear, hold the chassis securely in position in the frame.
Attach the front brackets to the frame with two (2) rack screws per bracket for 19-in and ETSI frames. For 23-in frames, install the rack screws in the top and bottom holes of the bracket. Do not over-tighten the screws.
The following figure shows an example of a the Waveserver 5 chassis installed in a 4-post 19-in rack at a 100 mm setback position.

Figure 40 Installing the Waveserver 5 chassis in the frame — 19-in bracket kit shown



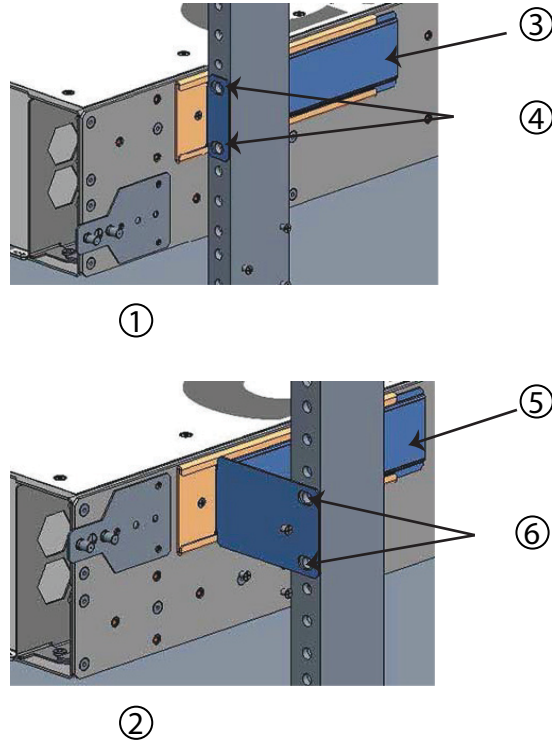
Legend

- 1 Two rack screws per bracket

- 4 Install the rear brackets and secure them to the equipment frame.
Slide a rear frame mounting bracket into the rear bracket guide on the chassis, and attach the bracket to the frame with two (2) rack screws. Do not over-tighten the screws.

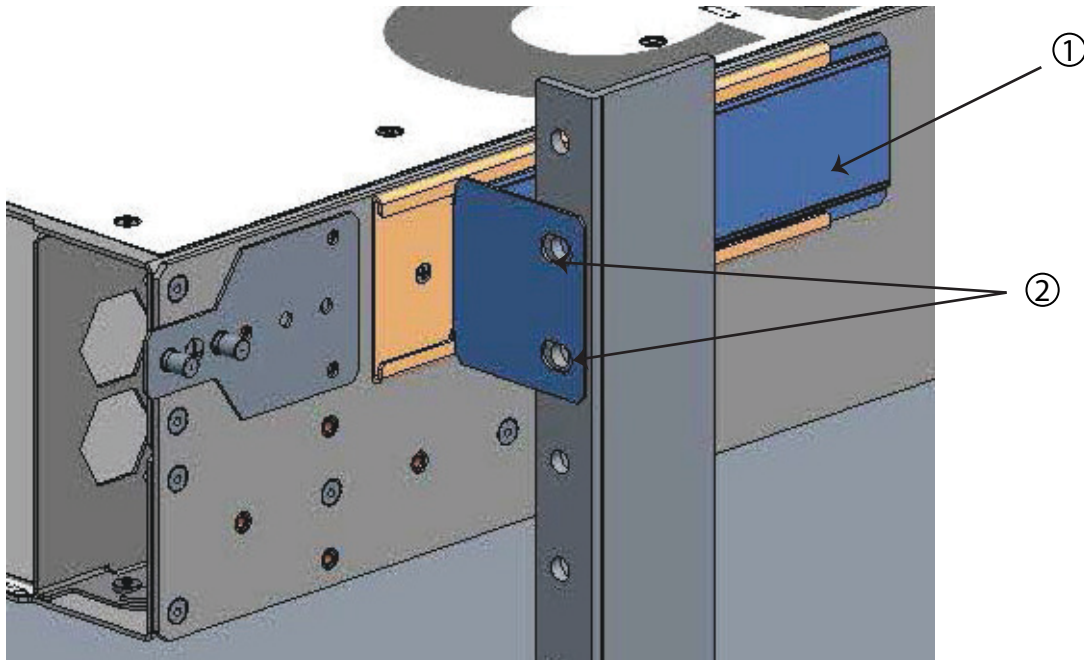
The following figures show the rear brackets installed on the equipment frame. The ground bracket is shown for illustration purposes only and is installed in a later step.

Figure 41 Rear bracket installation — 19-in and 23-in racks



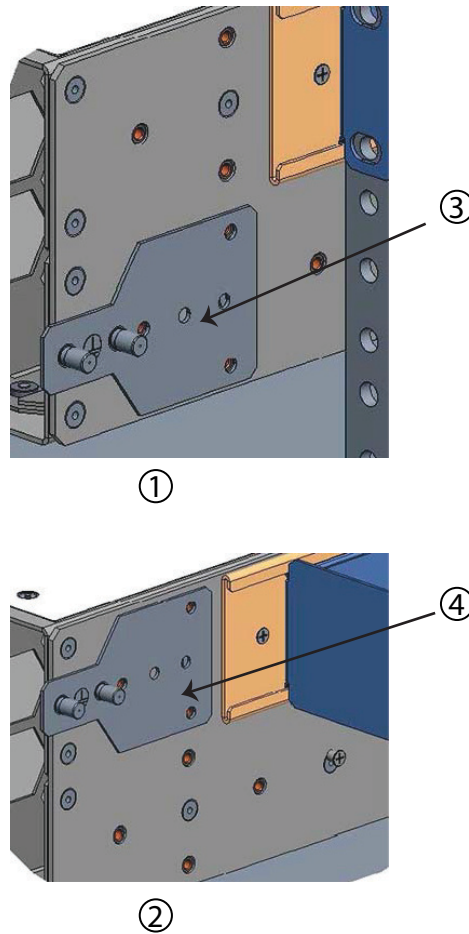
Legend

1	19-in rack	4	Fastened with 2 rack screws
2	23-in rack	5	Rear mounting bracket 410-7036-001
3	Rear mounting bracket 410-7036-001	6	Fastened with 2 rack screws

Figure 42 Rear bracket installation — ETSI rack**Legend**

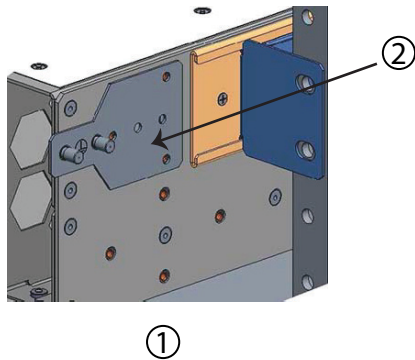
- | | | | |
|---|------------------------------------|---|-----------------------------|
| 1 | Rear mounting bracket 410-7036-003 | 2 | Fastened with 2 rack screws |
|---|------------------------------------|---|-----------------------------|
- 5 If required, install the ground bracket (410-6903-002).
- Note:** A chassis ground is required for all DC installations and AC installations requiring GR-1089-CORE compliance. For AC installations that do not require GR-1089-CORE compliance, the chassis ground is recommended but not required.

If you are installing the chassis ground	Then
on a 4-post 19-in. telecom equipment frame	• Align the pilot holes on the ground bracket with the pilot holes on the chassis rear on either the right or left side and fasten the bracket using 3 wafer head machine screws (312-3730-CS1). See the following figure. • The ground bracket can be mounted in the upper or lower position.
on a 4-post 23-in. telecom equipment frame	• Align the pilot holes on the ground bracket with the pilot holes on the chassis rear on either the right or left side and fasten the bracket using 3 wafer head machine screws (312-3730-CS1). See the following figure.
on an ETSI equipment frame	• Align the pilot holes on the ground bracket with the pilot holes on the chassis rear on either the right or left side and fasten the bracket using 3 wafer head machine screws (312-3730-CS1). See the following figure. • The ground bracket can be mounted in the upper or lower position.

Figure 43 Ground bracket installed — 19-in and 23-in racks**Legend**

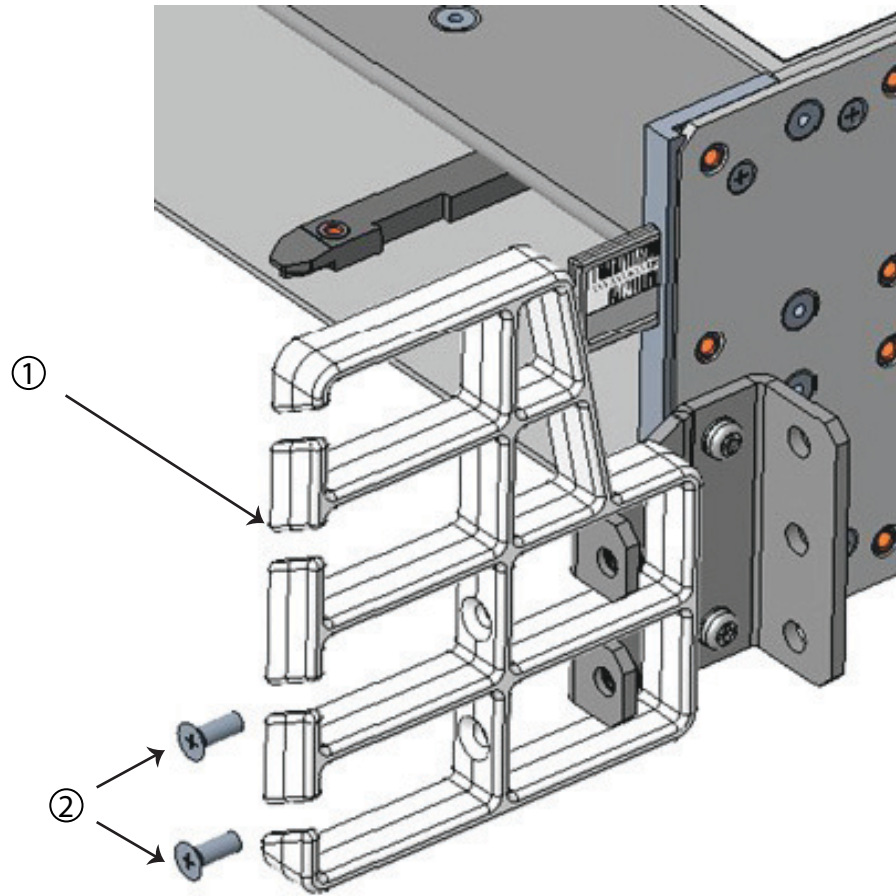
- | | | | |
|---|--|---|-----------------------------|
| 1 | Ground bracket installed on 19-in rack | 3 | Ground bracket 410-6903-002 |
| 2 | Ground bracket installed on 23-in rack | 4 | Ground bracket 410-6903-002 |

Figure 44 Ground bracket installed — ETSI rack



Legend

- | | | | |
|---|--|---|-----------------------------|
| 1 | Ground bracket installed on ETSI frame | 2 | Ground bracket 410-6903-002 |
|---|--|---|-----------------------------|
- 6** Install the chassis ground wire:
- For a 4-post 19-in. bracket, attach the lug to the ground bracket using one or two Keps nuts (N0041806) and torque to 29 lb-in (3.3 N-m).
 - For 4-post 23-in. bracket, fasten a two-hole lug to the ground bracket using one or two Keps nuts (N0041806) and torque to 29 lb-in (3.3 N-m).
 - For a 4-post ETSI bracket, fasten a two-hole lug to the ground bracket using one or two Keps nuts (N0041806) and torque to 29 lb-in (3.3 N-m).
- Route the ground wire and fasten it to an equipment frame ground location.
- 7** Install the fiber managers on the chassis, if required.
- Fasten the fiber managers to the brackets on left and right sides of the chassis. Use the two flat head screws (312-1030-M08), included in the bag with the fiber managers and shipped with the chassis. Torque to 3 lb-in (.339 N-m).

Figure 45 Fiber manager installation**Legend**

1 Fiber manager

2 Flat head screws

You have completed this procedure. Next, perform [Procedure 8, “Installing the power and fan modules and connecting power cables to Waveserver 5”](#).

Procedure 6 Installing and grounding Waveserver 5 in a data center 19-in. wide, 24-36-in. deep equipment frame

Use this procedure to install and ground Waveserver 5 in a 19-in wide, 24-in. to 36-in. deep equipment frame. One person is required to perform this procedure.

Requirements

Make sure that

- the installation site meets all safety and electrical requirements.
- for power disconnection after installation, the male end of each AC power cable must remain accessible at the socket-outlet connection to the power source.
- the power and fan modules have been removed from the Waveserver 5 chassis and placed on a flat, stable surface. See [Procedure 3, “Unpacking Waveserver 5 equipment”](#) for details.
- all tools and bracket kit parts required for installation are available, see [“Required materials for installation, powering, and removal”](#) on page 27.
- 6 AWG (16 mm²) wire for the chassis ground.
- dust covers are installed on all Waveserver 5 optical ports.
- you have the following bracket kit available for installation:
 - Data Center 19-in, 24 - 36-in deep Bracket Kit (186-1306-900), see [“Contents of Waveserver Data Center Bracket Kit, 19-in wide x 24-36-in deep \(186-1306-900\)”](#) on page 35

Note: Ciena recommends installing the brackets in the flush mount position to maximize the amount of clearance at the front of the frame and prevent accidental damage to the fibers. A minimum of 3.1-in clearance is required at the front of the frame uprights for fiber exit.

Precautions



DANGER

Risk of laser radiation exposure

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



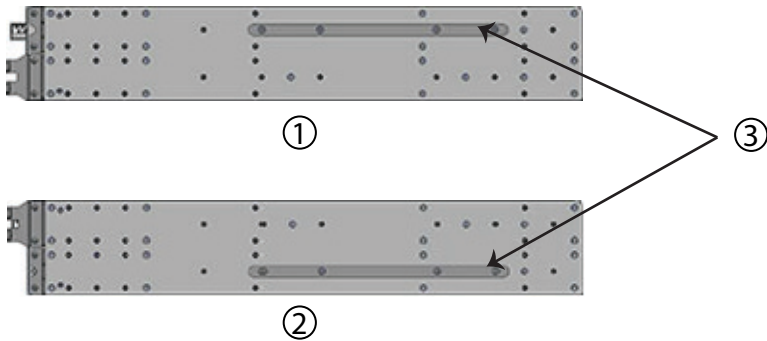
CAUTION
Risk of damage to the chassis, modules and pluggable transceivers
Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and modules or transceivers.

Steps

- 1 Waveserver 5 has two identical sets of mounting holes that are the mirror image of one another, for upper or lower mounting positions. You can use either the upper or lower mounting points, however Ciena recommends using the upper row for chassis installation and the lower row for the installation of other accessories.
Select your next step.

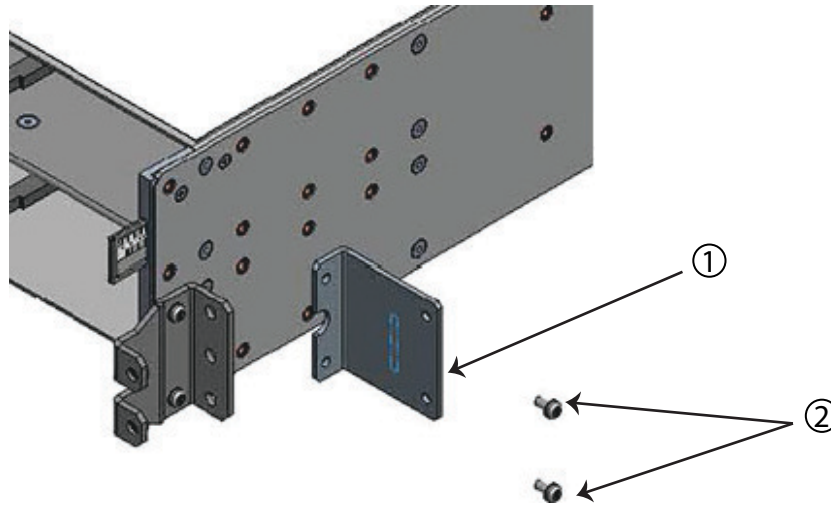
If you want to mount the chassis	Then go to
the upper mounting location	step 2
the lower mounting location	step 4
- 2 Install the T-Bars (42T-2970-001) on each side of the Waveserver 5 chassis in the upper mounting location.
Position a T-Bar against the side of the chassis so that the pilot holes on the T-Bar align with the upper row of pilot holes on the chassis, and secure the T-Bar using four flathead screws (312-1030-M05) on each side.

Figure 46 T-Bar installation



Legend

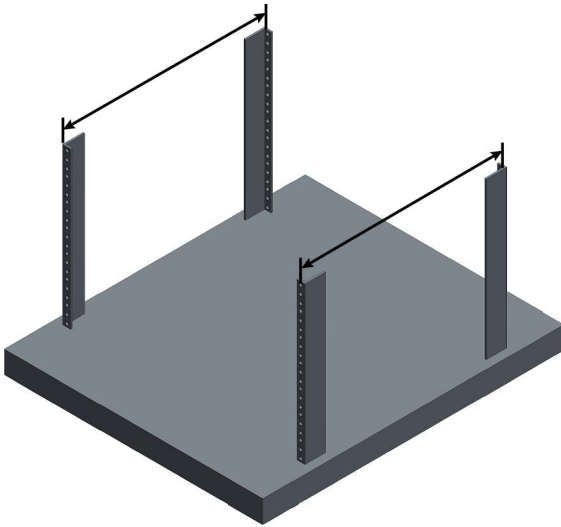
- | | | | |
|---|-----------------------------------|---|----------------------|
| 1 | T-Bar installed in upper position | 3 | T-Bar (42T-2970-001) |
| 2 | T-Bar installed in lower position | | |
- 3 Install the front frame mounting brackets on each side of the Waveserver 5 chassis.
Position a front bracket (410-7588-001) at the front of the chassis so that the tab with the two large pilot holes is facing the back of the chassis, and secure the bracket using two pan head SEMS screws (312-0630-M06) on each side.

Figure 47 Front frame-mounting bracket installation)**Legend**

- | | | | |
|---|---|---|-------------------------------------|
| 1 | Front frame mounting bracket (410-7588-001) | 2 | Pan head SEMS screws (312-0630-M06) |
|---|---|---|-------------------------------------|

Go to [step 8](#).

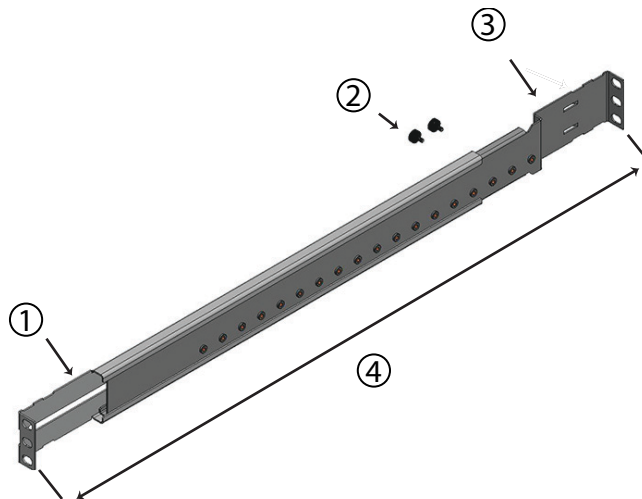
- 4 Install the T-Bars (42T-2970-001) on each side of the Waveserver 5 chassis in the lower mounting location.
Position a T-Bar against the side of the chassis so that the pilot holes on the T-Bar align with the lower row of pilot holes on the chassis, and secure the T-Bar using four flathead screws (312-1030-M05) on each side.
- 5 Fasten the fiber manager utility bracket to the upper mounting location on the chassis.
- 6 Rotate the fiber manager 180 degrees and assemble the utility bracket using the flat head screws provided in the bag with the fiber managers. Repeat this step on the other side of the chassis.
- 7 Fasten the two front mounting brackets (410-7588-001) to the front of the chassis using two pan head external tooth washers (312-0630-M06) per bracket. Position the brackets so that the large tab is facing the back of the chassis.
- 8 Measure the rack upright mounting surface spacing distance (front to back) on both sides of the equipment frame.

Figure 48 Measured frame upright depth

9 Assemble the front and rear bracket guides.

Insert the end of a rear bracket guide (410-8190-001) into the end of a front bracket guide (410-8189-001). Adjust the length so that the distance between the front flange and rear flange is approximately the same as the distance measured in [step 8](#) (24-in - 36-in depth).

Loosely fasten using two pan head external lock washers (312-0630-M06) on each rail assembly. For shorter rack depths, position the screws as far apart as possible.

Figure 49 Front and rear rail assembly

Legend

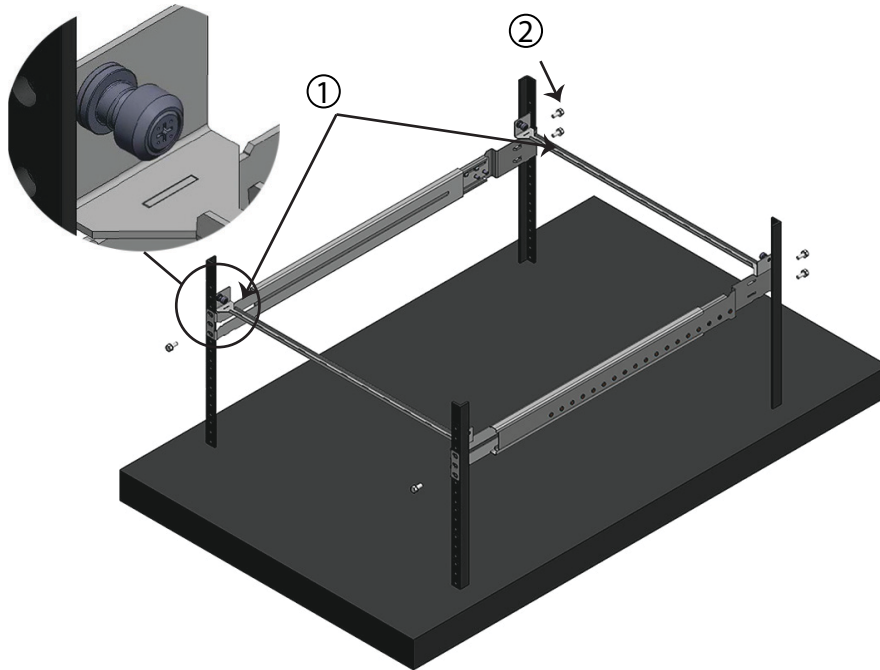
- | | | | |
|---|---|---|-----------------------------------|
| 1 | Front Bracket Guide (410-8189-001) | 3 | Rear Bracket Guide (410-8190-001) |
| 2 | Pan Head External Lock Washers (312-0630-M06) | 4 | Set to frame upright depth |

- 10** Install and secure the rail assemblies to the frame uprights. Use the rear cable management bar to stabilize the uprights and ensure that they remain square. Use the rear cable management bar to ensure spacing for the front and back brackets is correct. Position the rear cable management bar (410-8191-001) at the back of the frame. Use the keying points provided in the uprights to ensure that the spacing is correct.

Loosely install the rail assemblies to the uprights using the rack manufacturer's recommended hardware. Use one fastener in the middle pilot hole of the front bracket, and two fasteners in the top and bottom pilot holes of the rear bracket. Tighten the front rail screws on the front and rear rail assemblies so that the rails are firmly retained while still allowing for some final adjustment.

Note: Some final minor adjustments of the rail assemblies may be required before you install the Waveserver 5 in the frame.

Figure 50 Rail assemblies installed on frame uprights

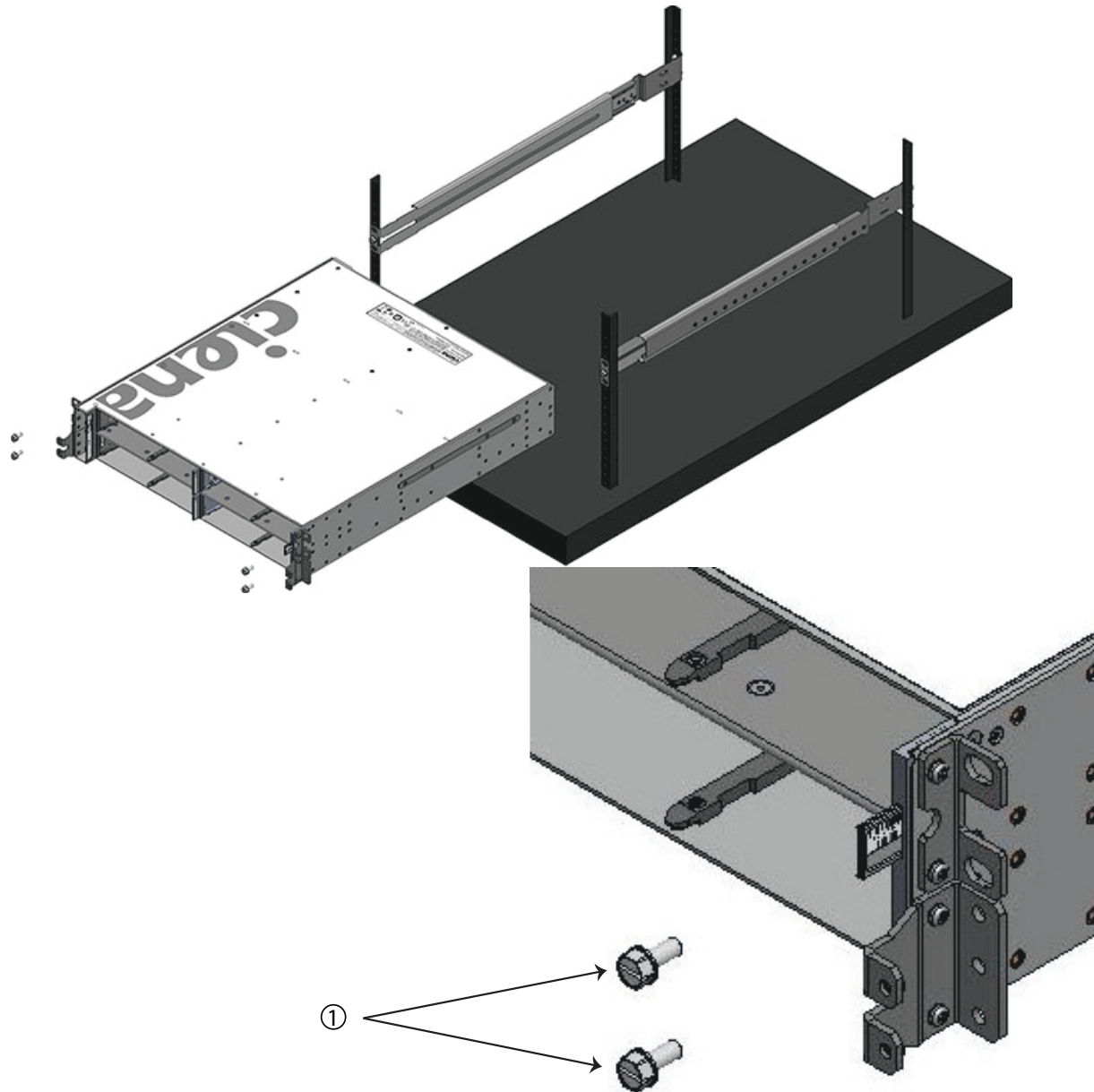


Legend

- | | | | |
|---|---|---|---|
| 1 | Rear Cable Management Bar
(410-8191-001) | 2 | Rack manufacturer's
recommended hardware |
|---|---|---|---|

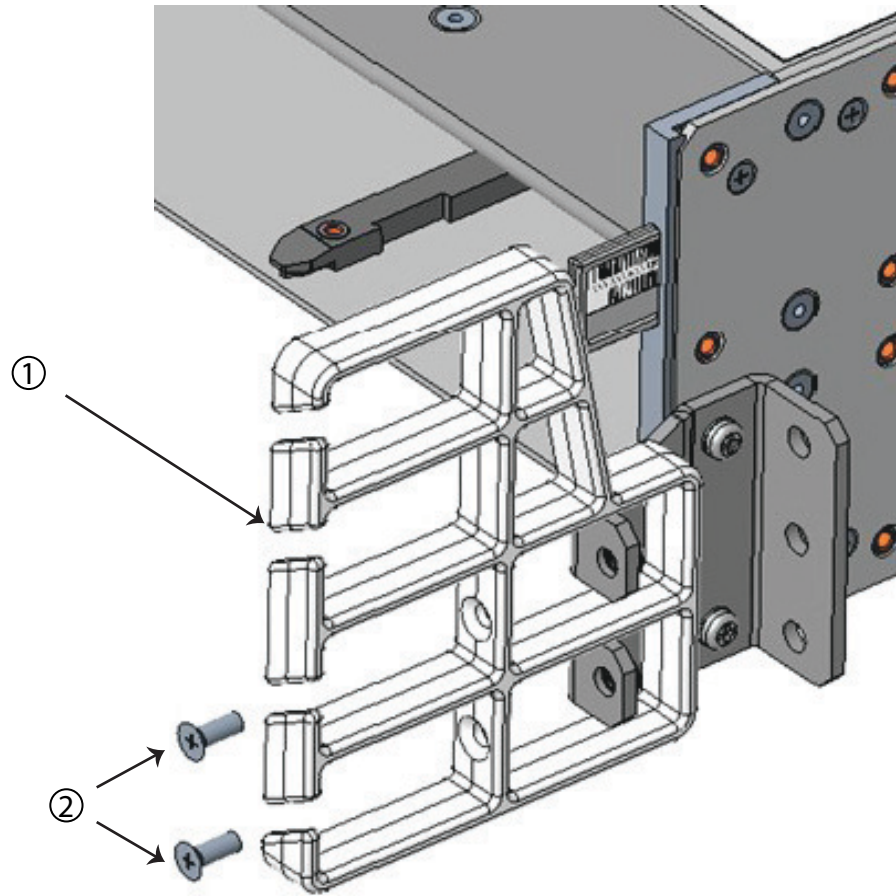
- 11** Install the Waveserver 5 chassis in the equipment frame.
- Holding the chassis securely at the front of the frame, align the T-Bars with the pilot slot on each front rail assembly. Carefully slide the chassis into position in the frame until the frame mounting bracket at each side of the chassis comes into contact with the bracket on the front rail. Install the rack mounting hardware in the holes provided.

Figure 51 Waveserver 5 chassis positioned for installation



Legend

- 1 Rack manufacturer's recommended hardware
- 12 Install the fiber managers on the chassis, if required.
Fasten the fiber managers to the brackets on left and right sides of the chassis. Use the two flat head screws (312-1030-M08), included in the bag with the fiber managers and shipped with the chassis. Torque to 3 lb-in (.339 N-m).

Figure 52 Fiber manager installation**Legend**

1 Fiber manager

2 Flat head screws

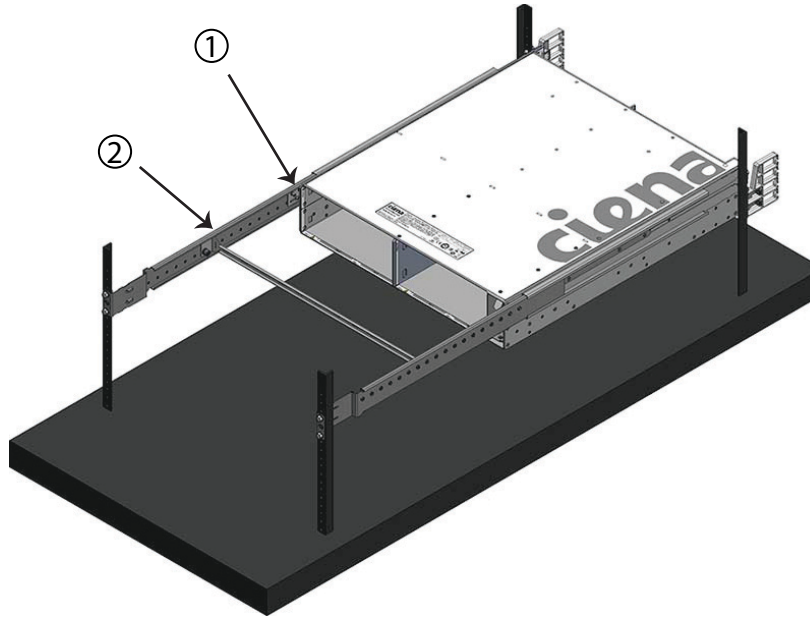
You have completed this procedure. Next, perform [Procedure 8, “Installing the power and fan modules and connecting power cables to Waveserver 5”](#).

- 13** Install the cable bracket (rear cable management bar) (410-8191-001).

Note: The cable management bar cannot be installed in equipment frames with depths of 26-in or less. Lances located on the rear bracket are provided for cable management.

With the cable-management bar tab-side up in the frame, insert the tab at each end of the guide on the front rail. Tighten the thumb screw on each side of the rear cable management bar to secure it to the rail assemblies. Do not over-tighten the thumb screws.

Figure 53 Rear cable-management bar installation



Legend

- | | | | |
|---|---------------------------------------|---|---------------------------|
| 1 | 10-32 studs for ground lug attachment | 2 | Rear cable management bar |
|---|---------------------------------------|---|---------------------------|

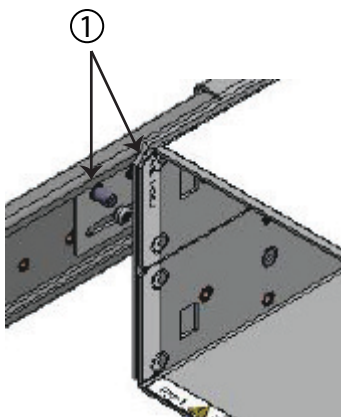
Note: A chassis ground is required for all DC installations and AC installations requiring GR-1089-CORE compliance. For AC installations that do not require GR-1089-CORE compliance, the chassis ground is recommended but not required.

14 Install the chassis ground.

Crimp the ground wire into a single-hole or two-hole ground lug. Fasten the lug to the 10-32 studs provided at the rear of the frame using Keps nuts (N0041806) and torque to 29 lb-in (3.3 N-m). See the following figure.

Route the ground wire from the studs and fasten to an equipment frame ground location.

Note: For 19-in. brackets, the ground cable must be formed to 90°.

Figure 54 Chassis ground studs**Legend**

- 1 10-32 studs for ground lug attachment

Procedure 7 Installing and grounding Waveserver 5 in a data center 23-in. wide, 24 - 36-in. deep equipment frame

Use this procedure to install and ground Waveserver 5 in a data center 23-in. wide, 24-in. to 36-in. deep equipment frame. One person is required to perform this procedure.

Requirements

Make sure that

- the installation site meets all safety and electrical requirements.
- for power disconnection after installation, the male end of each AC power cable must remain accessible at the socket-outlet connection to the power source.
- the power and fan modules have been removed from the Waveserver 5 chassis. See [Procedure 3, “Unpacking Waveserver 5 equipment”](#) for details.
- all tools and bracket kit parts required for installation are available, see [“Required materials for installation, powering, and removal”](#) on page 27.
- 6 AWG (16 mm²) wire for the chassis ground.
- dust covers are installed on all Waveserver 5 optical ports.
- you have the following bracket kit available for installation:
 - Data Center 23-in., 24 - 36-in. deep Bracket Kit (186-1307-900), see [“Contents of Waveserver Data Center Bracket Kit, 23-in wide, 24 - 36-in deep \(186-1307-900\)”](#) on page 37.

Note: Ciena recommends installing the brackets in the flush mount position to maximize the amount of clearance at the front of the frame and prevent accidental damage to the fibers. A minimum of 3.1-in. clearance is required at the front of the frame uprights for fiber exit.

Precautions



DANGER

Risk of laser radiation exposure

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.

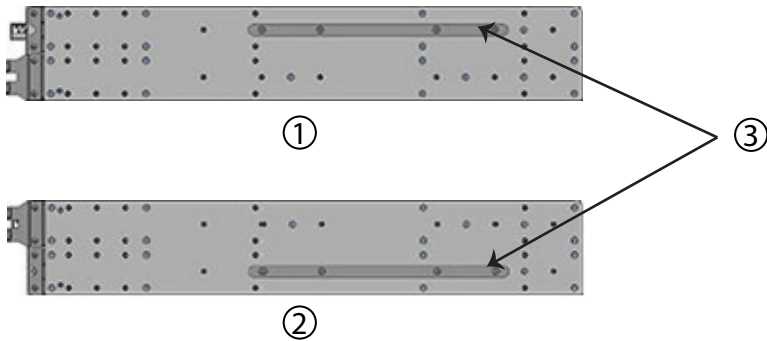


CAUTION
Risk of damage to the chassis, modules, and pluggable transceivers
Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and modules or transceivers.

Steps

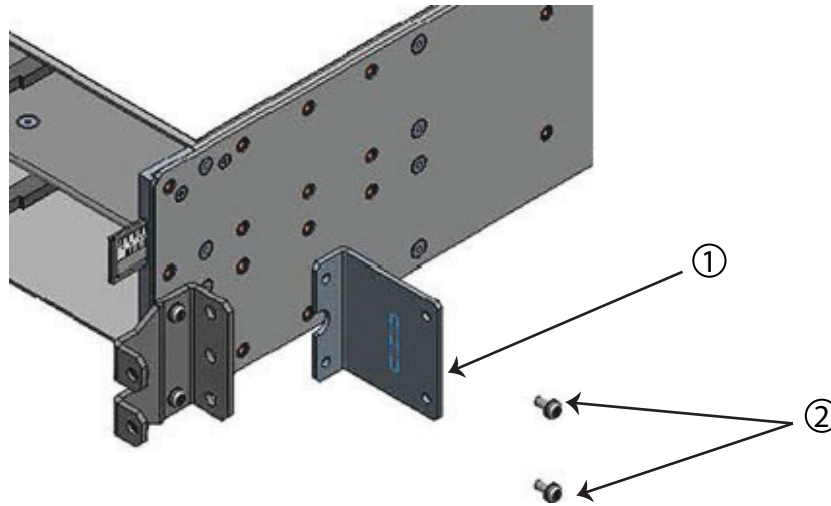
- 1 Waveserver 5 has two identical sets of mounting holes that are the mirror image of one another, for upper or lower mounting positions. You can use either the upper or lower mounting points, however Ciena recommends using the upper row for chassis installation and the lower row for the installation of other accessories.
- Select your next step
- | If you want to mount Waveserver 5 in | Then go to |
|--------------------------------------|------------------------|
| the upper mounting location | step 2 |
| the lower mounting location | step 4 |
- 2 Install the T-Bars (42T-2970-001) on each side of the Waveserver 5 chassis in the upper mounting location.
- Position a T-Bar against the side of the chassis so that the pilot holes on the T-Bar align with the upper row of pilot holes on the chassis, and secure the T-Bar using four flathead screws (312-1030-M05) on each side.

Figure 55 T-Bar installation



Legend

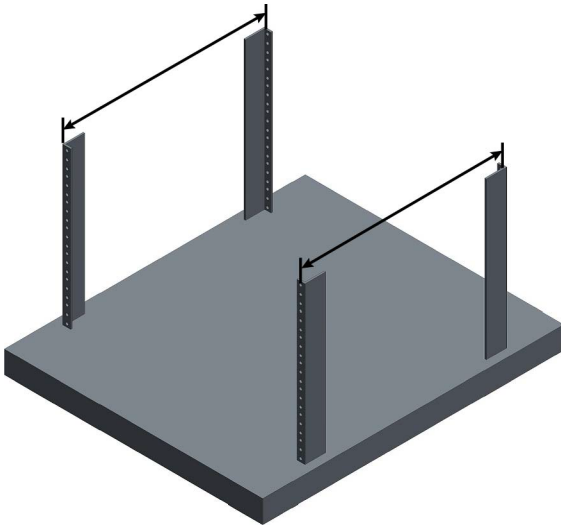
- | | | | |
|---|-----------------------------------|---|----------------------|
| 1 | T-Bar installed in upper position | 3 | T-Bar (42T-2970-001) |
| 2 | T-Bar installed in lower position | | |
- 3 Fasten the two front mounting brackets (410-7588-001) to the front of the chassis using two pan head external lock washers (312-0630-M06) per bracket. Position the brackets so that the large tab is facing the back of the chassis.

Figure 56 Frame-mount bracket installation**Legend**

- | | | | |
|---|---|---|-------------------------------------|
| 1 | Front frame mounting bracket (410-7588-001) | 2 | Pan head SEMS screws (312-0630-M06) |
|---|---|---|-------------------------------------|

Go to [step 8](#).

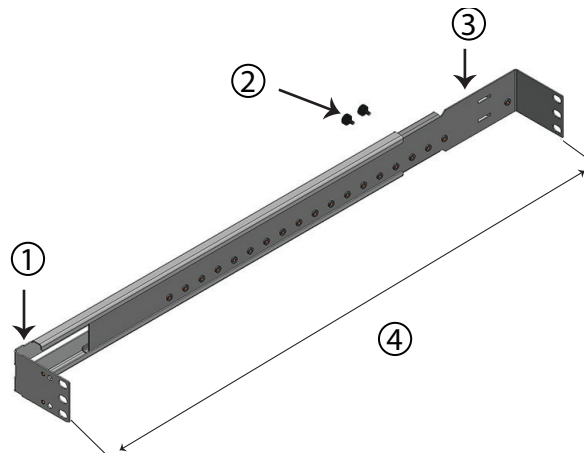
- 4 Install the T-Bars (42T-2970-001) on each side of the Waveserver 5 chassis in the lower mounting location.
Position a T-Bar against the side of the chassis so that the pilot holes on the T-Bar align with the lower row of pilot holes on the chassis, and secure the T-Bar using four flathead screws (312-1030-M05) on each side. See [Figure 56 on page 98](#) for details.
- 5 Fasten the fiber manager utility bracket to the upper mounting location on the chassis.
- 6 Rotate the fiber manager 180 degrees and assemble the utility bracket using the flat head screws provided in the bag with the fiber managers. Repeat on the other side of the chassis.
- 7 Fasten the two front mounting brackets (410-7588-001) to the front of the chassis using two pan head external lock washers (312-0630-M06) per bracket. Position the brackets so that the large tab is facing the back of the chassis. See [Figure 57 on page 99](#) for details.
- 8 Measure the rack upright mounting surface spacing distance (front to back) on both sides of the equipment frame.

Figure 57 Measured frame upright depth

9 Assemble the front and rear bracket guides.

Insert the end of a rear bracket guide (410-8194-001) into the end of a front bracket guide (410-8193-001). Adjust the length so that the distance between the front flange and rear flange is approximately the same as the distance measured in [step 8](#) (24-in - 36-in depth).

Loosely fasten the rails together using two pan head external lock washers (312-0630-M06) on each rail assembly. For shorter rack depths, position the screws as far apart as possible.

Figure 58 Front and rear rail assembly — 23-in bracket kit

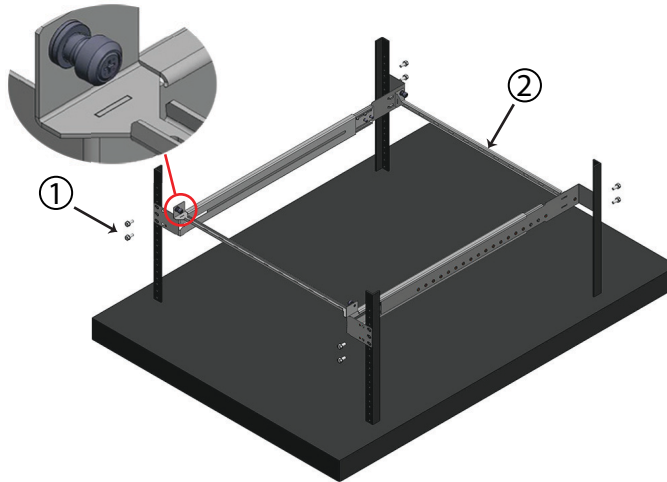
Legend

- | | | | |
|---|---|---|-----------------------------------|
| 1 | Front Bracket Guide (410-8193-001) | 3 | Rear Bracket Guide (410-8194-001) |
| 2 | Pan Head External Lock Washers (312-0630-M06) | 4 | Set to frame upright depth |

- 10** Install and secure the rail assemblies to the frame uprights. Use the rear cable management bar (410-8191-001) to stabilize the uprights and ensure that they remain square.

Use the rear cable management bar to ensure spacing for the front and back brackets is correct. Position the rear cable management bar (410-8191-001) at the back of the frame. Use the keying points provided in the uprights to ensure that the spacing is correct. You can also use the spacer (410-7782-001) to set the front brackets by sliding it into the groove for shelf mounting.

Figure 59 Rail assemblies installed on frame uprights

**Legend**

- | | | | |
|---|--|---|--|
| 1 | Rack manufacturer's recommended hardware | 2 | Rear Cable Management Bar (410-8191-001) |
|---|--|---|--|

- 11** Loosely install the rail assemblies to the uprights using the rack manufacturer's recommended hardware. Use one fastener in the middle pilot hole of the front brackets, and two fasteners in the top and bottom pilot holes of the rear brackets. Tighten the front rail screws on the front and rear rail assemblies so that the rails are firmly retained while still allowing for some final adjustments.

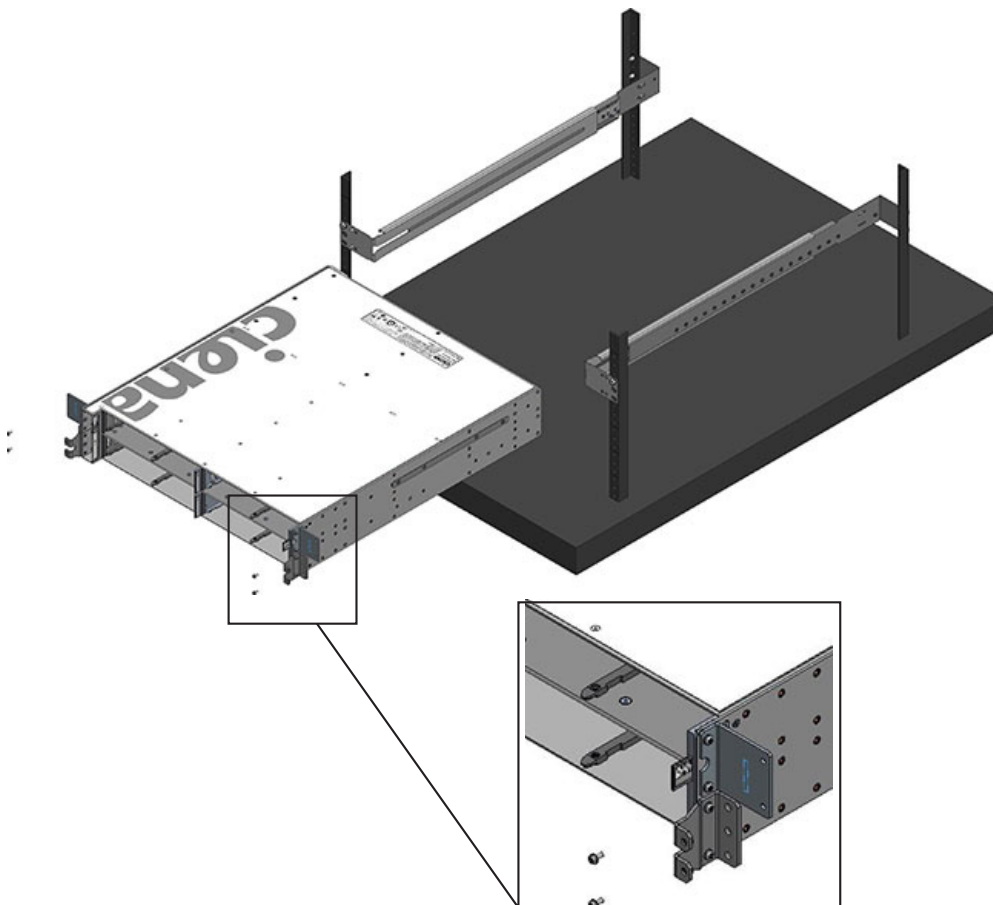
Note: Some final minor adjustments of the rail assemblies may be required before you install Waveserver 5 in the frame.

- 12** Install the Waveserver 5 chassis in the equipment frame.

Holding the chassis securely at the front of the frame, align the T-Bars with the pilot slot on each front rail assembly. Carefully slide the chassis into position in the frame until the frame mounting bracket at each side of the chassis comes into contact with the bracket on the front rail.

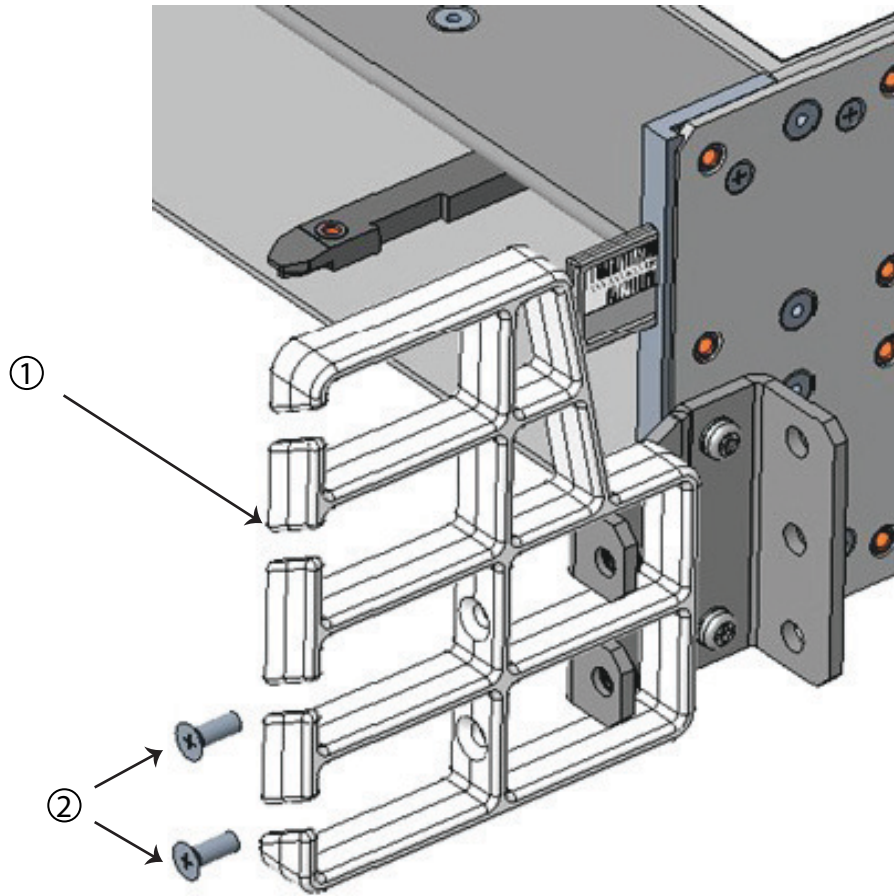
- 13 Secure each frame mounting bracket to the frame using two pan head external lock washers (312-0630-M06) inserted in the top and bottom pilot holes of the front frame mounting bracket (see the following figure). Fully tighten the middle screw used to position the rails in [step 10](#).

Figure 60 Waveserver 5 chassis positioned for installation



- 14 Install the fiber managers on the chassis, if required.
Assemble the utility bracket on the lower mounting location. Rotate the fiber manager 180 degrees and assemble the fiber manager to the bracket using the two flat head screws (312-1030-M08) that are shipped in the plastic bag with the fiber managers. Torque to 3 lb-in (.339 N-m).

Figure 61 Fiber manager installation



Legend

1 Fiber manager

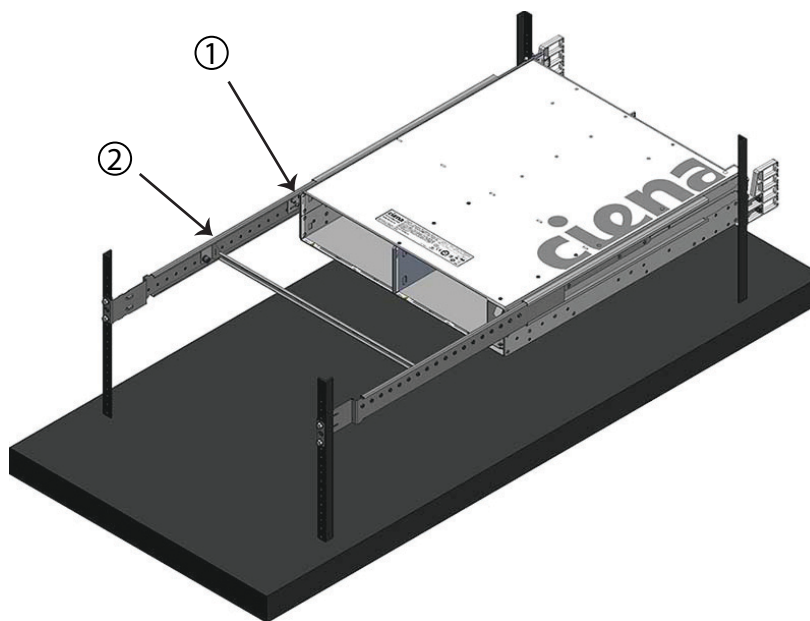
2 Flat head screws

You have completed this procedure. Next, perform [Procedure 8, “Installing the power and fan modules and connecting power cables to Waveserver 5”](#).

- 15 Install the rear cable management bar.

Note: The cable management bar cannot be installed in equipment frames with depths of 26-in. or less. Lances located on the rear bracket are provided for cable management.

With the cable-management bar tab-side up in the frame, insert the tab at each end of the guide on the front rail. Tighten the thumb screw on each side of the rear cable management bar to secure it to the rail assemblies. Do not over-tighten the thumb screws.

Figure 62 Rear cable-management bar installation**Legend**

- | | | | |
|---|---------------------------------------|---|---------------------------|
| 1 | 10-32 studs for ground lug attachment | 2 | Rear cable management bar |
|---|---------------------------------------|---|---------------------------|

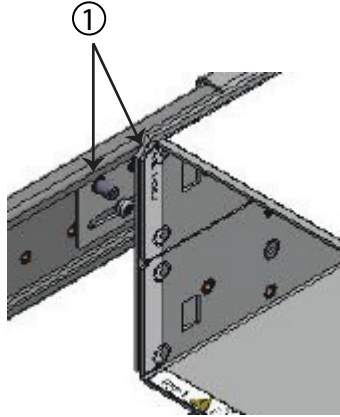
Note: A chassis ground is required for all DC installations and AC installations requiring GR-1089-CORE compliance. For AC installations that do not require GR-1089-CORE compliance, the chassis ground is recommended but not required.

16 Install the chassis ground.

Crimp the ground wire into a single-hole or two-hole ground lug. Fasten the lug to the 10-32 studs provided at the rear of the frame using Keps nuts (N0041806) and torque to 29 lb-in (3.3 N-m). See the following figure for the location of the 10-32 studs.

Route the ground wire from the studs and fasten to an equipment frame ground location.

Figure 63 Chassis ground studs



Legend

- 1 10-32 studs for ground lug attachment

Procedure 8 Installing the power and fan modules and connecting power cables to Waveserver 5

Use this procedure to install power and fan modules in the Waveserver 5 chassis and then connect power cables.

Requirements

Make sure that

- the Waveserver 5 chassis is installed and grounded in the equipment frame.
- for power disconnection after installation, the male end of each AC power cable must remain accessible at the socket-outlet connection to the power source.
- you have the following tools and materials, as required for your installation:
 - DC or AC power and fan modules, two for each shelf
 - a 7-mm socket
 - insulating sheet fiber
 - non-oxidizing compound
 - heat-shrink tubing and heat gun for attaching ring lugs (DC power modules)



DANGER

Risk of electrocution

Working with live circuits is dangerous. Before you continue, make sure that the Waveserver 5 power is turned off at the power distribution panel and cannot be turned on by accident. Use a multimeter to verify that no potential exists.

- the appropriate AC or DC power cables, lugs, and crimp tool. AC power modules are equipped with a unique Saf-D-Grid connector and are compatible with the following AC power cables, which must be ordered from Ciena:
 - AC Power Cord, Saf-D-Grid to C20, 12AWG, 250V, 2M (NTK955GA)
 - AC Power Cord, Saf-D-Grid to Open, 12AWG, 250V, 3M (NTK955GB)
 - AC Power Cord, Saf-D-Grid to C20, 12AWG, 250V, 2M, white (NTK955GC)
 - AC Power Cord, Saf-D-Grid to C14, 14 AWG, 250V, 2m, black (NTK955GD)

- AC Power Cord, Saf-D-Grid to CEE7/7, 12 AWG, 250V, 2m, black (NTK955GE)
- AC Power Cord, Saf-D-Grid, left routed to C20, 12 AWG, 250V, 2m, black (NTK955GF)
- AC Power Cord, Saf-D-Grid, right routed to C20, 12 AWG, 250V, 2m, black (NTK955GG)
- AC Power Cord, Saf-D-Grid, left routed to C20, 12 AWG, 250V, 2m, white (NTK955GH)
- AC Power Cord, Saf-D-Grid, right routed to C20, 12 AWG, 250V, 2m, white (NTK955GJ)

For more information about supported power cables and lugs, refer to the “Ordering” section of the *Waveserver 5 User Guide*, 323-3001-100. For more information about the supported AC power cables, see the “Power and grounding specifications” section of the *Waveserver 5 User Guide*, 323-3001-100.

- For DC installations, Ciena recommends using the lugs listed in [“Recommended lugs for DC Waveserver 5 installations” on page 19](#) to connect the DC power module to the DC power plant.

Note: You must use 90-degree power lugs to prevent cable interference with equipment installed directly below Waveserver 5.

Make sure that you:

- observe all the safety requirements described in [“Observing product and personnel safety guidelines” on page 3](#).
- have the EDP, IDP, or equivalent site/network engineering plans
- have a secure ladder for installations above shoulder level
- have the appropriate personal grounding device to dissipate electrostatic charges

ATTENTION

All AC inputs to an AC-powered shelf or to any rectifier supplying DC power to a shelf must be equipped with an external surge protection device (SPD not supplied) to meet the Telcordia criteria for lightning criteria for equipment interfacing with AC power port(s).

ATTENTION

The DC power module return lead, L+, must be earthed (connected to ground) at some point within the building consistent with applicable power infrastructure (DC-C or DC-I) to satisfy safety requirements.

ATTENTION

Refer to the “Power and grounding specifications” section of the *Waveserver 5 User Guide*, 323-3001-100 for information about the selection of the appropriate power circuit protection.

Overview

Power and fan modules are mated, with two fan modules installed in the power module housing. Individual fan modules are field-replaceable. When replacing a power module, you must replace the entire assembly (power and fan modules). Keep a power and fan module (AC or DC, as appropriate) in your inventory to be installed immediately in the event of a power module failure.

Two variants of the power and fan module are available:

- Waveserver 5 100-277Vac/240Vdc power and fan module (186-3040-900)
- Waveserver 5 -48Vdc power and fan module (186-3041-900)

Figure 64 100-277VAC/240Vdc power and fan module (186-3040-900)

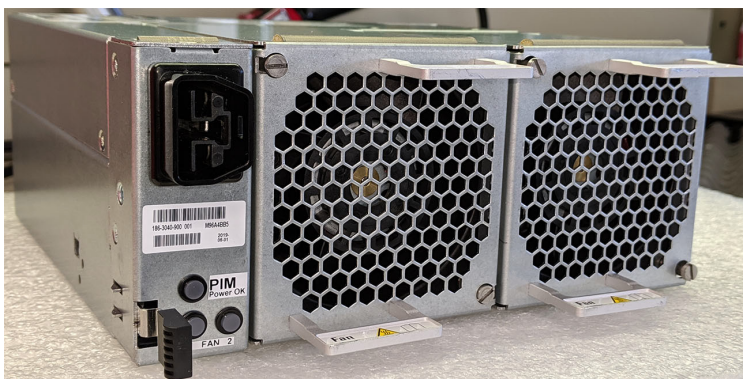
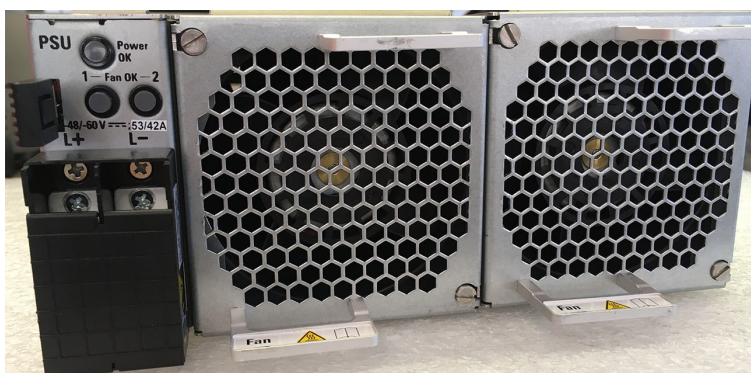


Figure 65 -48Vdc power and fan module (186-3041-900)



The power and fan modules are installed at the rear of the chassis. The following figures show the location of the power and fan modules at the rear of the chassis.

Figure 66 Example of Waveserver 5 with AC power and fan modules installed—rear view



Figure 67 Example of Waveserver 5 with DC power and fan modules installed—rear view



See *Waveserver 5 User Guide*, 323-3001-100 for module specifications, supported power cables, and ordering information.

Precautions



DANGER

Risk of electrical shock and equipment damage

Ground the rack to the common building network (CBN), isolated bonding network (IBN), or ETSI Mesh Bonding Network ground/protective earth. For details, see [“Grounding requirements” on page 15](#) and [Procedure 2, “Connecting the rack ground to the office ground”](#).

Correct grounding of the equipment rack is mandatory in accordance to electrical safety standards and mitigates the electrostatic discharge (ESD) and operational concerns described in [“Observing product and personnel safety guidelines” on page 3](#), the section on consequences of incorrect grounding. For additional details, see [“Working with power”](#) section.

Also, ground every shelf and peripheral tray as described in the related procedure.



DANGER

Risk of electrical shock

When working with the 100-277 VAC power module, disconnect from the power source before extracting the module, or partially extract the module and allow at least two seconds for hazardous voltage to dissipate before removing completely.

Removing a fan module from a 100-277 VAC power module exposes hazardous voltages in the empty slot of the module. Replacing a fan module must only be performed by skilled personnel familiar with the safety standards of working with live power.

Do not insert hands, fingers, or tools into a live power module. Power modules are safe to handle two seconds after the power has been disconnected or the power module has been partially extracted.

**DANGER****Risque de décharge électrique**

Lorsque vous travaillez avec le module d'alimentation 100-277 VAC, déconnectez-le de la source d'alimentation avant de l'extraire, ou extrayez-le partiellement et attendez deux secondes pour que la tension dangereuse se dissipe avant de le retirer complètement.

Le retrait d'un module de ventilation d'un module d'alimentation 100-277 VAC expose des tensions dangereuses dans l'emplacement vide du module. Le remplacement d'un module de ventilation ne doit être effectué que par du personnel qualifié connaissant les normes de sécurité du travail sous tension.

N'insérez pas les mains, les doigts ou les outils dans un module d'alimentation sous tension. Vous pouvez manipuler un module d'alimentation de façon sécuritaire deux secondes après sa mise hors tension ou son extraction partielle.

**DANGER****Risk of electrical shock**

Removing a fan module from a -48 Vdc power and fan module exposes 60V of live power in the empty slot of the PSU. Replacing a fan module must only be performed by qualified personnel familiar with the safety standards of working with live power.

**CAUTION****Risk of equipment damage**

Electrostatic discharge (ESD) can damage electrostatic sensitive devices. Use anti-static protection to avoid damaging modules.

**DANGER****Risk of electrocution**

Working with live circuits is dangerous. Before you continue, make sure that the shelf power is turned off at the power distribution panel and cannot be turned on by accident. Use a multimeter to verify that no potential exists.

Steps

- 1 Verify that the equipment rack is grounded. Also, you must ground every shelf (and any peripheral trays as described in the applicable installation procedure). Grounding is mandatory.
Follow the instructions in the precautionary messages listed above.

- 2 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an anti-static wrist strap, connect the cord to the ESD jack on the shelf installed and grounded in a grounded rack or clip to a suitable ground point.

Making sure that power to the system is off

- 3 Make sure that power is off at the secondary power distribution panel (any upstream protector).
- 4 Select your next step.

If you are installing	Then go to
DC power and fan modules	step 5
AC power and fan modules	step 17

Installing the DC power and fan modules

- 5 Slide the module latch to the right and insert the power and fan module in its assigned slot.
- 6 Push the module into the slot by its handles or faceplate until it fully engages in the backplane, indicated by an audible click.
- 7 Repeat [step 5](#) and [step 6](#) for the second DC power and fan module, if required.

Installing the DC power cables

- 8 Prepare the appropriate DC power cables with the appropriate 90-degree lugs based on industry standards according to the EDP, IDP, or equivalent site/network engineering plans.
- 9 Remove the plastic safety covers from the terminals.
- 10 Check that the nut/washer assemblies are not installed on the terminals.
- 11 Lightly coat with non-oxidizing compound as required by your company's practices and connect the prepared DC power cables as applicable (based on the configuration) to the appropriate terminals on the power modules.
- 12 Reinstall the nut/washer assembly on each of the DC power cables. Verify that the torque is 20 lb-in. (2.26 N-m) using a torque driver. If required, lightly coat non-oxidizing compound on the terminals and the two-hole lugs.
- 13 After you attach the cables, align the holes on the plastic cover with the threaded studs (diagonal) of the power terminals and push the safety cover until it secures in place.
- 14 Route and secure the DC power cables as follows:
 - From the left module (facing the chassis rear), route the cable out and to the left. Secure the cables to the cable management bar.
 - From the right module (facing the chassis rear), route the cable out and to the right. Secure the cable to the cable management bar.

**DANGER****Risk of equipment damage and electrical shock**

When routing power cables over metallic edges, wrap and secure an insulating sheet fiber around the power cables to protect them against damage and to prevent personal injury from electrical shock.

15 Replace the plastic cover on the terminal block.

16 Connect the DC power cable to a power source.

If you are installing DC power and fan modules, you have completed this procedure. Next perform [Procedure 9, “Testing power to Waveserver 5”](#).

Installing the AC power and fan modules

17 Slide the module latch to the right and insert the power and fan module in its assigned slot.

18 Push the module into the slot by its handles or faceplate until it fully engages in the backplane, indicated by an audible click.

19 Repeat [step 17](#) and [step 18](#) for the second AC power and fan module, if required.

Installing the AC power cables

20 Connect the AC power cables to the power modules. Route and secure the power cables as follows:

- From the left module (facing the chassis rear), route the cable out and to the left. Secure the cables to the cable management bar.
- From the right module (facing the chassis rear), route the cable out and to the right. Secure the cable to the cable management bar.

**DANGER****Risk of equipment damage and electrical shock**

When routing power cables over metallic edges, wrap and secure an insulating sheet fiber around the power cables to protect them against damage and to prevent personal injury from electrical shock.

21 Connect the AC power cables to a power source.

You have completed this procedure. Next perform [Procedure 9, “Testing power to Waveserver 5”](#).

Procedure 9 Testing power to Waveserver 5

Use this procedure to test power to the Waveserver 5. This procedure instructs you to first test power to each power module individually and then with both power modules together. You then test power by removing each fan module to make sure that the remaining fan is operational and the shelf is still powered.

Requirements

Make sure that you

- have a digital voltmeter (DVM)



DANGER

Risk of electrocution

Working with live circuits is dangerous. Before you continue, make sure that the shelf power is turned off at the power distribution panel and cannot be turned on by accident. Use a multimeter to verify that no potential exists.

- observe all the safety requirements described in [“Observing product and personnel safety guidelines” on page 3](#)
- have a flathead screwdriver
- have the appropriate personal grounding device to dissipate electrostatic charges

Precautions



DANGER

Risk of electrical shock, energy hazard

Removing a fan module from the AC and/or DC power and fan module exposes 500A of live power in the empty slot of the power module. Replacing a fan module must only be performed by qualified personnel familiar with the safety standards of working with live power.



DANGER

Risk of electrical shock

Removing a fan module from a -48Vdc power and fan module exposes 60V of live power in the empty slot of the module. Replacing a fan module must only be performed by qualified personnel familiar with the safety standards of working with live power.

**DANGER****Risk of electrical shock**

When working with the 100-277 VAC power module, disconnect from the power source before extracting the module, or partially extract the module and allow at least two seconds for hazardous voltage to dissipate before removing completely.

Removing a fan module from a 100-277 VAC power module exposes hazardous voltages in the empty slot of the module. Replacing a fan module must only be performed by skilled personnel familiar with the safety standards of working with live power.

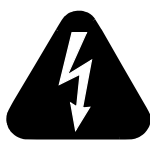
Do not insert hands, fingers, or tools into a live power module. Power modules are safe to handle two seconds after the power has been disconnected or the power module has been partially extracted.

**DANGER****Risque de décharge électrique**

Lorsque vous travaillez avec le module d'alimentation 100-277 VAC, déconnectez-le de la source d'alimentation avant de l'extraire, ou extrayez-le partiellement et attendez deux secondes pour que la tension dangereuse se dissipe avant de le retirer complètement.

Le retrait d'un module de ventilation d'un module d'alimentation 100-277 VAC expose des tensions dangereuses dans l'emplacement vide du module. Le remplacement d'un module de ventilation ne doit être effectué que par du personnel qualifié connaissant les normes de sécurité du travail sous tension.

N'insérez pas les mains, les doigts ou les outils dans un module d'alimentation sous tension. Vous pouvez manipuler un module d'alimentation de façon sécuritaire deux secondes après sa mise hors tension ou son extraction partielle.

**DANGER****Risk of electrical shock and equipment damage**

Ground the rack to the common building network (CBN), isolated bonding network (IBN), or ETSI Mesh Bonding Network ground/protective earth. For details, see [“Grounding requirements” on page 15](#) and [Procedure 2, “Connecting the rack ground to the office ground”](#).

Correct grounding of the equipment rack is mandatory in accordance to electrical safety standards and mitigates the electrostatic discharge (ESD) and operational concerns described in [“Observing product and personnel safety guidelines” on page 3](#), the section on consequences of incorrect grounding. For additional details, see the section on working with power.

Also, ground every shelf and peripheral tray as described in the related procedure.

**CAUTION****Risk of equipment damage**

Electrostatic discharge (ESD) can damage electrostatic sensitive devices. Use anti-static protection to avoid damaging modules.

**DANGER****Risk of electrical shock**

The battery feeds can be at a high enough potential to constitute a shock hazard. Read and understand the power procedures you are performing. Take necessary precautions and use the appropriate insulated tools when working with power. For additional details, see [“Working with power” on page 14](#).

**CAUTION****Hazardous moving parts**

Fan modules contain hazardous moving parts. Keep fingers and other body parts away.

Steps

- 1 Verify that the equipment rack is grounded. Also, you must ground every shelf (and any peripheral trays as described in the applicable installation procedure). Grounding is mandatory.
Follow the instructions in the precautionary messages listed above.

- 2 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an anti-static wrist strap, connect the cord to the ESD jack on the shelf installed and grounded in a grounded rack or clip to a suitable ground point.

- 3 Select your next step:

If the shelf is equipped with	Then go to
-------------------------------	------------

AC power and fan module(s)	step 4
----------------------------	------------------------

DC power and fan module(s)	step 23
----------------------------	-------------------------

Testing power to the shelf equipped with AC power modules

Note: You must test power to each power module individually, and then test both power modules at the same time.

- 4 Connect the AC power cable to the AC power module you are testing.
- 5 Connect the other end of the AC power cable to a live AC power outlet.
- 6 Check that the “Power OK” LED on the power module turns on.

The power modules are equipped with a “Power OK” LED that indicates when the AC power module is operating normally.

If the “Power OK” LED is off, possible causes are: no power at the AC power outlet, a faulty AC power cable, or a faulty AC power module.

Additional “Power OK” LED scenarios are as follows:

- Flashing green: indicates a missing or improperly seated fan module.
- Flashing red: warning of a potential fault in the AC power and fan module.
- Solid red: indicates a faulty AC power module.

- 7 Make sure to immediately disconnect the AC power cable from the AC power outlet.

- 8 With a single power module running, remove one of the fan modules by removing the two screws on the fan module faceplate using a flat head screwdriver.

- 9 Place the fan module on a flat, stable surface.

- 10 Ensure that the fan module corresponding power module turns off and the shelf is unpowered. The fan module “Fan OK” LED on the power module is off and the power module “Power OK” LED is flashing green, to indicate that the fan module is missing and the power module is turned off.

Additional “Fan OK” LED scenarios are as follows:

- Off: indicates that there is a missing fan module or there is no power to the fan module.
- Flashing red: indicates that the fan module is running above the expected speed.
- Solid red: indicates that the fan module is running below the expected speed.
- Solid green: indicates that the fan is operating at the expected speed.

- | | If | Then |
|----|--|---|
| 11 | the power module does not turn off after the fan module is removed (the shelf remains powered) | return the entire power and fan module to Ciena |
| | the power module turns off after the fan module is removed | proceed to step 12 |
| 12 | Re-install the fan module by sliding the module into the power module housing and then fastening the two screws on the fan module faceplate with a flat head screwdriver. | |
| 13 | Repeat step 8 to step 12 for the second fan module. | |
| 14 | Repeat step 8 to step 13 for the second feed/AC power module. After you have completed testing the AC power and fan modules individually, proceed to the next step. | |
| 15 | Connect AC power cables to both AC power modules. | |
| 16 | Connect the AC power cables to a live AC power outlet. | |
| 17 | Check that the “Power OK” LEDs on the power modules turn on.
The power modules are equipped with a “Power OK” LED that indicates when the AC power module is operating normally.
If the “Power OK” LED is off, possible causes are: no power at the AC power outlet, a faulty AC power cable, or a faulty AC power module. | |
| 18 | With both power modules running, remove one of the fan modules by removing the two screws on the fan module faceplate using a Phillips screwdriver. | |
| 19 | Place the fan module on a flat, stable surface. | |
| 20 | Ensure that the fan module corresponding power module turns off but the shelf remains powered. The fan module corresponding “Fan OK” LED on the power module is off and the power module “Power OK” LED flashes green, to indicate that the fan module is missing and the power module is turned off. | |
| 21 | Re-install the fan module by sliding the module into the power module housing and then fastening the two screws on the fan module faceplate with a flathead screwdriver. | |
| 22 | Repeat step 18 to step 21 for each of the three remaining fan modules. | |

Testing power to the shelf equipped with DC power modules

Note: You must test power to each power module individually, and then test both power modules at the same time.

- 23 Turn on the power source for the DC power module you are testing.
- 24 Check that the “Power OK” LED on the power module turns on.
The LED indicates that the feed is live (has a minimum voltage applied).
If the LED does not turn on, possible causes are ground short-circuit or reverse polarity. You must power down the system, then remove the power cable before you troubleshoot further.

Additional “Power OK” LED scenarios are as follows:

- Flashing green: indicates a missing or improperly seated fan module.
- Flashing red: indicates a warning of a potential fault in the DC power and fan module.
- Solid red: indicates a faulty DC power module.

25 With a single power module running, remove one of the fan modules by removing the two screws on the fan module faceplate using a flat head screwdriver.

26 Place the fan module on a flat, stable surface.

27 Ensure that the fan module corresponding power module turns off and the shelf is unpowered. The fan module “Fan OK” LED on the power module is off and the power module “Power OK” LED is flashing green, to indicate that the fan module is missing and the power module is turned off.

Additional “Fan OK” LED scenarios are as follows:

- Off: indicates that there is a missing fan module or there is no power to the fan module.
- Flashing red: indicates that the fan module is running at above the expected speed.
- Solid red: indicates that the fan module is running at below the expected speed.
- Solid green: indicates that the fan is operating at the expected speed.

28	If	Then
	the power module does not turn off after the fan module is removed (the shelf remains powered)	return the entire power and fan module to Ciena
	the power module turns off after the fan module is removed	proceed to step 29
29	Re-install the fan module by sliding the module into the power module housing and then fastening the two screws on the fan module faceplate with a flat head screwdriver.	
30	Repeat step 25 to step 29 for the second fan module.	
31	Repeat step 25 to step 30 for the second feed/DC power module. After you have completed testing the DC power and fan modules individually, proceed to the next step.	
32	Turn on the power source for both power modules.	
33	Check that the “Power OK” LEDs on the power modules turn on. The LED indicates that the feed is live (has a minimum voltage applied).	
34	With both power modules running, remove one of the fan modules by removing the two screws on the fan module faceplate using a Phillips screwdriver.	
35	Place the fan module on a flat, stable surface.	

- 36** Ensure that the fan module corresponding power module turns off but the shelf remains powered. The fan module corresponding “Fan OK” LED on the power module lights red, to indicate that the fan module is missing.

Additional “Fan OK” LED scenarios are as follows:

- Off: indicates that there is a missing fan module or there is no power to the fan module.
- Flashing red: the fan module is running above the expected speed.
- Solid red: the fan module is running below the expected speed.
- Solid green: indicates that the fan is operating at the expected speed.

If the power module does not turn off after the fan module is removed (the shelf remains powered), return the entire power and fan module to Ciena.

- 37** Re-install the fan module by sliding the module into the power module housing and then fastening the two screws on the fan module faceplate with a Phillips screwdriver.

- 38** Repeat [step 34](#) to [step 37](#) for the three remaining fan modules. When you have tested both fan modules, you have completed this procedure.

Next, perform [Procedure 10, “Installing the access panel”](#).

Procedure 10 Installing the access panel

Use this procedure to install the Waveserver 5 access panel (AP) (186-3020-900) in the chassis.

Requirements

Make sure that you

- have a secure ladder for installations above shoulder level.
- observe all the safety requirements described in [“Observing product and personnel safety guidelines” on page 3](#).
- have the appropriate personal grounding device to dissipate electrostatic charges.

Note 1: DCN and ILAN ports on the AP support Category 5 enhanced Unshielded Twisted Pair (Cat 5e UTP) Ethernet cables.

Note 2: DCN4 and DCN5 ports are not supported in the current release.

Overview

For AP technical specifications, including ports, LEDs, and alarm details, see the *Waveserver 5 User Guide*, 323-3001-100

The following figure shows the location of the access panel at the rear of the chassis.

Figure 68 Example of Waveserver 5 chassis—rear view, equipped



Precautions



DANGER

Risk of electrical shock and equipment damage

Grounding is mandatory to satisfy local electrical codes/regulations for the safe use of the equipment.

Ground the rack to the common building network (CBN), isolated bonding network (IBN), or ETSI Mesh Bonding Network ground/protective earth. For details, see [“Grounding standards” on page 12](#) and [Procedure 2, “Connecting the rack ground to the office ground”](#).

Correct grounding of the equipment rack is mandatory in accordance to electrical safety standards and mitigates the electrostatic discharge (ESD) and operational concerns described in [“Consequences of incorrect grounding” on page 17](#). For additional details, see [“Working with power” on page 14](#).

Also, ground every shelf and peripheral tray as described in the related procedure.



CAUTION

Risk of equipment damage

Electrostatic discharge (ESD) can damage electrostatic sensitive devices. Use anti-static protection to avoid damaging modules.



CAUTION

Risk of equipment damage

Make sure that your hands or tools do not come into contact with the circuitry of the access panel (AP) or backplane pins. Avoid contact until the shelf is grounded to the rack.



CAUTION

Risk of equipment damage

The shelf is suitable for connection to intrabuilding or unexposed wiring or cabling only.

Steps

- 1 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an anti-static wrist strap, connect the cord to the ESD jack on the shelf installed and grounded in a grounded rack/cabinet or clip to a suitable ground point.

- 2** Select your first step.

If the Waveserver 5	Then go to
is not a packs-in-place kit	step 3
is a packs-in-place kit	step 4

- 3** Position the access panel in the track of the slot at the rear of the chassis and slide it into the slot.

- 4** Push firmly on the access panel faceplate until you hear a click, which indicates that the AP is fully engaged in the backplane.

You have completed this procedure.

Next, perform [Procedure 11, "Installing the control processor"](#).

Procedure 11 Installing the control processor

Use this procedure to install the Waveserver 5 control processor (CP) (186-3010-900).

Requirements

Make sure that

- the system is powered down.
- you have a secure ladder for installations above shoulder level.
- you observe all the safety requirements described in [“Observing product and personnel safety guidelines” on page 3](#).
- you have the appropriate personal grounding device to dissipate electrostatic charges.
- you have the EDP, IDP, or equivalent site/network engineering plans for the slots to use for your application.

Note: DCN ports on the CP support Category 5 enhanced Unshielded Twisted Pair (Cat 5e UTP) Ethernet cables.

Overview

The CP is located at the front of the chassis (see the following figure).

Figure 69 Example of Waveserver 5 chassis—front view, equipped



Legend

- 1 Control Processor

For CP technical specifications, including ports, LEDs, and alarm details, see the *Waveserver 5 User Guide*, 323-3001-100.

Precautions



DANGER

Risk of laser radiation exposure

Do not look into the ends of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



CAUTION

Risk of equipment damage

Electrostatic discharge (ESD) can damage electrostatic sensitive devices. Use anti-static protection to avoid damaging modules.



CAUTION

Risk of module damage

Handle modules with care at all times. Make sure that you do not drop any module. Otherwise, you will damage it. Also, do not remove the connector protectors until you are ready to install the module.



CAUTION

Risk of module damage

Do not force a module all the way to the back of its slot if it resists insertion. Before installing a module, make sure that you understand the detailed procedure for inserting modules.

Steps

- 1 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an anti-static wrist strap, connect the cord to the ESD jack on the shelf installed and grounded in a grounded rack/cabinet or clip to a suitable ground point.
- 2 Select your first step.

If the CP	Then go to
is not in a packs-in-place kit	step 3
is in a packs-in-place kit	step 4
- 3 Position the control processor in the track of the slot at the front of the chassis and slide it into the slot.
- 4 Push firmly on the CP faceplate until you hear a clicking noise, which indicates that the CP is fully engaged in the backplane.
You have completed this procedure.

Next, perform [Procedure 12, “Installing or removing a filler module”](#) and/or [Procedure 13, “Installing or removing Service Modules \(SMs\)”](#).

Procedure 12 Installing or removing a filler module

Use this procedure to install or remove a filler module.

Requirements

Make sure that you

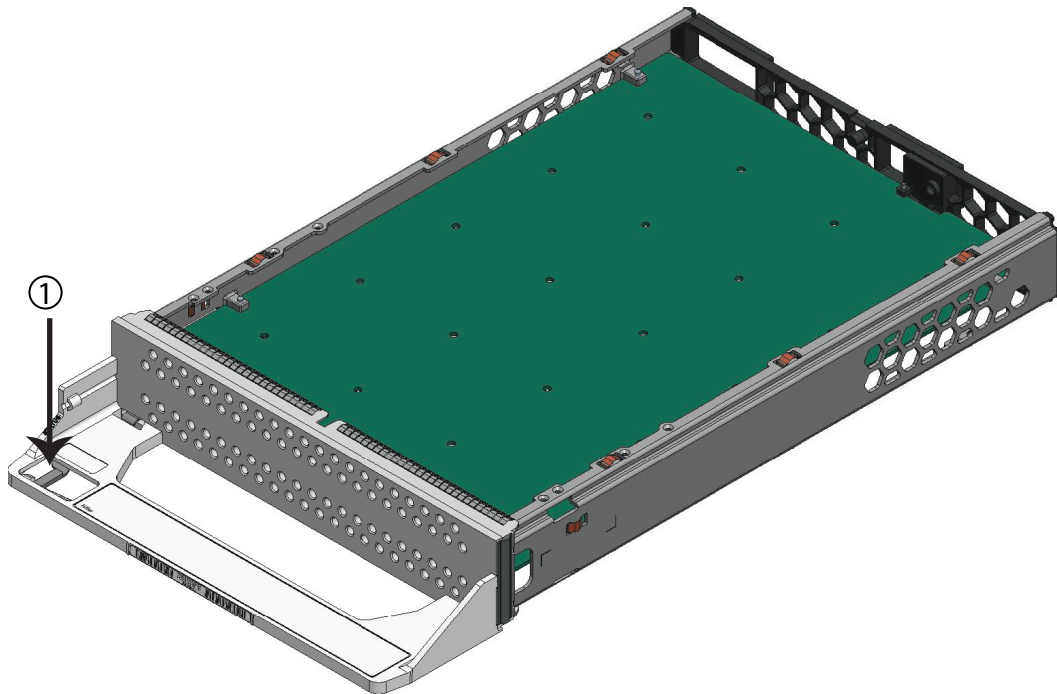
- have a secure ladder for installations above shoulder level
- observe all of the safety requirements described in [“Observing product and personnel safety guidelines” on page 3](#)
- have the appropriate personal grounding device to dissipate electrostatic charges
- have the EDP, IDP, or equivalent site/network engineering plans with the information on which slots should be equipped with filler modules

Overview

If your configuration includes empty module slots, you must install filler modules in the respective slots. All chassis slots must be filled to ensure adequate cooling of the Waveserver 5 components. Use this procedure to install or remove a filler module (186-3031-900).

The following figure provides an example of a filler module.

Figure 70 Waveserver 5 filler module (186-3031-900)



Legend

- 1 Module latch

Precautions**DANGER****Risk of electrical shock and equipment damage**

Grounding is mandatory to satisfy local electrical codes/regulations for the safe use of the equipment.

Ground the rack to the common building network (CBN), isolated bonding network (IBN), or ETSI Mesh Bonding Network ground/protective earth. For details, see [“Grounding standards” on page 12](#) and [Procedure 2, “Connecting the rack ground to the office ground”](#).

Correct grounding of the equipment rack is mandatory in accordance to electrical safety standards and mitigates the electrostatic discharge (ESD) and operational concerns described in [“Consequences of incorrect grounding” on page 17](#). For additional details, see [“Working with power” on page 14](#).

Also, ground every shelf and peripheral tray as described in the related procedure.

**CAUTION****Risk of equipment damage**

Electrostatic discharge (ESD) can damage electrostatic-sensitive devices. Use anti-static protection to avoid damaging modules.

Steps

- 1 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an anti-static wrist strap, connect the cord to the ESD jack on the shelf installed and grounded in a grounded rack/cabinet or clip to a suitable ground point.

- 2 Select your next step.

If you are**Then go to**

installing a filler module

[step 2](#)

removing a filler module

[step 8](#)

Installing a filler module

- 3 Check the rear of the module. If a connector protector is present, unclip and remove the protector.

- 4 Refer to the EDP, IDP, or equivalent site/network engineering plans to make sure that you are installing the module in the correct slot.
- 5 Lift the module by the edges of the faceplate.
- 6 Carefully guide the back end of the module into the required slot, making sure that the edges of the module correctly enter the slot guide rails dedicated to modules.

7

**CAUTION****Risk of module damage**

Do not force a module all the way to the back of its slot if it resists insertion.

**CAUTION****Risk of incorrect installation**

Make sure that the module lock/eject latches are locked into position. The module cannot autoprovision if the latches are not locked.

Slide the module handle's inner latch to the right and push the module towards the back of the shelf until it engages with the backplane. Push on the faceplate to fully seat it.

The latches click into place and lock into the shelf as you push the faceplate towards the backplane.

If the module does not lock into place, remove the module and examine the connector at the back of the module. Look for bent pins or damage.

You have completed the installation of the filler module.

Next, perform [Procedure 13, "Installing or removing Service Modules \(SMs\)"](#).

Removing a filler module

- 8 Slide the filler module handle's inner latch towards the middle until the module connector disengages from the backplane. Pull the module out of the slot.

You have completed the removal of the filler module. If you were referred to this procedure from another procedure, return to the referring procedure.

Procedure 13 Installing or removing Service Modules (SMs)

Use this procedure to install or remove SMs. Refer to the “Module descriptions, equipping rules, and specifications” section in *Waveserver 5 User Guide*, 323-3001-100” for a list of supported SMs.

Requirements

Make sure that you

- observe all safety requirements described in [“Observing product and personnel safety guidelines” on page 3](#)
- have the appropriate personal grounding device to dissipate electrostatic charges
- have the EDP, IDP, or equivalent site/network engineering plans

Overview

All chassis slots must be filled to ensure adequate cooling of the components. Install filler modules in any unused slots. See [Procedure 12, “Installing or removing a filler module”](#) for details.

Waveserver 5 supports the following SMs:

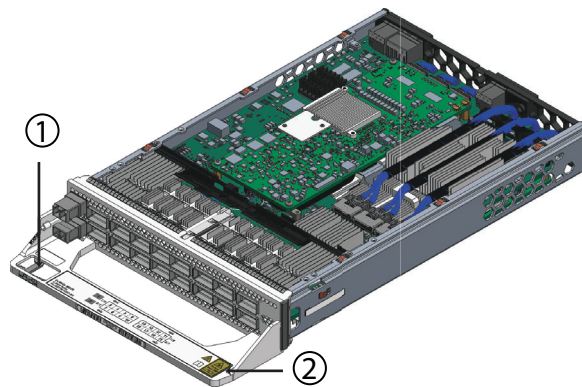
- Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-900)
- Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-901)
- Waveserver WL5e 2x800G L-Band Submarine 16xQSFP28/QSFP-DD Module (186-3110-901)
- Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-900)
- Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-901)
- Waveserver WL5e 2x800G L-Band Premium 16xQSFP28/QSFP-DD Module (186-3111-901)
- Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-900)
- Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-901)
- Waveserver WL5e 2x800G L-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3112-901)

- Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD Module (186-3302-900)
- Waveserver WL5e 2x800G C-Band Multi-Baud Submarine 12xQSFP28/QSFP-DD Module (186-3120-900)
- Waveserver WL5e 2x800G C-Band Multi-Baud Premium 12xQSFP28/QSFP-DD Module (186-3121-900)

For SM details, see the “Module descriptions, equipping rules, and specifications” section of the *Waveserver 5 User Guide*, 323-3001-100.

SMs are installed at the front of the chassis. The following figure shows an example of a Waveserver 5 service module.

Figure 71 Example of Waveserver WL5e 2x800G C-Band and L-Band Submarine 16XQSFP28/QSFP-DD Modules



Legend

- | | | | |
|---|--------------|---|--------------------|
| 1 | Module latch | 2 | Laser hazard level |
|---|--------------|---|--------------------|

Precautions



CAUTION

Risk of equipment damage

Electrostatic discharge (ESD) can damage electrostatic sensitive devices. Use anti-static protection to avoid damaging modules.



CAUTION

Risk of module damage — High

Handle modules with care at all times. Make sure that you do not drop any module, otherwise you will damage it. Also, do not remove connector protectors until you are ready to install the module.

Steps

- 1 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an anti-static wrist strap, connect the cord to the ESD jack on the shelf installed and grounded in a grounded rack/cabinet, or clip to a suitable ground point.
- 2 Select your next step.

If you want to	Then go to
install an SM	step 3
remove an SM	step 10

Installing a service module

- 3 Refer to the EDP, IDP, or equivalent site/network engineering plans to make sure that you are installing the module in the correct slot.
- 4 If applicable, remove the SM from its packaging.
- 5 Check the rear of the module; if a connector protector is present, unclip and remove it before installing the SM.
- 6 Lift the module by the side rail edges.
- 7 Carefully guide the back end of the module into the required slot. Slide the module latch to the right and push the module towards the back of the shelf until it fully engages with the backplane. Push on the faceplate to fully seat it and release the module latch.
- 8 If required, install pluggables in the SM. Refer to [Procedure 14, "Installing or removing pluggable transceivers"](#) for details. Ensure that dust covers are installed on any empty pluggable cages to prevent equipment damage and to facilitate shelf cooling.
- 9 Make sure you have installed all the required SMs and pluggables.
You have completed this procedure. Next, perform [Procedure 14, "Installing or removing pluggable transceivers"](#) or [Procedure 15, "Routing cables from/to modules on the Waveserver 5"](#).

Removing a service module

- 10 Make sure that the green Status LED on the module faceplate is off.
If the green Status LED is on, do not remove the module.
- 11 If the SM you are removing has cables or fibers connected to it, make sure that all cables and fibers are labeled for reconnection. and disconnect any cables or fibers from the module faceplate as applicable. Place protective caps on the optical connectors and optical fiber cables to protect them from contamination and damage.
- 12 Slide the module handle's inner latch towards the middle until the module until the connector disengages from the backplane. Carefully pull the module out of the slot.
You have completed the removal of the SM. If you were referred to this procedure from another procedure, return to the referring procedure.

Procedure 14 Installing or removing pluggable transceivers

Use this procedure to install or remove QSFP28 or QSFP-DD pluggable transceivers.

Requirements

Make sure that

- you have dust covers for any empty pluggable cages. Dust covers are required to prevent damage to empty cages.
- you have the correct type of pluggable and the required number of pluggables. For the list of pluggables supported by each SM, refer to the “Pluggables and specifications” section of the *Waveserver 5 User Guide* 323-3001-100.
- you observe all safety requirements in [“Observing product and personnel safety guidelines” on page 3](#)
- you have the appropriate personal grounding device to dissipate electrostatic charges
- you have the LC fiber tool (174-0099-900). Although optional, this tool assists in installing and removing LC fibers or removing pluggables when space is restricted. When not in use, this tool can be stored in its container.

Overview

The following figures illustrate examples of supported pluggable transceivers.

Figure 72 Example of a QSFP28 pluggable transceiver



Figure 73 Example of a QSFP-DD pluggable transceiver**Precautions****DANGER****Risk of laser radiation exposure**

Do not look into the ends of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.

**CAUTION****Risk of equipment damage**

Electrostatic discharge (ESD) can damage electrostatic sensitive devices. Use anti-static protection to avoid damaging modules.

Steps

- 1 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an anti-static wrist strap, connect the cord to the ESD jack on the shelf installed and grounded in a grounded rack, or clip to a suitable ground point.
- 2 Select your next step.

If you are	Then go to
installing a pluggable transceiver	step 3
removing a pluggable transceiver	step 9

Installing a pluggable transceiver

- 3 If a dust cover is installed on the pluggable cage, remove the dust cover.
- 4 Remove the pluggable transceiver from its packaging and ensure that its optical surfaces are clean. Make sure that the latch on the pluggable transceiver is closed.

- 5 Hold the pluggable transceiver by its sides and orient it so that the connector faces the pluggable cage on the chassis. Orient the pluggable as follows:
 - If you are installing the pluggable transceiver in the top row of the module, the plastic hinge (or tab) is facing up (above the fiber).
 - If you are installing the pluggable transceiver in the bottom row of the module, the plastic hinge (or tab) is facing down (below the fiber).
 - 6 Carefully insert the pluggable transceiver into the cage until it is fully seated.
 - 7 Clean the bulkhead of the pluggable transceiver and the ends of the fiber-optic cables. Refer to [Procedure 19, “Cleaning the optical connector ports of pluggable transceivers”](#) for details.
 - 8 Connect the fiber-optic cables to the pluggable transceiver. Refer to the “Ordering” section of the *Waveserver 5 User Guide*, 323-3001-100 for a list of supported cables.
- You have completed this procedure. Repeat [step 3](#) to [step 8](#) for additional pluggable transceivers, if required.

Removing a pluggable transceiver

- 9 Select your next step.

If	Then
you are using the LC fiber tool	go to step 10
you are not using the LC fiber tool	disconnect the fiber-optic cables from the pluggable. Then go to step 12 .

Figure 74 LC fiber tool (174-0099-900) — two views



Legend

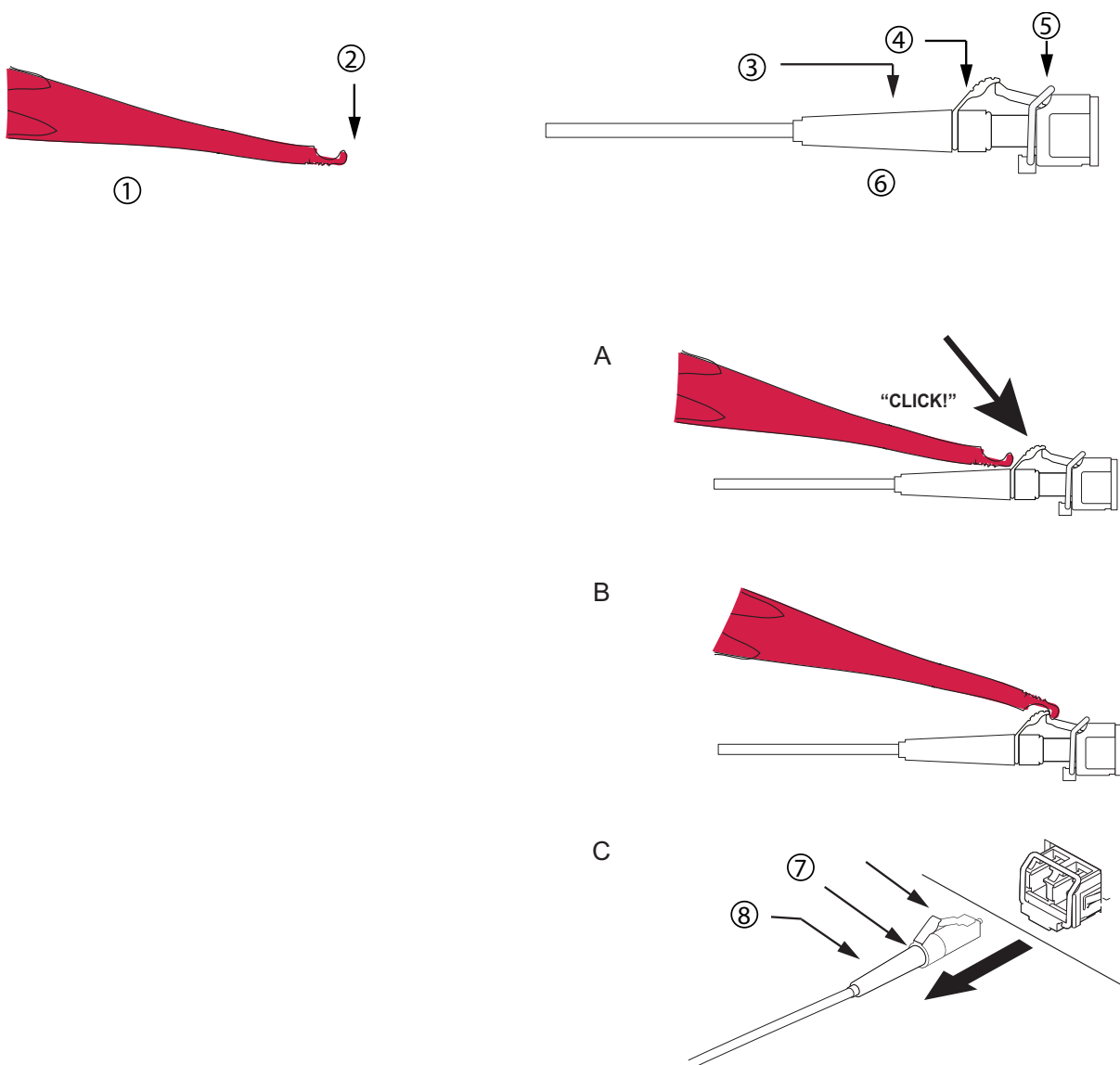
- | | |
|----------|--------|
| 1 Cradle | 2 Hook |
|----------|--------|

- 10 Disconnect the fiber-optic cables from the pluggable transceiver using the LC fiber tool as follows:

The following figure provides an illustration of the steps used, including a close-up of the LC fiber tool (partial view) and a close-up of the fiber [side view] clicked into position on the pluggable.

- a. Hold the LC fiber tool with the hook end pointing up and press down on the LC connector trigger. An audible click indicates that the LC connector is ready to be disengaged.
- b. With the hook end of the LC fiber tool pointing down, carefully disengage the LC connector from the latch of the pluggable.
- c. Grip the boot of the LC fiber between your thumb and index finger, or use the cradle end of the LC fiber tool to grip the boot. Then remove the LC fiber from the pluggable.

Figure 75 Disconnecting fiber-optic cables using the LC fiber tool (174-0099-900)

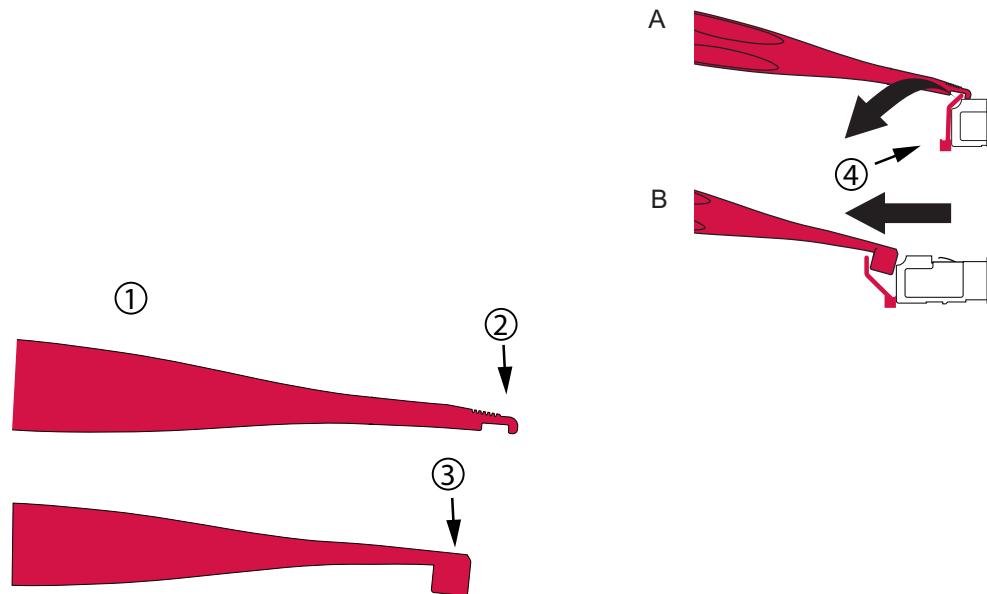


Legend

- | | | | |
|---|--|---|--|
| A | With the hook end of the LC fiber tool pointing up, press down on the LC connector trigger. You will hear a click. | C | Then grip the boot of the LC fiber between your thumb and index finger or use the cradle end of the LC fiber tool to grip and remove the LC fiber from the pluggable module. |
| B | With the hook end of the LC fiber tool pointing down, carefully disengage the LC connector trigger from the latch of the pluggable module. | | |
| 1 | Close-up of LC fiber tool (partial view) | 5 | Latch |
| 2 | Hook | 6 | Side view - close-up of LC fiber, clicked into position in the pluggable module |
| 3 | Boot | 7 | Boot |
| 4 | LC connector trigger | 8 | Cradle of LC fiber tool |

Removing a pluggable transceiver

- 11** Remove a pluggable transceiver using the LC fiber tool as follows.
- The following fiber provides an illustration of the following steps, including close-ups of the LC fiber tool (partial views).
- With the LC fiber tool pointing downwards, open the latch of the pluggable transceiver using the hook end of the tool.
 - With the cradle end pointing downwards, carefully pull the latch to remove the pluggable from the SM.
- Then go to [step 14](#).

Figure 76 Removing a pluggable using the LC fiber tool (174-0099-900)**Legend**

- | | |
|---|--|
| <p>A Open the latch of the pluggable module using the hook (pointing down as shown) of the LC fiber tool.</p> | <p>B Then using the cradle (pointing down as shown) carefully pull the latch to remove the pluggable module from the circuit pack.</p> |
|---|--|

- | | |
|---|--|
| <p>1 Close-ups of LC fiber tool (partial views)</p> <p>2 Hook pointing down</p> | <p>3 Cradle pointing down</p> <p>4 Latch</p> |
|---|--|

- 12** Swing open the latch on the pluggable transceiver to disengage it from the SM. Open the bail latch until it is perpendicular to the face of the pluggable.
- 13** Use the bail latch to carefully pull the pluggable out of its cage on the SM. Make sure that you apply the extraction force in a direction parallel to the long dimension of the pluggable.
- 14** If the pluggable cage will remain empty, insert a dust cover into the cage. The dust cover prevents contamination of the electrical connector.
- 15** Place the pluggable transceivers back into their ESD-protective packaging. You have completed this procedure.

Procedure 15 Routing cables from/to modules on the Waveserver 5

Use these guidelines to route cables to or from modules.

Overview

Use the examples in the following figures to route cables from/to modules on the Waveserver 5.

Ciena recommends that

- cables exiting the CP are routed to the left side (facing the front of the chassis) through the fiber manager
- cables exiting the AP are routed to the right side (facing the front of the chassis) through the fiber manager
- cables exiting the SMs installed on the left side of the chassis are routed to the left, through the fiber manager
- cables exiting the SMs installed on the right side of the chassis are routed to the right, through the fiber manager

Bundle fibers using hook and loop fasteners, such as Velcro.

Figure 77 Fibering example



Procedure 16 Connecting Active Optical Cables (AOC)

Use this procedure to install a QSFP28-QSFP28 Active Optical Cable (AOC) (160-9460-xxx) /160-9660-xxx: 400GE, QSFP-DD - QSFP-DD to connect a Waveserver 5 client port to subtending equipment. See “Pluggables and specifications” section in the *Waveserver 5 User Guide*, 323-3001-100.

Requirements

Make sure you

- observe all safety requirements described in [“Observing product and personnel safety guidelines” on page 3](#)
- have the appropriate personal grounding device to dissipate electrostatic charges
- have the EDP, IDP, or equivalent site/network engineering plans
- have the required number of QSFP28-QSFP28 Active Optical Cables (160-9460-xxx) of the correct length
- have the required number of 160-9660-xxx: 400GE, QSFP-DD - QSFP-DD of the correct length

Precautions



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit, modules, and cables.

Steps

- 1 Remove the Active Optical Cables from the protective packaging.
- 2 Remove the dust caps from the ends of the cables.
- 3 Insert one cable plug end into the Waveserver 5 client port. For ports in the bottom row of the module, ensure that the pull-tabs are facing downward. In the top row of the module, ensure that the pull-tabs are facing upward.
- 4 Insert the other cable plug into the port of the subtending equipment.
- 5 Enable the Waveserver 5 client port:

```
port enable port <slot/port>
```

 You have completed this procedure.

CHAPTER 5

Cleaning connectors

This section describes how to clean optical fiber connectors. Ciena recommends that you always clean optical fiber connectors before connecting to transmission equipment, test equipment, patch panels, or other connectors.

This section contains the following topics:

- [“Precautions” on page 141](#)
- [“List of procedures” on page 143](#)

Connector cleaning:

- Ensures that fiber-optic assemblies work at their correct power level, by removing any dust or particles that could lead to optical power attenuation.
- Prevents damage to alignment sleeves and adapter housings.

Precautions

**CAUTION****Risk of shelf malfunction**

Ciena recommends that you do not use cellular phones at any Waveserver 5 site. The use of cellular phones in proximity to equipment can cause equipment malfunction.

**DANGER****Risk of personal injury**

Ensure the optical fiber patch cord that you want to scope does not carry optical power.

**DANGER****Risk of laser radiation exposure**

Do not look into the ends of optical fiber connectors or into faceplate connectors of installed modules. The light source used in fiber-optic devices can damage your eyes.



DANGER

Risk of personal injury

Wear eye protection such as safety goggles or safety glasses with side guards when you work with canned compressed gas to avoid getting dust in your eyes.

Wear eye protection when working with isopropyl alcohol. Isopropyl alcohol can cause eye irritation on contact.

To avoid extended inhalation of alcohol vapors, use only small amounts of alcohol and work in a clean and ventilated area.

To avoid direct skin contact with the isopropyl alcohol and to avoid contaminating the ferrules with skin oils, it is preferable to wear chemical-resistance gloves (latex, vinyl/PVC or nitrile type).



CAUTION

Risk of connector contamination

To reduce the risk of oil contamination on connectors, do not use commercial compressed air or hose air when cleaning connectors and receptacles. Always use canned compressed gas to remove excess dust and dirt.

Follow the appropriate cleaning process before re-mating connectors with their receptacles.

To prevent contamination, make sure you cover the optical faceplate connectors of all installed modules and all the unconnected fiber connector ferrules with dust caps.

To avoid applying any oil from your fingers or other contaminants to the end face of the ferrule, handle connectors carefully.



CAUTION

Risk of signal degradation

Do not allow the solvent to dry on the ferrule. Dried solvent leaves a residual film on ferrule surfaces that prevents the fiber-optic assembly from working correctly.

**CAUTION****Risk of equipment damage**

Clean the port of the module or patch panel as required. Clean the associated optical connector as required.

Then immediately after cleaning, insert the clean optical connector into its port (on the module or patch panel), making sure that the optical fiber does not come into contact with any object or surface. Do not touch the port or optical fiber.

If the optical connector was not inserted into its port immediately after cleaning, you must re-inspect and clean the optical connector.

**CAUTION****Risk of equipment damage**

Electrostatic discharge (ESD) can damage electrostatic sensitive devices. Use antistatic protection to avoid damaging modules.

List of procedures

The following table provides a list of procedures for cleaning and inspecting optical fiber connectors.

Table 14 Optical fiber connectors — cleaning and inspection procedures

Procedure	Comment
Procedure 17, “Inspecting and cleaning optical interface connectors”	Perform as required or when referred to by another procedure.
Procedure 18, “Cleaning optical connectors and adapters on patch cords”	Perform as required or when referred to by another procedure.
Procedure 19, “Cleaning the optical connector ports of pluggable transceivers”	Perform as required or when referred to by another procedure.
Procedure 20, “Inspecting and cleaning MPO connectors”	Perform as required or when referred to by another procedure.

Procedure 17 Inspecting and cleaning optical interface connectors

Use this procedure to inspect and clean optical interface connector sleeves on modules. Internal fibers are fully verified and cleaned as a last step in module manufacturing.

Requirements

Make sure that you

- read the precautions in the section on handling optical fibers in [“Handling optical fibers” on page 12](#) and in [“Precautionary messages” on page 4](#).
- have the following:
 - fiber inspection tool equipped with a fiber inspection probe with 1.25 mm adapter for LC connectors
 - canned compressed gas
 - rubber or chemical-resistance gloves (latex, vinyl/PVC or nitrile type)
 - eye protection such as safety goggles or safety glasses
 - single-action fiber cleaning tool (refer to the following figure)

Figure 78 Example of single-action fiber cleaning tool



Legend

1 Tool wand cap

2 Tool wand

- have the LC fiber tool (174-0099-900), optional.

The LC fiber tool is available to assist in installing and removing LC fibers or removing pluggables when space is restricted. When not in use, this tool can be stored in its container.

Precautions

Before you disconnect optical fiber patch cords and remove modules from the shelf, make sure that you do not affect traffic.



CAUTION

Risk of equipment damage

Electrostatic discharge (ESD) can damage electrostatic sensitive devices. Use anti-static protection to avoid damaging modules.

Steps

- 1 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an antistatic wrist strap, connect the cord to the ESD jack on another shelf installed in a grounded bay/cabinet or clip to a suitable ground point.
- 2 Wear eye protection such as safety goggles or safety glasses.



DANGER

Risk of laser radiation exposure

Do not look into the end of optical fiber connectors or into faceplate connectors of installed modules. The light source used in fiber-optic devices can damage your eyes.

- 3 Select your next step.

If

you are inspecting (and cleaning, if required) LC bulkhead connectors on modules
otherwise

Then go to

[step 4](#)

[step 11](#)

Inspecting (and cleaning, if required) LC bulkhead connectors on modules

4 Select your next step.

If optical patch cords

are connected to the module

Then

before you remove the module from its slot, make sure that:

- the optical patch cords are labeled
- traffic is routed away from the path you are inspecting
- there is no optical signal power present on the module and the optical patch cords you are handling

Then go to [step 5](#).

are not connected to the module

remove the dust cap from the bulkhead connector. Store the dust cap.

Then go to [step 7](#).

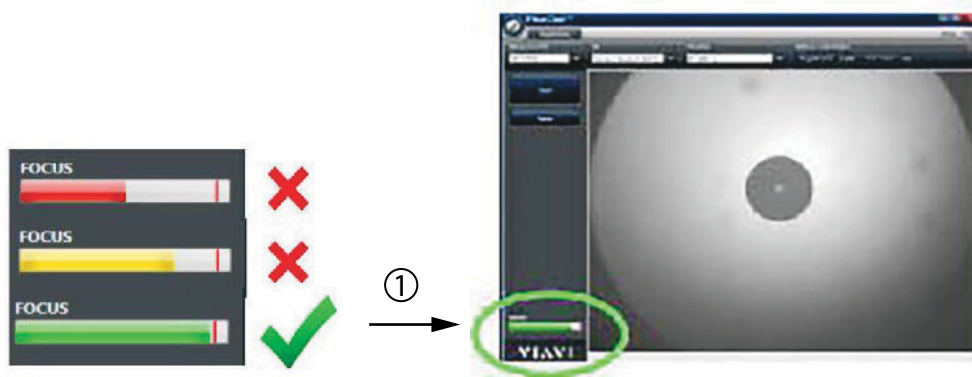
5 Loosen the optical patch cords to provide enough slack.

6 Disconnect the optical patch cords. See the procedure on connecting or disconnecting fiber-optic cables.

7 Do not proceed directly to cleaning. First inspect the module connector to determine whether cleaning is required.

- Use the appropriate adapter on the fiber scope and insert onto the bulkhead connector.
- Turn on the light in the fiber scope and adjust the focus (see the following figure).

Figure 79 Focusing the fiber scope—clean fiber example



Legend

1 Focus

- 8 Identify the inspection zones (see, [Figure 82 on page 152](#)).
- 9 Select your next step.
- | | |
|---|---|
| If the module connector is | Then |
| clean (Figure 84 on page 153) | go to step 19 |
| not clean (Figure 83 on page 153) | use the single-action fiber cleaning tool to clean the connector, go to step 10 |
- 10 If the module connector is not clean ([Figure 83 on page 153](#)), use the single-action fiber cleaning tool to clean the connector:
- a. Remove the protective cap from the tool wand.
 - b. Insert the wand into the female optical surface with a firm, steady motion until the tool makes an audible click, then release. Repeat as necessary until the endface inspection shows a clean mating surface. Then go to [step 19](#).
- 11 Select your next step.
- | | |
|--|--|
| If optical patch cords | Then go to |
| are connected to the module (including to pluggable modules on the module) | loosen the optical patch cords from the shelf fiber-optic guides to provide enough slack. Then go to step 12 . |
| are not connected to the module | step 13 |
- 12 Disconnect the optical patch cords from the module or pluggable module. You can use the optional LC fiber tool to disconnect an LC fiber patch cord.
- Figure 80** LC fiber tool (174-0099-900)



Legend

1 Cradle

2 Hook

- 13** Remove the module from the shelf.

Cleaning the LC connector sleeve (dry optical swab)

- 14** Select your next step.

If there are	Then go to
protective caps on the connector sleeve	step 15
no protective caps on the connector sleeve	step 16

- 15** Remove the protective caps from the adapter housing. Store the protective caps in a clean ESD plastic bag until you have completed cleaning the adapter housing and alignment sleeve.
- 16** Insert a new, clean optical swab into the connector sleeve. Use a 1.25 mm optical swab.
- 17** While rotating the swab in one direction, remove the swab from the connector sleeve. If necessary, repeat this step with a new, clean optical swab.
- 18** If you are not using the connector sleeve immediately, cover the connector with a clean dust cap.
- 19** Insert the module in the shelf.
- 20** Clean and inspect the optical connectors and adapters on the external patch cords before you re-connect them to the module. See [Procedure 18](#), "Cleaning optical connectors and adapters on patch cords".

Procedure 18 Cleaning optical connectors and adapters on patch cords

Use this procedure to clean LC optical connectors or adapters on patch cords.

Requirements

Make sure that you

- read the precautions on handling optical fibers in the “Observing product and personnel safety guidelines” section in this document.
 - clean all optical connectors and universal adapters inside and outside the module before installation.
 - To clean optical connectors and adapters inside modules, see [Procedure 17, “Inspecting and cleaning optical interface connectors”](#).
 - To clean optical connector ports on pluggable modules, see [“Cleaning the optical connector ports of pluggable transceivers” on page 157](#).
 - have the following:
 - fiber microscope with 1.25 mm adapter for LC connectors
 - cleaning cassette (as required)
 - lint-free tissues
 - isopropyl alcohol (only as last resort)
 - canned compressed gas (to clean the adapter on the fiber microscope)
 - eye protection such as safety goggles or safety glasses
 - have the LC fiber tool (174-0099-900), optional
- The LC fiber tool is available to assist in installing and removing LC fibers or removing pluggable transceivers (SFP) when space is restricted. When not in use, this tool can be stored in its container.

Precautions



CAUTION

Risk of equipment damage

Electrostatic discharge (ESD) can damage electrostatic sensitive devices. Use anti-static protection to avoid damaging modules.

Steps

- 1 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an antistatic wrist strap, connect the cord to the ESD jack on another shelf installed in a grounded bay/cabinet or clip to a suitable ground point.

- 2

**CAUTION****Risk of equipment damage**

After you inspect and clean a port by performing [Procedure 17, "Inspecting and cleaning optical interface connectors"](#) and then inspect and clean its associated optical connector (by performing the steps in this procedure), you must immediately re-insert the optical connector into its port.

Do not touch the optical fiber. Make sure that the optical fiber does not come into contact with any surface or object. Otherwise, you must repeat this procedure (re-inspect and clean).

Disconnecting the optical fiber patch cord

- 3 Wear eye protection such as safety goggles or safety glasses.

**DANGER****Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors of installed modules. The light source used in fiber-optic devices can damage your eyes.

- 4

**DANGER****Risk of personal injury**

Ensure the optical fiber patch cord that you want to scope does not carry optical power.

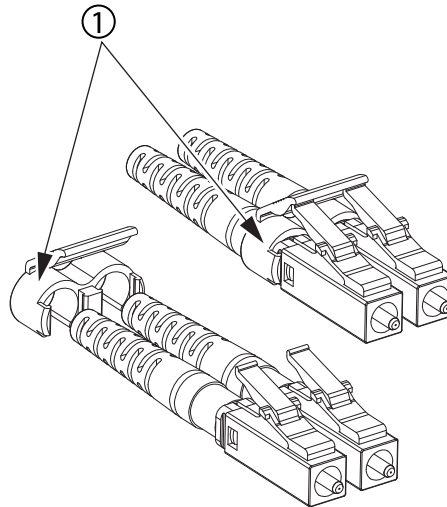
- 5 Loosen the optical patch cord you want to clean from the shelf fiber-optic guides to provide enough slack.
- 6 Make sure that the adapter on the fiber microscope is clean. Clean as required with compressed gas.

Hold the compressed gas can upright and position the tip of the nozzle as close as possible, but not in physical contact, to the adapter on the fiber microscope.

If you do not hold the can upright, pressurizing inert gas can be released and leave contamination that you cannot remove.
- 7 Apply short blows of canned gas to remove dust.
- 8 Disconnect the optical patch cord from the module.

The connectors are LC, the plastic snap-in type with RJ style latch (see the following figure).

Figure 81 Duplex clip on optical fiber patch cords



Legend

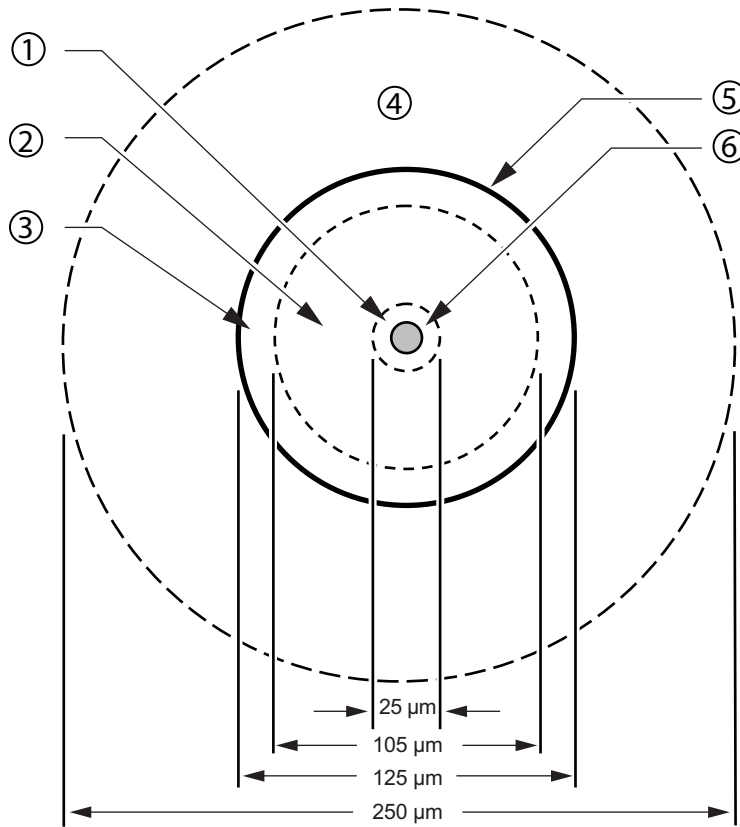
1 Duplex Clip

You can use the optional LC fiber tool to disconnect an LC fiber patch cord.

Inspecting the optical connector

- 9** Insert the optical fiber into the fiber microscope adapter.
- 10** Turn on the light in the fiber microscope.
- 11** Adjust the focus so that you can identify the four inspection zones of the optical fiber end. For the definition of the inspection zones, refer to the following figure.

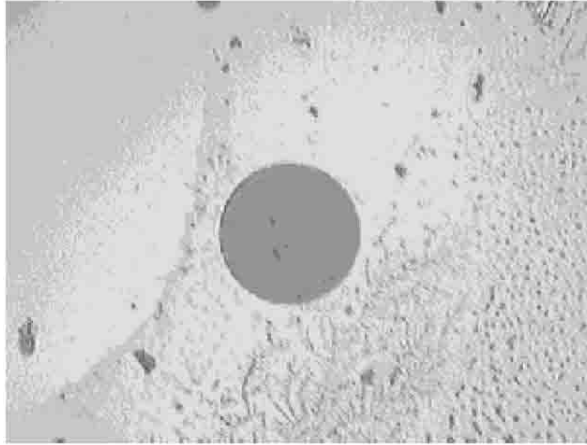
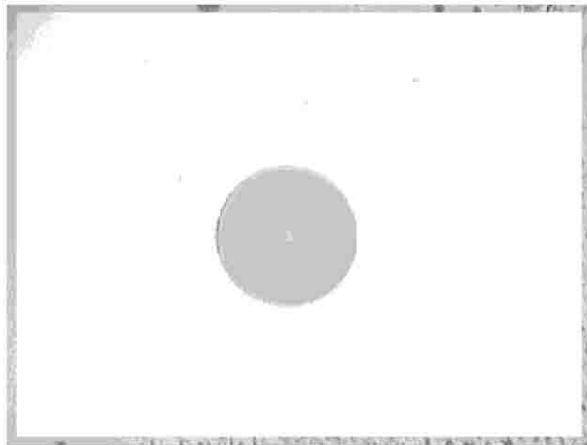
Figure 82 Definition of inspection zones (single-mode fiber)



Legend

- | | | | |
|---|--------|---|----------|
| 1 | Zone A | 4 | Zone D |
| 2 | Zone B | 5 | Cladding |
| 3 | Zone C | 6 | Core |

For examples of dirty and clean fibers, see the following two figures.

Figure 83 Example of a dirty optical fiber**Figure 84** Example of a clean optical fiber***Cleaning the connector ferrule (dry cleaning)***

- 12** Put the end face connector in the center of a new, lint-free tissue. Applying moderate pressure, wipe the ferrule or plug by rotating the connector along the container and the ferrule or plug tip. Perform four or five rotations to remove any dirt on the ferrule or plug.

Use a new, lint-free tissue for each connector. Avoid contamination of the tissue from dirty surfaces.

13

**DANGER****Risk of personal injury**

Ensure the optical fiber patch cord that you want to scope does not carry optical power.

- 14** Insert the connector into a fiber microscope and examine the connector. For examples of dirty and clean fibers, see [Figure 83 on page 153](#) and [Figure 84 on page 153](#).

- 15** Select your next step.

If the connector ferrule is	Then
still dirty after the first attempt of dry cleaning	repeat dry cleaning (step 12 to step 14)
still dirty after two attempts of dry cleaning	go to step 16 to perform wet and dry cleaning
clean	go to step 20

Cleaning the connector ferrule (wet and dry cleaning)

- 16**



DANGER

Risk of vapor inhalation

Always work in a clean and ventilated area to avoid extended inhalation of alcohol vapors. Use small amounts of alcohol to clean optical connectors.

Wet one corner of a lint-free cloth with alcohol or open a pre-moistened cloth. Wipe across the surface of the ferrule.

Always use a new cloth.

- 17**

While the ferrule is still wet with solvent, dry as follows:

- Put a clean, optical grade, lint-free cloth (or equivalent) on a flat surface.
- Gently press the ferrule on the cloth and wipe in one direction.
- Rotate the connector one quarter of a turn and wipe again.

Do not allow the alcohol to dry on the ferrule. The alcohol can leave a film on the surface.

Always use a new, clean cloth for each connector. Make sure you use only the untouched sections of the cloth.

If a flat surface is not available or if the ferrule is not clean enough, use a cleaning cassette according to manufacturer instructions.



DANGER

Risk of personal injury

Ensure the optical fiber patch cord that you want to scope does not carry optical power.

- 18**

Insert the optical connector into a fiber microscope and inspect.

- 19**

Select your next step.

If	Then
still dirty	repeat step 16 to step 18
still dirty or scratched after multiple attempts of cleaning	polish or replace the optical patch cord. Then inspect and clean the new optical patch cord as required. Go to step 9 .
clean	go to the next step

Connecting the optical connector

- 20 Select your next step.

If you want to	Then go to
inspect and clean the optical interface internal connectors and fiber	step 21
connect the optical connector to the module	step 23

- 21 Place the connector in a clean adapter housing to prevent the ferrule from coming into contact with any dirt.
Do not place a clean connector in an adapter that has not been cleaned.
- 22 Inspect and clean the optical interface internal connectors and fiber. See [Procedure 17, "Inspecting and cleaning optical interface connectors"](#). Then go to [step 24](#).

Connecting the optical connector to the module

- 23 Follow this caution message for all modules.



CAUTION

Risk of equipment damage

Immediately after cleaning, insert the clean optical connector into the required port of the module, making sure that the optical fiber does not come into contact with any object or surface. Do not touch the port or optical fiber.

If the optical connector was not inserted into its port immediately after cleaning, you must re-inspect and clean the optical connector, starting with [step 9](#). Otherwise, go to [step 24](#).

- 24 Insert the ferrule into the sleeve so the guide pin enters the guide slit.
- 25 Hold the adapter retainer and lock it in by pushing it into the connector.
- 26 Repeat [step 23](#) to [step 25](#) for the second connector and adapter, as applicable.
- 27 Identify the patch panels (at the site) that are associated with the fiber of the optical connector, which you inspected and cleaned.

There can be more than one patch panel associated with a fiber that is exiting a site.

- 28** Make sure to inspect and clean the applicable ports and optical connectors of the patch panels.



CAUTION

Risk of equipment damage

After you inspect and clean a port on the patch panel and inspect and clean its associated optical connector, you must re-insert the optical connector immediately into its port on the patch panel.

Do not touch the optical fiber. Make sure that the optical fiber does not come into contact with any surface or object. Otherwise, you must re-inspect and clean the optical connector.

Procedure 19 Cleaning the optical connector ports of pluggable transceivers

Use this procedure to clean the optical connector ports of SFP transceivers.

Requirements

Make sure that

- you read the precautions on handling optical fibers in [“Observing product and personnel safety guidelines” on page 3](#) and in [“Precautionary messages” on page 4](#)
- before disconnecting optical fiber patch cords and removing pluggables from the host modules, you
 - make sure not to affect traffic
 - verify that the connectors on the optical fiber patch cords are labeled
- you have the following fiber cleaning materials:
 - 1.25 mm optical cleaning swabs
 - canned compressed gas
- you have the LC fiber tool (174-0099-900), optional.

The LC fiber tool is available to assist in installing and removing LC fibers or removing pluggables (SFP) when space is restricted. When not in use, this tool can be stored in its container.

Overview

Perform the cleaning of optical pluggable transceivers in accordance with the specific pluggable supplier's recommendations. If no recommended cleaning procedures are available from the specific component supplier, cleaning may be performed assuming a degree of risk.

ATTENTION

There is no guarantee that all the dirt present on the connector port of the pluggable will be removed and there is a risk of damaging the connector's end-face while performing this procedure.

It is recommended that you clean the optical connectors on pluggable transceivers only:

- during initial installation
- when you suspect performance is reduced due to contamination

Precautions



CAUTION
Risk of equipment damage
Electrostatic discharge (ESD) can damage electrostatic sensitive devices. Use anti-static protection to avoid damaging modules.

Steps

- 1

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an anti-static wrist strap, connect the cord to the ESD jack on another shelf installed in a grounded bay/cabinet or clip to a suitable ground point.
- 2

Select your next step.
- | If | Then go to |
|--|------------------------|
| optical patch cords are connected to the pluggable | step 3 |
| there are no optical patch cords connected to the pluggable | step 5 |
| the pluggable is not inserted in a host module and dust caps or optical terminators are placed in the module receptacles | step 7 |
- 3

Loosen the optical patch cords from the shelf fiber-optic guides to provide enough slack.
- 4

Disconnect the optical patch cords from the pluggable.
You can use the optional LC fiber tool to disconnect an LC fiber patch cord.

Figure 85 LC fiber tool (174-0099-900)



Legend

- 1 Cradle
- 2 Hook

- 5

Remove the pluggable from the host module.
- 6

Select your next step.

		If there are	Then go to
		dust caps placed in the module receptacles	step 7
		no dust caps placed in the module receptacles	step 8
7	Remove the dust caps or optical terminators from the module receptacles and place them in a clean, resealable container.		
8	Select your next step.		
		If a bulkhead inspection tool is	Then go to
		available	step 9
		not available	step 10
9	Inspect the port.		
		If the port looks like	Then
		Figure 86	identify the inspection zones (see, Figure 88 on page 160) and if cleaning is required go to step 10
		Figure 87	do not clean this port. Go to step 15 .

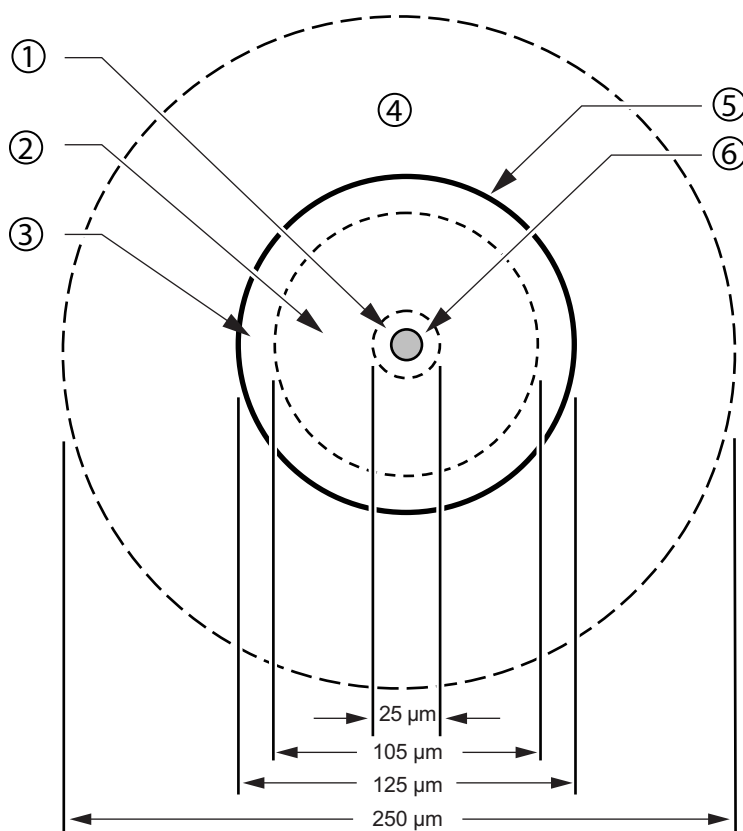
Figure 86 Example of transceiver port with ferrule



Figure 87 Example of transceiver port without ferrule



Figure 88 Definition of inspection zones (single-mode fiber)



Legend

1	Zone A	4	Zone D
2	Zone B	5	Cladding
3	Zone C	6	Core

- 10** Hold the compressed gas can upright and position the tip of the nozzle as close as possible, but not in physical contact, to the inside of the receptacle. If you do not hold the can upright, pressurizing inert gas can be released and leave contamination that you cannot remove.
- 11** Apply short blows of canned gas into the receptacle to remove mild contaminants.
- 12** If contaminant presence is still suspected, carefully insert the tip of a dry cleaning swab into the receptacle until you apply light, perpendicular force to the end surface of the fiber stub.
The pressing force must be less than the maximum spring force of the mating plug (0.7 kgf for LC connectors).
- 13** Gently rotate the swab three to five full turns.
- 14** Remove and discard the swab.

15 Select your next step.

If	Then
you will not use the cleaned ports immediately	place a clean dust cap or optical terminator, as applicable, on the ports.
you removed the pluggable from a host module and there were no optical fiber patch cords connected to the pluggable	• If you will not use the ports immediately, place clean dust caps or optical terminator, as applicable, on the ports. • If you are using the ports, insert the pluggable. • Before you connect optical patch cords to the pluggable, make sure that the connectors on the optical fiber patch cords are clean (see Procedure 18, “Cleaning optical connectors and adapters on patch cords”).
you removed the pluggable from a host module and there were optical fiber patch cords connected to the pluggable	• Insert the pluggable. • Make sure that the connectors on the optical fiber patch cords are clean (see Procedure 18, “Cleaning optical connectors and adapters on patch cords”). • Reconnect the fibers into their original location on the pluggable (refer to the labels). • Verify that the system performance is not affected by contaminants anymore.

Procedure 20 Inspecting and cleaning MPO connectors

Use this procedure to clean Multifiber Push-On (MPO) connectors.

Requirements

Make sure that

- you read the precautions on handling optical fibers in [“Handling optical fibers” on page 12](#) section in this document.
- before you disconnect optical fiber patch cords, make sure not to affect traffic, and verify that the connectors on the optical fiber patch cords are labeled
- you have the following tools and materials:
 - MPO adapters (required to inspect female MPO connectors, see [step 6](#))
 - inspection tool. The following figure illustrates a fiberscope example (with angled MPO adapter).
 - MPO connector cleaners; refer to the following figure for an example.

Figure 89 Fiberscope example



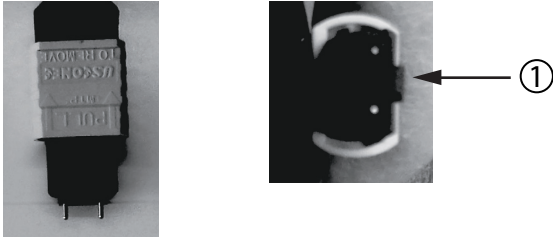
Figure 90 Example of MPO connector cleaner



Overview

The following illustrations provide examples of MPO connectors.

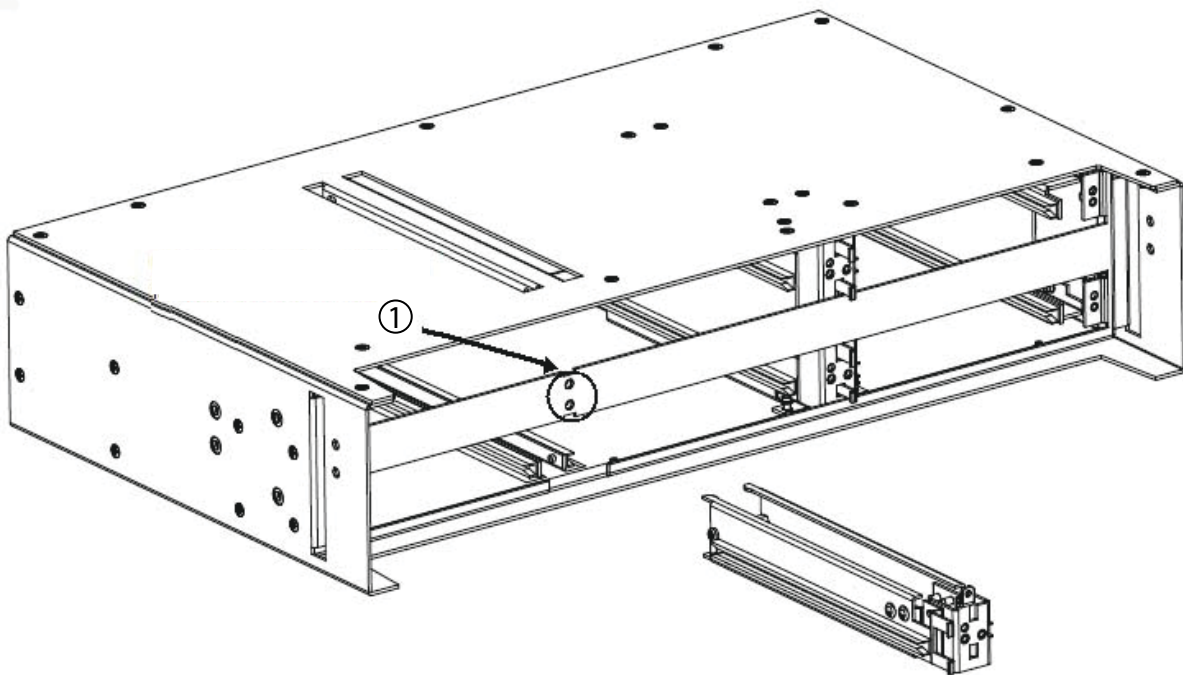
Figure 91 MPO connector example (male)



Legend

- 1 Key

Figure 92 MPO connector example (female)



Legend

- 1 Unscrew to remove partitions for full-width filter modules

Precautions



CAUTION

Risk of equipment damage

Electrostatic discharge (ESD) can damage electrostatic sensitive devices. Use anti-static protection to avoid damaging circuit packs.



CAUTION

Risk of affecting performance

You must inspect and clean the MPO connectors after removing the MPO cable from its original packaging or if the MPO cable has been removed from the faceplate for any reason.

Always inspect the MPO connectors before making a connection.

Steps

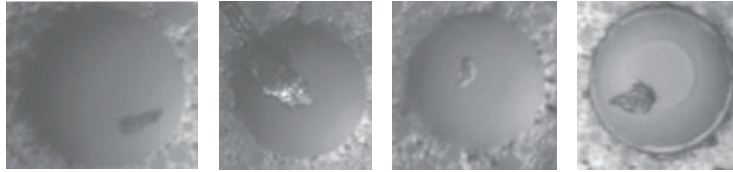
- 1 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an anti-static wrist strap, connect the cord to the ESD jack on another shelf installed in a grounded rack/cabinet or clip to a suitable ground point.
- 2 Select your next step.

If there are	Then go to
dust caps or MPO loopbacks placed on the applicable ports	step 3
no dust caps or MPO loopbacks placed on the applicable ports	step 4
- 3 Remove the dust caps or MPO loopbacks from the applicable ports and place them in a clean resealable container.
- 4 Select your next step:

If you are inspecting	Then go to
a male MPO connector	step 5
a female MPO connector	step 6
- 5 The connector has a key that indicates the correct alignment and mating position. Make sure you are placing the key in the correct position and apply the fiberscope inspection tip to the male connector. Then go to [step 7](#).
- 6 To inspect a female connector, you must use an external MPO adapter. The connector and the adapter have keys that indicate the correct aligning and mating position. Make sure you are placing the keys in the correct position, then insert the connector into one side of the adapter and the inspection tip into the other side of the adapter.

- 7 Inspect the connector. See the following figure for examples.

Figure 93 Examples of dirty optical fibers



ATTENTION

Always clean the connectors on MPO cables before you connect the cables to equipment.

- 8 Select your next step.

If you are cleaning	Then go to
a male connector	step 9
a female connector	step 10

- 9 To clean a male connector:

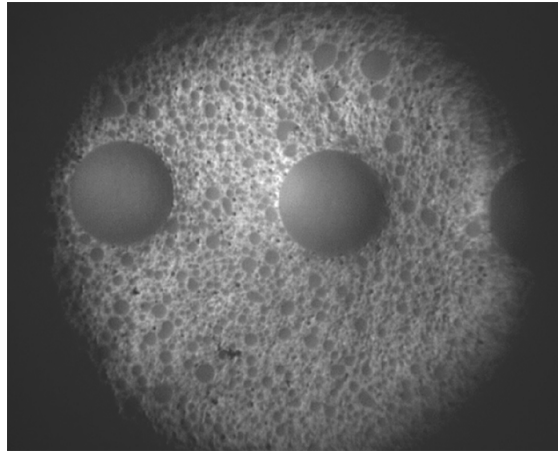
- Remove the cleaner attachment.
- Insert the cleaner tip with the key up directly into the MPO adapter.
- Press until you hear a click and remove straight out.

Then go to [step 11](#).

- 10 To clean a female connector:

- Insert the female MPO connector into the cleaner attachment with the key up until it clicks.
- Insert the cleaner tip with the key up directly into the attachment.
- Press until you hear a click and remove straight out.

- 11 Re-inspect the connector after cleaning. See the following figure for an example of clean optical fiber.

Figure 94 Example of a clean optical fiber

12 Select your last step.

If	Then
you will not use the cleaned connector immediately	place a clean dust cap (or an MPO loopback, as applicable) on the connector
an MPO loopback must be installed	connect the MPO loopback
you are connecting a cable	return to the referring procedure or go to the procedure on connecting cables

CHAPTER 6

List of terms

The following table provides a list of terms used in this technical publication.

AC	alternating current
AOC	active optical cable
AP	Access Panel
ASIC	Application Specific Integrated Circuit
AWG	American Wire Gauge
BW	bandwidth
COLAN	central office local area network
CP	control plane
DAC	direct attach copper cable
DC	direct current
DCN	Data Communications Network
EDP	engineering documentation package
EIA	Electronics Industries Alliance
ESD	electrostatic discharge
ETSI	European Telecommunications Standards Institute
FRU	Field Replaceable Unit
GE	Gigabit Ethernet
IDP	installation drawing package
IEC	International Electrotechnical Commission
ILAN	integrated local area network
LAN	local area network

LED	light-emitting diode
MOTR	Multiplexing Optical Transponder
PCB	Printed Circuit Board
PEC	Product Engineering Code
PSU	Power Supply Unit
PTE	Physical Telecommunication Environment
QSFP28	Enhanced Quad (4-channel) Small Form-factor Pluggable 28
QSFP-DD	Enhanced Quad (4-channel) Small Form-factor Pluggable 28 double density
RTFD	Return to Factory Default
RX	receive
SM	Service Module
TX	transmit
USB	Universal Serial Bus
XFP	large form-factor pluggable

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